SUSQUEHANNA × UNSW FINTECH SOCIETY

ALGOTHON 2026 · TECHNICAL RESEARCH REPORT

Forecasting Under a Limited Effective Sample

Sequential Strategy Development under Staged Data Release

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Post-competition Research Report

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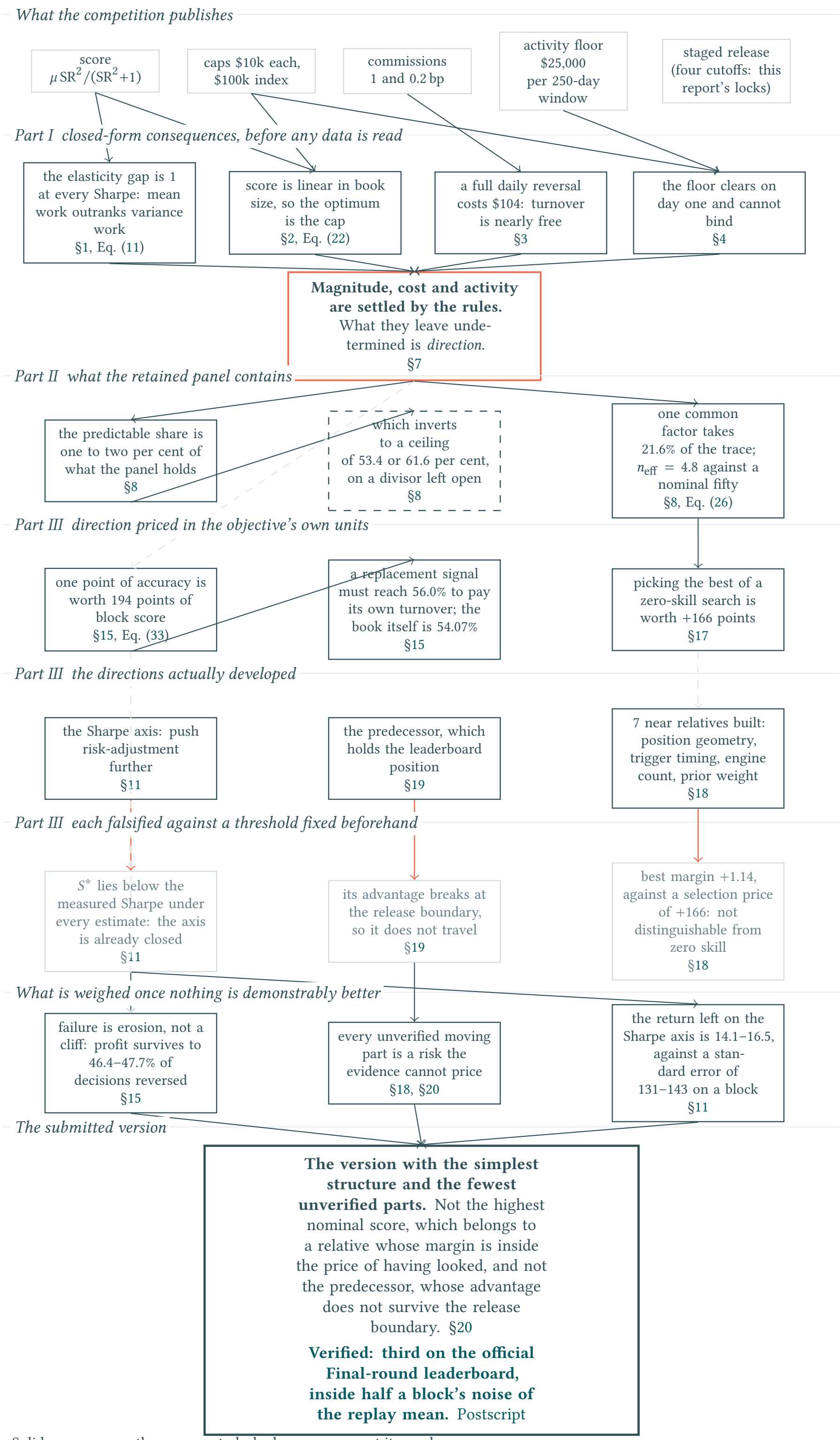
Abstract

At the final model-development cutoff, the available panel contains 1,500 daily observations for each of 51 assets—76,500 cells nominally, but not 76,500 independent observations. It is one realised multivariate market history with serial dependence, cross-asset and common-factor dependence, and possible non-stationarity. Repeated reuse of that same history creates adaptive selection and multiple-testing bias in the research process; it does not create dependence in prices. This report asks how a forecasting and trading strategy should be developed when market history is released in stages but the effective quantity of independent information remains limited.

The study reconstructs official rules and evaluator mechanics before examining data or candidate strategies. It then fixes contemporaneous information sets at cutoffs of 500, 750, 1,000, and 1,500 observed days, and adopts the rule that hypothesis formation, validation design, candidate comparison, and model locking use only evidence available at the cutoff, later observations informing the next stage without rewriting an earlier decision. That rule is a standard this report holds itself to and evidences only through its dated artefact record: no contemporaneous narrative of the cutoffs was kept, and the report says so rather than reconstructing one after the fact. Official facts, reproducible local replays, secondary transcriptions, and unverified or unknown claims are reported as distinct evidence classes.

The resulting methodological position favours stability across time, cost-aware, leakage-controlled time-series validation, shrinkage or simpler specifications where supported, and explicit rejection of complexity that fails to survive out-of-sample checks. Nominal panel dimensions therefore do not remove the limited-effective-sample problem. Local replays are not official Final-round evidence, and the body of this report makes no verified claim about Days 1,501–2,000 performance, submission acceptance, or final rank; the one outcome verified after the event — third place on the official Final-round technical leaderboard, within half a block's standard error of the public replay mean — is confined to a dated postscript. Its contribution is a disciplined, auditable protocol for sequential model development rather than an unsupported assertion of final competition performance.

Keywords: algorithmic trading; limited effective sample; time-series validation; staged data release; model selection; risk-adjusted performance.



Solid arrows carry the argument; dashed arrows support it; coral arrows carry a falsification. Grey boxes are directions the evidence closed; the dashed box is a step the report withdrew; the coral border marks the one question the rules leave open. Every node names what establishes it, and every number is read from a retained record.

Figure 1: The argument, from the published constants to the submitted version. Read downwards: the rules settle three of the four axes a strategy could work on and leave direction open; Part II prices what direction is worth and how much of it this panel can hold; Part III develops the directions that remain and falsifies each against a threshold fixed before any candidate was examined; what survives is chosen on structure rather than on score. Every node names the equation or section that establishes it.

Contents

1 The Score Function: Avoid Loss First, Then Maximise the Mean

The published rule. Write PL for the sequence of daily net dollar profit and loss the evaluator records over the active scoring window, read as a whole. The competition publishes the following score [2]:

$$\text{Score} = \begin{cases} \mu \dfrac{\mathrm{SR}^2}{\mathrm{SR}^2 + 1}, & \mu \geq 0 \text{ and } \sigma \geq 10^{-10}, \\ \mu, & \mu < 0 \text{ or } \sigma < 10^{-10}. \end{cases} \tag{1}$$

$$\textit{where} \qquad \mu = \mathrm{Mean}(PL), \quad \sigma = \mathrm{StdDev}(PL), \quad \mathrm{SR} = \sqrt{250}\,\frac{\mu}{\sigma}.$$

How it works. If $\mu < 0$, the published score is the mean daily loss. If $\mu \geq 0$ and $\sigma \geq 10^{-10}$, the mean daily profit is multiplied by $\mathrm{SR}^2/(\mathrm{SR}^2 + 1)$, so that "Strategies with high returns but massive volatility (low SR) will see their scores severely discounted" [2]. If $\sigma < 10^{-10}$, Sharpe scaling is skipped and the score is μ directly, which "mainly matters for strategies sitting right at the minimum trading activity threshold" [2].

1.1 Branching on the sign of the mean

Conditioning on the sign of μ splits the rule in two. The split is published; how unequally it treats a profit and an equal loss is not. How large the discount on the profitable branch is at any point is the subject of Section 1.2.

Non-negative mean ($\mu \geq 0$). On the regular branch only the fraction $\mathrm{SR}^2/(\mathrm{SR}^2 + 1)$ of μ is retained. The near-zero-variance guard returns Score $= \mu$, which is the limit of that same branch: as $\sigma \to 0$ the fraction tends to one. The guard completes the branch where floating-point arithmetic would fail.

Negative mean ($\mu < 0$). The score equals μ regardless of Sharpe. The multiplier is confined to the profitable side, so dispersion never reaches a losing score at all.

A loss is scored at μ and an equal profit at μM, so the two stand in the ratio

$$\frac{\text{penalty}}{\text{reward}} = \frac{1}{M} = 1 + \frac{1}{\mathrm{SR}^2}, \tag{2}$$

unbounded as $\mathrm{SR} \to 0$: two to one at $\mathrm{SR} = 1$, forty to one at $\mathrm{SR} = 0.16$. A weakly profitable strategy is not a cautious version of a profitable one; on this objective it scores as though it were not there. Later multiplier results are therefore conditional on $\mu > 0$ and $\sigma \geq 10^{-10}$.

1.2 The positive branch: multiplier geometry

On the positive, non-guarded branch,

$$S = \mu M, \qquad M = \frac{\mathrm{SR}^2}{\mathrm{SR}^2 + 1}, \qquad \frac{\mathrm{d}M}{\mathrm{dSR}} = \frac{2\,\mathrm{SR}}{(\mathrm{SR}^2 + 1)^2}, \qquad 0 < M < 1. \tag{3}$$

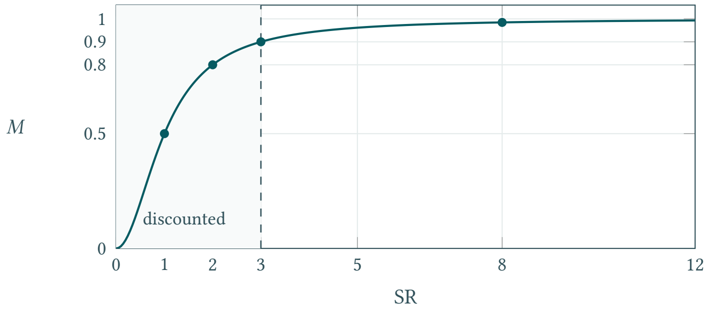


Figure 2: The multiplier. The curve is already flat by SR $= 3$; the dashed line there marks the boundary adopted in Section 1.7.

M is monotone and bounded above by one, with $M(1) = 0.5$, $M(2) = 0.8$, $M(3) = 0.9$ and $M(8) = 64/65 = 0.9846$: by SR $= 3$ only a tenth of the mean is still withheld and by SR $= 8$ only a sixty-fifth. The marginal return falls faster still. From its peak of 0.65 at SR $= 1/\sqrt{3}$ it is 0.060 by SR $= 3$ and 0.0038 by SR $= 8$, so the last sixty-fifth costs about sixteen times as much Sharpe per point recovered as the last tenth did.

What a dollar of mean profit is worth, in dollars. The elasticities below are dimensionless, and the rest of the report works in dollars, so the bridge between them is worth deriving once. Differentiating the composed score with respect to the mean at fixed dispersion gives

$$g(\mathrm{SR}) \;=\; \frac{\partial S}{\partial \mu} \;=\; \frac{\mathrm{SR}^2}{\mathrm{SR}^2+1} + \frac{2\,\mathrm{SR}^2}{(\mathrm{SR}^2+1)^2}, \tag{4}$$

a quantity with four landmarks and no free parameters. It vanishes as SR $\to 0$, so **the first dollar of mean profit is worth nothing**, against a slope of exactly one on the loss branch — the same asymmetry Equation (2) states in levels, now in slopes. It equals 1 exactly at SR $= 1$. It reaches a global maximum of $g(\sqrt{3}) = 9/8$, at the same SR $= \sqrt{3}$ where Equation (18) changes sign, so the curvature landmark and the point where a dollar of mean buys the most score are one point, not two. And it falls back towards 1 from above: $g(3) = 1.08$, and by SR $= 8$ it is 1.015.

Over the whole competitive range — SR ≥ 3, formalised as the research range in Section 1.7 — a dollar of mean profit is a dollar of score to within eight per cent, and at any qualifying Sharpe to within two. That is what licenses the rest of this report to quote dollars and score points interchangeably, which it does throughout and which would otherwise be an unexamined convenience.

What is left on the table, in closed form. The question a reader asks next — if the multiplier is already 0.9846, what is the rest of the Sharpe axis worth? — does not need the two inputs opened up, and does not need any data. What a strategy forgoes by not raising its Sharpe further is exactly what the multiplier withholds, and pushing SR to infinity recovers all of it and no more:

$$\mu - S = \mu\bigl(1 - M\bigr) = \frac{\mu}{1+\mathrm{SR}^2}, \tag{5}$$

so the *entire* remaining value of every possible improvement in risk-adjustment, from the present operating point to a perfect one, is a single closed-form quantity that falls as the square of the Sharpe already achieved. At SR $= 3$ it is a tenth of the mean; at SR $= 8$ a sixty-fifth; at SR $= 11$ a hundred-and-twenty-second. Moving from SR $= 8$ to SR $= 11$ therefore collects $\frac{1}{65} - \frac{1}{122}$ of the mean, which is under half of a remainder that was already under two per cent.

This is a bound, not an estimate. It is stated here because it costs nothing to state — it uses only the multiplier of Equation (3) — and because it converts every later proposal on the Sharpe axis into a question with a ceiling. Section 11 evaluates it at the measured operating point and compares the result with the dispersion of a single scored block; the comparison is not made here, because μ and SR are not known until a strategy exists.

Where the two inputs *do* matter is in how they trade against each other, so the next step is to open SR up.

1.3 What the intermediate variable is made of

The multiplier is written in SR, but SR is not something a strategy sets: it is the ratio μ/σ magnified by $\sqrt{250} \approx 15.81$. Taking logarithms separates the three factors,

$$\ln \mathrm{SR} = \ln \sqrt{250} + \ln \mu - \ln \sigma, \tag{6}$$

and differentiating gives the two elasticities outright, as constants:

$$\frac{\partial \ln \mathrm{SR}}{\partial \ln \mu} = +1, \qquad \frac{\partial \ln \mathrm{SR}}{\partial \ln \sigma} = -1, \qquad \text{summing to zero.} \tag{7}$$

Read the second with its sign: a one per cent rise in σ lowers SR by one per cent, so a one per cent *fall* in σ raises it by one per cent — exactly what a one per cent rise in μ does. Through SR the two inputs are interchangeable. The annualiser enters Equation (6) additively, so it fixes where a book sits on the multiplier but disappears under differentiation, and neither elasticity depends on it.

Why a *sum* is the quantity to read requires one observation. Elasticities add: for independent moves, $\mathrm{d} \ln f = E_\mu \,\mathrm{d} \ln \mu + E_\sigma \,\mathrm{d} \ln \sigma$, so if both inputs move by the *same* percentage the response is $(E_\mu + E_\sigma)$ times it. Moving both by the same percentage is precisely rescaling the book, so the sum of the elasticities is the book's response to being rescaled — and here that response is zero. For SR the exact statement follows without any limit argument, since scaling numerator and denominator of a ratio cancels:

$$\mathrm{SR}(c\mu,\, c\sigma) = \mathrm{SR}(\mu, \sigma) \qquad \text{for every } c > 0. \tag{8}$$

Double a book — twice the mean and twice the dispersion — and SR is untouched: it sees shape and not size. An arbitrary book makes the point concrete.

Table 1: One book at four scales. The ratio does not move; the score moves with the scale factor.

Scale	μ	σ	SR	M	Score
×1	\$1,000	\$5,000	3.162	0.909	\$909
×1.01	\$1,010	\$5,050	3.162	0.909	\$918
×2	\$2,000	\$10,000	3.162	0.909	\$1,818
×10	\$10,000	\$50,000	3.162	0.909	\$9,091

Following the scaling through the composition locates the asymmetry without any further argument. M depends on μ and σ only through SR, so rescaling leaves it fixed; the outer factor is not so protected, and

$$S(c\mu,\, c\sigma) = c\mu \cdot M = c\, S(\mu, \sigma). \tag{9}$$

Every bit of the score's dependence on size sits in the outer μ, and none of it in the multiplier. The mean acts twice and the deviation once: both set the multiplier through their ratio, and μ alone sets the level. That is where the two inputs part company, and Section 1.4 measures the distance by substituting SR back into S.

1.4 Composed score and sensitivity

Substituting $\mathrm{SR} = \sqrt{250}\,\mu/\sigma$ returns the score to the two quantities a strategy moves, and logarithmic differentiation of $\ln S$ then gives both sensitivities at once:

$$S = \frac{250\mu^3}{250\mu^2 + \sigma^2}, \qquad |E_\sigma| = \frac{2}{1 + \mathrm{SR}^2}, \qquad E_\mu = 1 + |E_\sigma|. \tag{10}$$

Hence

$$E_\mu - |E_\sigma| = 1, \qquad 1 < E_\mu < 3, \qquad 0 < |E_\sigma| < 2. \tag{11}$$

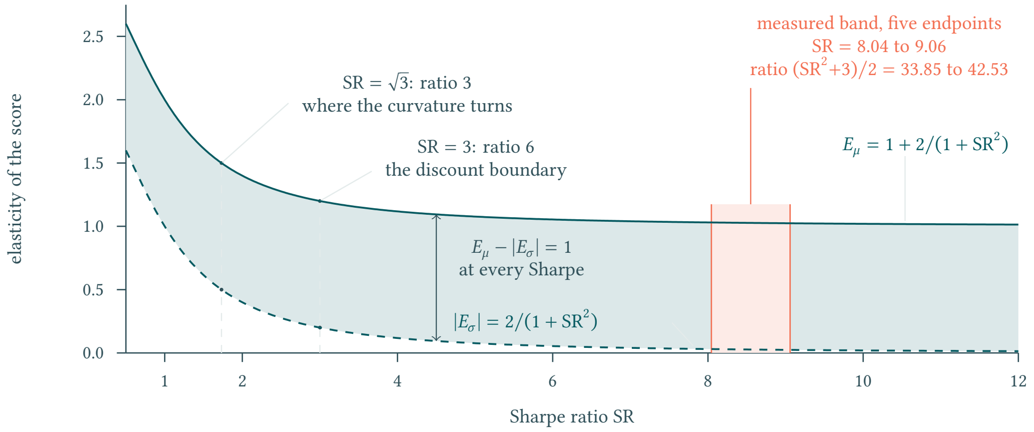


Figure 3: The two elasticities of the published score on one axis: the shaded ribbon between them has the same height at every Sharpe, because $E_\mu - |E_\sigma| = 1$ exactly, while their ratio $(\mathrm{SR}^2+3)/2$ runs from 3 at $\mathrm{SR} = \sqrt{3}$ to 6 at the discount boundary and to 33.9–42.5 across the five measured endpoints. The lower dashed curve is $|E_\sigma|$, all but spent by the measured band while E_μ is still a little above one.

Neither vanishes, so the score is strictly increasing in μ, strictly decreasing in σ, and has no interior optimum in either input: there is no risk level to search for. Set beside Equation (7) the contrast is exact: the elasticities of SR sum to zero, those of S sum to one. That difference of one is the whole of the input asymmetry, and the next paragraph shows where it enters.

The input asymmetry promised in Section 1.3 is now visible in the count of terms. Writing $s \equiv \mathrm{SR}^2$, so that $S = \mu s/(s+1)$ with $\partial s/\partial\mu = 2s/\mu$ and $\partial s/\partial\sigma = -2s/\sigma$,

$$\frac{\partial S}{\partial \mu} = \frac{s}{s+1} + \frac{2s}{(s+1)^2}, \qquad \frac{\partial S}{\partial \sigma} = -\frac{2\mu}{\sigma} \cdot \frac{s}{(s+1)^2}. \tag{12}$$

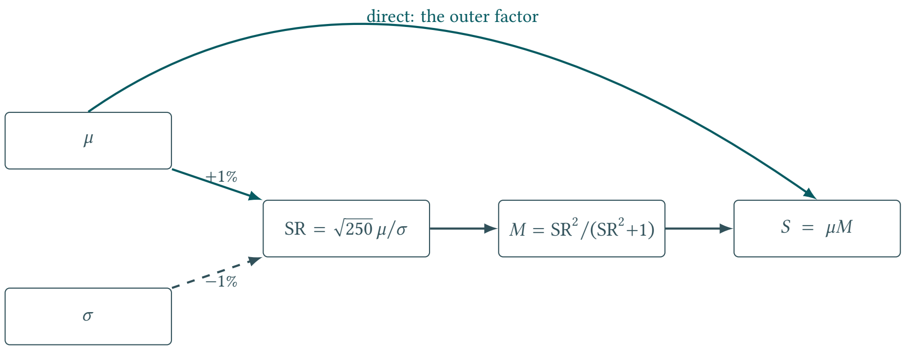


Figure 4: How each input reaches the score. A one per cent rise in μ and a one per cent fall in σ move SR identically, so the shared route treats them alike; μ alone also arrives directly, and that arc is the whole of the asymmetry between the inputs.

σ reaches the score by one route only, through s; μ reaches it by that same route *and* directly, as the outer factor. In elasticity terms the shared routes are exactly equal, and the direct term contributes exactly the one by which E_μ exceeds $|E_\sigma|$.

Their ratio $E_\mu/|E_\sigma| = (\mathrm{SR}^2 + 3)/2$ grows quadratically, so the floor of 1.5 is reached only as $\mathrm{SR} \to 0$ and describes nowhere a real book sits:

$$\mathrm{SR} = 1,\ 3,\ 8 \quad \Longrightarrow \quad E_\mu/|E_\sigma| = 2,\ 6,\ 33.5. \tag{13}$$

At SR = 8, where the multiplier is all but spent, one per cent on the mean is worth more than thirty times one per cent off the dispersion. The preference for mean work is not marginal; it is close to a total ordering.

The exponent and the scaling law are different statements, and confusing them inverts the conclusion. With σ held fixed the score follows $S \propto \mu^{E_\mu}$; with both inputs moved by a common factor it follows Equation (9) instead. Raising the mean at unchanged dispersion is rewarded more than proportionally; enlarging the whole book is rewarded exactly proportionally, because it leaves SR untouched (Section 2).

The scaling law also explains the gap rather than leaving it a coincidence. The published score has the form

$$S = \mu\, f(\mu/\sigma), \qquad f(x) = \frac{250x^2}{250x^2 + 1}, \tag{14}$$

a dollar amount multiplied by a scale-free discount. Scaling the book leaves f alone and multiplies the outer μ, so a one per cent larger book earns one per cent more, and by the reading of Section 1.3 its two elasticities sum to one. **The unit gap is a property of the score being a dollar quantity that scales with the book, not of the particular multiplier chosen**: any rule of the form Equation (14) has it, whatever f is. The conclusion that mean work outranks dispersion work is therefore robust to the shape of the discount, and would survive a different one.

1.5 The score is a statistic, not a number

Everything above treats μ and σ as quantities a strategy has. The evaluator does not receive them; it receives N daily figures and computes them. Section 5 fixes $N = 250$, so the published score is a function of two estimates, and reading it as a property of the strategy requires knowing how much of it is the estimate.

The annualiser and the window are the same number, and that is not a coincidence to pass over. The published rule annualises by $\sqrt{250}$ and the published window is $N = 250$ days. With $\sqrt{N}$ in the numerator and σ in the denominator, the intermediate variable is

$$\mathrm{SR} = \frac{\sqrt{250}\,\mu}{\sigma} = \frac{\mu}{\sigma/\sqrt{N}} = \frac{\mu}{\mathrm{se}(\mu)}, \tag{15}$$

exactly the t-statistic of the mean, not merely something proportional to it. Substituting into the multiplier turns the published score into

$$S = \mu \cdot \frac{\mu^2}{\mu^2 + \mathrm{se}(\mu)^2}, \tag{16}$$

the mean multiplied by its own reliability — the share of its squared magnitude that is not sampling variance.

Three things that looked like separate design choices are one choice read three ways. The dollars withheld, $\mu - S = \mu/(1 + \mathrm{SR}^2)$, are to first order $\mathrm{se}(\mu)^2/\mu$: **the evaluator withholds the sampling-noise share of the mean and pays the rest**. The $\sigma < 10^{-10}$ guard is the limit in which reliability reaches one, so it is the correct terminus of the same formula rather than a numerical patch. And the loss branch is unshrunk because a loss needs no distinguishing from zero.

This is the strongest statement Part I can make about what the objective *is*, and it is available from the published constants alone.

Two consequences follow without any distributional assumption beyond finite variance. First, SR inherits sampling error from both arguments; for a mean estimated on N observations the standard error of μ alone is $\sigma/\sqrt{N}$, so

$$\mathrm{se}(\mu) = \frac{\sigma}{\sqrt{N}} \qquad \Longrightarrow \qquad \mathrm{se}(S) \;\approx\; E_\mu \frac{S}{\mu} \cdot \frac{\sigma}{\sqrt{N}}, \tag{17}$$

the second following from the elasticity of Section 1.4 applied to a small perturbation. The dispersion argument's own sampling error enters through E_σ, and over the measured range $|E_\sigma| \le 0.03$: its contribution is about one per cent of the μ term and is dropped. At $E_\mu \approx 1$ this says the score carries roughly the same relative error as the mean does, and the mean's relative error is $1/\mathrm{SR}$ scaled by $\sqrt{250/N}$ — which at $N = 250$ is exactly $1/\mathrm{SR}$. **A book scored at $\mathrm{SR} = 8$ over one window has a score whose standard error is about an eighth of itself.**

Second, and more usefully, this is the quantity Section 1.7 will need. The luck scale ε introduced there — the amount a score moves with the strategy untouched — is precisely $\text{se}(S)$, and Equation (17) says it can be read off a single window's own σ without any partition into blocks. Section 11 evaluates it both ways and they do not agree, which is a fact about the estimator rather than about the strategy.

What this does not license. The relation above is a first-order propagation and assumes the daily series is well enough behaved for a standard error to mean what it usually means. Section 8 finds mild serial dependence in the realised series, which inflates the true error above this figure; nothing here corrects for that, and no confidence statement in this report rests on Equation (17). It is used only to establish that ε has an estimate that needs no partition.

1.6 Curvature, and what it says about combining books

The elasticities of Section 1.4 are first-order. Two questions they cannot answer are whether the returns to each input accelerate or diminish, and what happens when two books are put together — the second being the architectural question any ensemble has to answer. Both follow from the second derivatives, and the two turn out to be governed by one factor:

$$\frac{\partial^2 S}{\partial \mu^2} = \frac{-500\,\mu\sigma^2\big(250\mu^2 - 3\sigma^2\big)}{(250\mu^2 + \sigma^2)^3}, \qquad \frac{\partial^2 S}{\partial \sigma^2} = \frac{-500\,\mu^3\big(250\mu^2 - 3\sigma^2\big)}{(250\mu^2 + \sigma^2)^3}. \tag{18}$$

The bracket is common to both, so both change sign together, at $250\mu^2 = 3\sigma^2$ — that is, at $\text{SR} = \sqrt{3}$. **Above** $\text{SR} = \sqrt{3}$ **the score is concave in each input; below it, convex in each.** The transition sits *inside* the discounted region, at $\text{SR} = 1.73$ against a discount boundary of $\text{SR} = 3$, so the two partitions do not coincide: a book between $\sqrt{3}$ and 3 is discounted by the multiplier and already on diminishing returns to both inputs, while accelerating returns require a Sharpe below 1.73 — far inside the discounted region and far below anything this report measures. **Any book at a competitive Sharpe is on the diminishing side of both inputs**, which is why the elasticity comparison rather than the curvature is what governs the research range.

Two books, combined. Take two books with the same mean and the same dispersion, whose daily series correlate at ρ, and hold them at equal weight. The combined mean is unchanged and the combined variance is $\frac{1}{2}(1 + \rho)\sigma^2$, so

$$\frac{S_{\text{combined}}}{S_{\text{single}}} = \frac{2\big(250\mu^2 + \sigma^2\big)}{500\mu^2 + (1 + \rho)\sigma^2}. \tag{19}$$

At the operating point of Section 11 this ratio is 1.0035 at $\rho = 0.5$ and 1.0070 at $\rho = 0$. **Combining two equally good and completely uncorrelated books is worth seven tenths of one per cent** — about eight points of score, which Section 11 measures to be about a seventeenth of the noise in a single block.

The reason is the multiplier. Diversification acts on σ, and Equation (12) already showed that σ reaches the score only through a factor that is spent by $\text{SR} = 8$. **An ensemble on this objective cannot be justified by risk reduction**; if it is worth building, it must be because averaging raises the mean, and that is a claim about the signals rather than about their covariance.

1.7 The research range

The lower boundary. The organiser declares that a penalised region exists and that it lies at low Sharpe [2]; the multiplier of Section 1.2 gives its extent; only the cut is ours, taken at the round figure on the vertical axis:

$$\text{SR} \geq 3 \quad \Longleftrightarrow \quad M \geq 0.9 \quad \Longleftrightarrow \quad \sigma \leq \frac{\sqrt{250}}{3}\,\mu = 5.27\,\mu. \tag{20}$$

Reading the slope instead of the level puts it in the same place, the marginal return falling to a tenth of its peak at $\text{SR} = 2.91$. Since $S = \mu M$, this is where the objective becomes the mean: above the line the score is μ to within ten per cent, and to within 1.6% by $\text{SR} = 8$. Below it a strategy fights a discount; above it the competition is mean maximisation and nothing else.

The upper boundary. The line above is a qualifying condition, not a target, and it says nothing about where attention to SR should stop: the multiplier approaches one smoothly and offers no natural terminus. A second line has to be brought in.

Everything a strategy could still recover by driving Sharpe to infinity is the shortfall $\delta = 1 - M = 1/(1 + \mathrm{SR}^2)$. Write ε for the luck scale of one scored block — the amount its score moves, with the strategy untouched, purely because different days were drawn. Once the headroom falls below ε, no amount of Sharpe work is measurable against it. Setting $\delta = \varepsilon/\mu$ closes the question:

$$S^* = \sqrt{\frac{\mu}{\varepsilon} - 1}, \qquad \sigma \leq \frac{\sqrt{250}\,\mu}{S^*} \approx \sqrt{250\,\mu\varepsilon}. \tag{21}$$

The approximation drops the -1 inside the radical and therefore holds only for $\mu/\varepsilon \gg 1$. It errs in the safe direction: the exact band exceeds the approximate one by $\sqrt{(\mu/\varepsilon)/(\mu/\varepsilon - 1)}$, which is 5.4% at $\varepsilon/\mu = 10\%$ and 1.0% at $\varepsilon/\mu = 2\%$, so a book inside the approximate band is inside the exact one as well.

Table 2: The upper boundary at four noise levels, and the resulting map

ε/μ	10%	5%	3%	2%
S^*	3.0	4.4	5.7	7.0
σ tolerance	$\leq 5.3\mu$	$\leq 3.6\mu$	$\leq 2.8\mu$	$\leq 2.3\mu$

Region	Range	Policy on dispersion
Discounted	$\mathrm{SR} < 3$	Must be escaped; but the same one per cent still buys more on μ
Diminishing	$3 \leq \mathrm{SR} < S^*$	Case by case, against the rate $(\mathrm{SR}^2 + 3)/2$
Closed	$\mathrm{SR} \geq S^*$	Ignore; anything inside the tolerance band is free

S^* is not a ceiling on SR: past it a higher Sharpe is simply no longer worth attention, not unattainable. Both μ and ε are measured only once a strategy exists, so the boundary is stated here as a function and evaluated in Section 11, where the daily series of each scored block supplies ε as a standard error and the boundary comes out between 2.7 and 2.8 — **the criterion precedes the strategy**. At that measured level the boundary sits below the discount edge at $\mathrm{SR} = 3$: the Diminishing band of the table is empty and the map degenerates to two regions, which changes no policy, since dispersion is ignored past the boundary either way.

1.8 What the objective restricts

Five constraints follow, and they are the whole of what this section hands to the rest of the report.

1. **Budget.** The score is μ times a factor whose controllable range is small, so engineering effort belongs on the mean. This is why Section 10 ranks mean net daily P&L above score and Sharpe, and why the data work of Part II looks for *directional* information rather than risk structure.
2. **Losses.** Losses are undiscounted, so no amount of volatility can wash one out; the only route to fewer losing days is a better hit rate.
3. **Dispersion, two lines.** Escape the discounted region, $\sigma \leq 5.27\mu$; once inside the tolerance band $\sigma \lesssim \sqrt{250\,\mu\varepsilon}$, stop.
4. **Path.** Between the two inputs there is no optimal intermediate point — the ratio $(\mathrm{SR}^2 + 3)/2$ never falls below 1.5 — so no route of the form "raise Sharpe first, then switch to the mean" exists.
5. **Parameterisation.** Every criterion above is a function of μ and ε alone, evaluated once a strategy exists. The claim is marginal, not absolute: M still fixes what fraction of the mean survives, so a large mean can be heavily discounted all the same.

Scope. Everything derived in this section — the multiplier landmarks, Equations (3)–(11), and Figure 2 — is an algebraic consequence of the published rule, and introduces no empirical working point, strategy performance, or official threshold. Three conventions matter when Equation (1) is reproduced rather than read: the displayed branch conditions follow the public Evaluation page, whereas the pinned evaluator instead tests $\mu > 0$ [5], although both formulations return zero at $\mu = 0$; σ is a population standard deviation, computed with `np.std` at `ddof=0`; and the annualiser is the competition convention $\sqrt{250}$, held fixed, so that `numTestDays` selects the scoring-window length but never replaces the fixed $\sqrt{250}$ annualiser.

2 Position Caps: The Rules Diversify for You

The published caps.

Table 3: Position limits and transaction costs [3]

Asset class	Absolute dollar-position cap	Commission on executed dollar turnover
ALGO (asset 0)	\$100,000	0.00002 = 0.2 bp
Assets 1–50	\$10,000 per asset	0.0001 = 1.0 bp

Dollar position means current price times absolute share size. The cap is a holding limit, applies long or short, and is rechecked at that day's price on every day the strategy is called. The evaluator's final iteration is a mark-only day: it makes no call, so it applies no clip, and a holding compliant when it was set can be carried over its cap by a price move on that day alone. That day is still scored. Commissions apply to every dollar traded on both buys and sells, using evaluator-recorded position changes. There is no starting-capital or aggregate portfolio budget and no separate daily trading-volume limit.

How large should a position be? The published caps answer the question almost completely, and they do so before any price is examined.

Scale has no interior optimum. Scale the whole book by a common factor $c \in (0, 1]$, so that every recorded holding $q_{i,t}$ becomes $c\, q_{i,t}$. Daily net dollar P&L then scales by c, so $\mu \mapsto c\mu$ and $\sigma \mapsto c\sigma$, which by Equation (8) leaves SR unchanged, and with it the multiplier M. On the positive branch the score is therefore

$$S(c) = c\,\mu M, \tag{22}$$

strictly linear in c. A linear objective on a closed interval attains its maximum at an endpoint, so the optimum is $c = 1$.

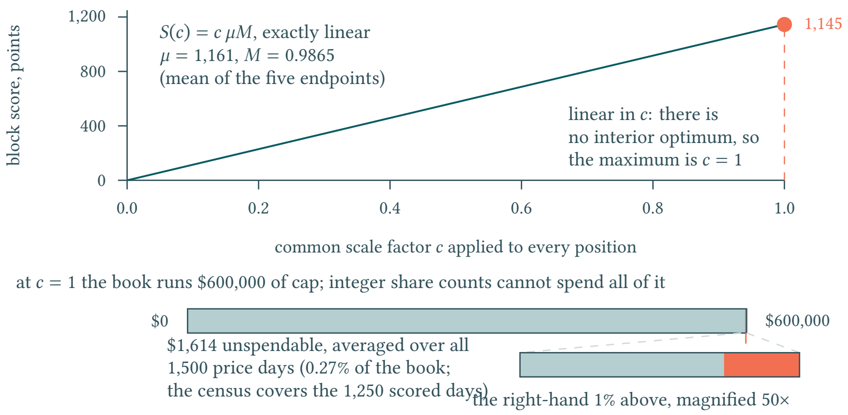


Figure 5: Scaling every position by a common factor c leaves the Sharpe ratio untouched and multiplies the score exactly by c, so the score is a straight line through the origin and its maximum on the closed interval sits at the endpoint c = 1; the lower panel draws, to the same scale as the \$600,000 of cap, the \$1,614 that integer share counts leave unspendable at c = 1, magnified fifty times because it is 0.27 per cent of the bar.

This is Equation (9) read as a statement about sizing: enlarging the whole book is rewarded exactly proportionally, whereas raising the mean at fixed dispersion is rewarded more than proportionally, by the elasticity of Equation (10). **Sizing is not a research dimension**: within the caps the extended book is optimal, and the remaining freedom is direction, not magnitude.

The submitted book takes that literally, and it is worth recording how literally. A census of the retained position path against the caps of the pinned rules contract, cell by cell, finds **all** 63,750 **decision cells**

sitting exactly on the cap: none flat, none strictly inside it, none above it, for the index instrument and for all fifty stocks alike. The book carries 31,808 long cells against 31,942 short, a long share of 49.89 per cent, and rebuilding the whole path from those signs and the published prices alone reproduces it in every one of the 63,750 cells.

Two things follow. The conclusion above is not merely available to the book; it has been taken to its limit, and **the book has no sizing degree of freedom at all**. And every statement this report makes about its "positions" is therefore a statement about *signs*: the magnitudes are a function of the caps and the day's prices, not a decision. That is a measurement against the retained record, and it belongs here rather than in Part III because it is what the rules of this section already imply.

The cap cannot be reached exactly, and the gap is calculable. Holdings are integers, so the evaluator's per-asset limit is $L_i = \lfloor C_i/p_i \rfloor$ and the dollar position actually attainable is $p_i L_i$, not C_i. The shortfall is whatever is left over after the last whole share, which for a price uniform against the cap averages half a price:

$$C_i - p_i \left\lfloor \frac{C_i}{p_i} \right\rfloor \in [0,\ p_i), \qquad \mathbb{E}\Big[\sum_i (C_i - p_i L_i)\Big] \approx \frac{1}{2} \sum_i p_i. \tag{23}$$

Evaluating that expectation needs prices, so it is the one quantity in this section that is a recomputation rather than a closed-form consequence. Two objects can be evaluated and they are not the same: the exact shortfall on the left of Equation (23), and the half-price approximation on its right. Averaged over the retained daily price vectors of the published panel, and summing all fifty-one legs because the permitted book is fifty caps of \$10,000 plus ALGO's \$100,000, the exact shortfall is \$1,614 and the approximation \$1,612. The approximation is therefore good to a part in eight hundred *in the mean*, and much worse day by day: the exact figure ranges from \$886 to \$2,834 across single days where the approximation ranges only from \$1,172 to \$2,249. It is the mean that matters here. Against a permitted book of \$600,000, \$1,614 is **0.27 per cent, and the closest any integer holding can come to the caps**. It is not a cost of the strategy but a property of the rule, and it is the reason a fully extended book is measured slightly under its own ceiling rather than exactly at it. Section 13 recovers \$1,626 from the replayed ledger — the \$12 difference is the evaluator's integer-clip path, traced there.

The caps perform the diversification. The published caps are \$10,000 per equity and \$100,000 for ALGO [3], so a fully extended book is worth

$$50 \times \$10{,}000 + \$100{,}000 = \$600{,}000,$$

and a single equity position can never exceed $\$10{,}000/\$600{,}000 = 1/60$ of it. No individual signal or instrument can dominate the book. Diversification here is not a modelling choice that has to be argued for; it is imposed by the rule, and any later ensemble inherits it.

Asset 0 holds a different seat. ALGO carries ten times the cap and one fifth the commission rate of any other instrument [3]. The rules single out exactly one instrument as both larger and cheaper to trade, which is a reason to analyse it separately rather than as the fifty-first column of a cross-section.

The caps bind holdings rather than trades and are rechecked daily at that day's price, so an extended book generates follow-the-price turnover as a by-product. That hands the next question to the commission clause.

Scope. Closed-form consequences of the published caps and the published score, using no price data and no replay: Equation (22) and the optimum at $c = 1$, the ten-to-one ratio between ALGO's cap and each stock's, and the one-in-sixty share a single stock can take of the extended book. Recomputed against the retained price panel, and therefore class B: the dollar evaluation of Equation (23), both the exact shortfall and the half-price approximation, and the quarter-of-a-per-cent ratio that divides it by the permitted book. The script that reproduces the second group is recorded against it in the register.

3 Commissions: At One Basis Point, Turnover Is Nearly Free

The published rates. Commission is charged on executed dollar turnover at 0.2 bp for ALGO and 1.0 bp for assets 1–50, on both buys and sells, using evaluator-recorded position changes (Table 3) [3].

Is trading cost worth optimising? Two closed-form quantities settle it.

The reversal threshold. A full reversal from $+q$ to $-q$ trades $2q$, so it costs twice the one-way rate: **2 bp** of the position for assets 1–50. Holding the wrong sign rather than the right one forfeits $2|\mathbb{E}[r]|$ per day, the return not earned plus the return taken against. A reversal therefore repays itself on its first day when

$$2|\mathbb{E}[r]| > 2\,\text{bp} \qquad \Longleftrightarrow \qquad |\mathbb{E}[r]| > 1\,\text{bp}. \tag{24}$$

Any directional signal whose daily expected return exceeds one basis point can afford to reverse *every day*. Daily repositioning is not something the cost structure discourages.

The cost ceiling. Even a book that reverses completely every day pays at most

$$\begin{aligned} \text{assets 1–50} \;&:\; 2 \times \$500{,}000 \times 1\,\text{bp} = \$100, \\ \text{ALGO} \;&:\; 2 \times \$100{,}000 \times 0.2\,\text{bp} = \$4, \end{aligned}$$

for a total of \$104 per day, using only published constants and the caps of Section 2. The figure is an order-of-magnitude ceiling rather than a strict bound: turnover on a reversal charges the pre-trade holding as well as the post-trade one, and a holding re-marked over its cap on a mark-only day (Section 2) enters that first term above the cap, which lifts the arithmetic ceiling slightly past \$104.

What matters is the order, not the last dollar. Cost is second-order only against a mean daily P&L an order of magnitude above it: at \$1,000 a day the ceiling is a tenth, while at \$200 a day it is half, and the direction "trade less to save cost" returns to the research agenda. **Once the measured mean is confirmed an order of magnitude above the ceiling, the research direction "trade less to save cost" is eliminated**, and the elasticity result of Section 1 had already directed that budget to the mean before any data was read.

The action space closes. Section 2 fixed magnitude at the cap and left only direction; Equation (24) prices changing it. Each instrument's daily action therefore reduces to three values, $\{+\text{cap},\ 0,\ -\text{cap}\}$, with zero chosen only where the expected edge falls below the one-basis-point threshold.

Boundary. The extended form is affordable only because the published rate is one basis point and the evaluator applies no slippage or market impact. Both are properties of this competition rather than of a traded market, so no claim about real-market value follows from the cost result.

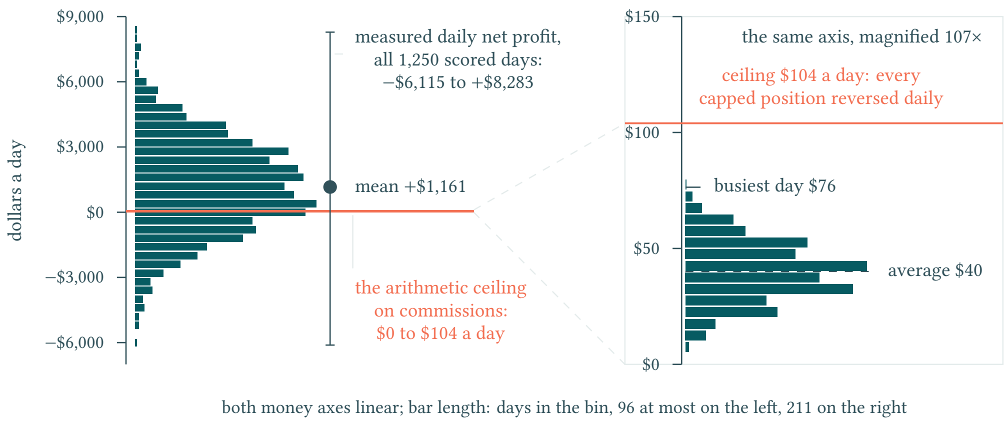


Figure 6: The published caps and commission rates bound the whole cost dimension: a book holding every position at its dollar cap and reversing all of them daily would pay \$104 a day, drawn here to scale as the marked band lying across the measured distribution of daily net profit and magnified 107 times in the right-hand panel. Against a mean of +\$1,161 a day and a range of -\$6,115 to +\$8,283, that ceiling is 9 per cent of the mean and under 1 per cent of the range, and the submitted book pays \$40 of it on an average day and \$76 on its busiest.

Scope. Equation (24) and the \$104 ceiling are arithmetic on published rates and caps; realised turnover and realised fees are reported later, from replay.

4 Minimum Volume: The Only Clause That Forbids Idling

The published floor. Testing requires at least \$25,000 cumulative dollar volume over its 250-day scoring window, defined as the sum of price times absolute shares traded; below the threshold gives an automatic score of zero [3].

Is there an activity floor that a submission must clear? There is exactly one, and for the extended book it is not binding.

Two quantities settle whether that floor binds. A book put on from flat trades its own gross exposure, because the opening trade charges price times absolute share size on every instrument; a book that then reverses trades twice its gross exposure again. At the caps of Section 2 those are

$$V_{\text{open}} = \$600{,}000 = 24\,V_{\min}, \qquad V_{\text{reverse}} = 2 \times \$600{,}000 = 48\,V_{\min}, \tag{25}$$

against the published floor $V_{\min} = \$25{,}000$ for the whole 250-day window. A single day of either clears it. The clause imposes no constraint on the form that Sections 2 and 3 arrived at.

Its function is exclusionary rather than restrictive, but the exclusion it achieves is narrower than it first appears. Establishing a book from flat is itself turnover: the opening trade charges price times absolute share size on every instrument, so a portfolio put on once and held generates cumulative volume equal to its gross exposure. A hold-only book therefore clears the \$25,000 floor for the whole window the moment its gross exposure reaches \$25,000, one twenty-fourth of the permitted book by Equation (25), and it trades nothing afterwards. **The floor does not exclude buy-and-hold at any competitive size**; what it excludes is a book too small to be a serious entry at all. The organiser draws the connection to the score explicitly: the $\sigma < 10^{-10}$ branch of Equation (1) "mainly matters for strategies sitting right at the minimum trading activity threshold" [2]. A submission trading just enough to clear the floor while earning a nearly constant small profit drives σ towards machine precision, and the published score carries a separate path for exactly that case. The clauses interlock by design: the floor creates the degenerate strategy, and the score disposes of it.

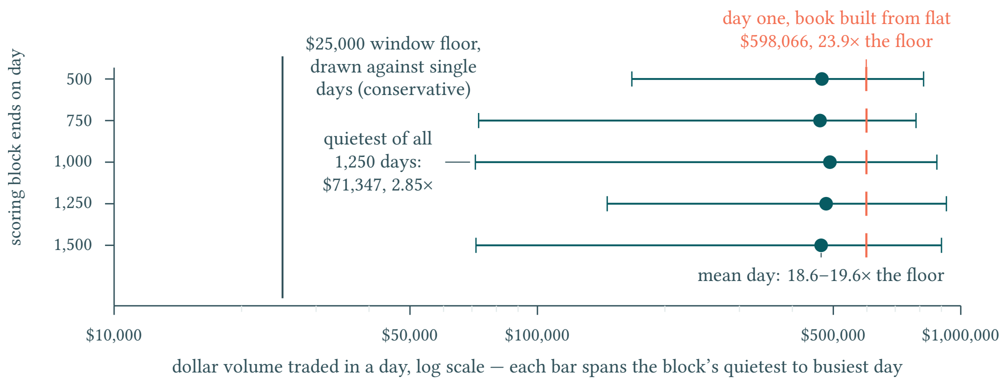


Figure 7: The activity floor against what the book actually traded: on every one of the five blocks the mean day turns over 18.6 to 19.6 times the \$25,000 minimum, and the quietest of all 1,250 days still clears it 2.85-fold. The marked day is the first of each block, when the book is put on from flat: \$598,066 of volume, 23.9 times the floor and 99.7 per cent of the \$600,000 the caps permit, so the clause is discharged on day one and never binds again.

Scope. The threshold and the 48-fold margin are arithmetic on published constants; the realised end-to-end volume is reported later, from replay.

5 The Interface and the Evaluator: What "Position" Means to the Scorer

5.1 Trading universe and submission interface

The universe comprises 51 assets: ALGO (asset 0) plus 50 synthetic instruments, with one price update per day; long and short positions are allowed. The submission entry file is `teamName.py`, with the global signature `def getMyPosition(prcSoFar):`. The argument `prcSoFar` is a NumPy array with shape `(51, numDays)` containing the full price history through the current day, earliest column first [3, 4].

The function returns, in the organiser's words, "a 1D NumPy vector of exactly 51 integers (positive or negative)" [3], representing desired total end-of-day share positions. The parenthesis records that both signs are admissible; it does not exclude zero, and a flat position in an instrument is expressed as zero. This return is a target holding, not an order. Requested holdings are not necessarily evaluator-recorded holdings because current-price caps can clip them; the clipping transformation is defined in Section 5.3.

The submitted vector is a requested portfolio state, not an executed order. The evaluator applies current-price position limits and integerisation before it records holdings, turnover, commissions, and daily net dollar P&L [5].

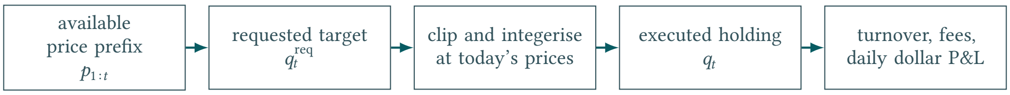


Figure 8: Evaluator processing path. An available price prefix produces a requested target; evaluator post-processing determines executed holdings before turnover, fees, and daily net dollar P&L are recorded [5].

5.2 System and activity constraints

Four further published lines bound the submission. None is a performance criterion; each forecloses a class of approach.

Table 4: Published system and activity constraints, and what each forecloses [3, 4]

Constraint	Published limit	What it forecloses
Runtime	Maximum total runtime of 10 minutes (600 seconds) over the full evaluation; a timeout or execution error is a failure	Refitting a heavy model inside each call. The limit is on the total, not on any one call, so a slow first call followed by fast subsequent ones is admissible
Sandbox	Graded inside an isolated, network-disabled sandbox; no external data, no websites, no local files beyond the passed input	Every signal must be reconstructible from the price prefix alone; no outside series may enter, at any stage
Packages	NumPy, pandas, SciPy, scikit-learn, statsmodels and Matplotlib accepted; anything else must be declared in `requirements.txt`, and the starter repo's `requirements-dev.txt` is not submitted	Exotic dependencies, and re-declaring an accepted package is itself an error; network-dependent packages remain unusable whether or not they are declared
Entry point	A global `def getMyPosition(prcSoFar):` in `teamName.py`, taking `prcSoFar` of shape `(51, numDays)`	Any interface other than a per-day target holding; the evaluator never sees orders

The runtime line is the only one of the four with a margin worth measuring, and that measurement belongs to Part II; the activity floor is analysed separately in Section 4, where the extended book clears it forty-eight times over on a single day.

5.3 Time-ordered input and execution post-processing

Write i for an instrument and t for a scored day, so that $p_{i,t}$ is the price the evaluator uses on day t and $q_{i,t}$ the holding it records as executed. The daily net dollar profit and loss assembled below is the day-t element PL_t of the sequence Equation (1) scores.

On each decision iteration, the strategy sees only the **available price prefix** of prices through the current day. It returns desired total integer holdings, not trade orders. The evaluator then computes price-dependent per-asset caps

$$L_{i,t} = \left\lfloor \frac{C_i}{p_{i,t}} \right\rfloor, \qquad q_{i,t} = \operatorname{int}\!\left(\operatorname{clip}(q_{i,t}^{\text{req}}, -L_{i,t}, L_{i,t})\right).$$

The request is therefore clipped and integerised before execution. Prices can change $L_{i,t}$ even when the model's request does not change. These execution semantics follow the official Evaluation page and pinned evaluator [2, 5].

5.4 Executed turnover, fees, and daily accounting

$$\begin{aligned}
\Delta q_{i,t} &= q_{i,t} - q_{i,t-1}, \\
D_t &= \sum_i p_{i,t} |\Delta q_{i,t}|, \\
c_t &= \sum_i \kappa_i \, p_{i,t} |\Delta q_{i,t}|, \\
PL_{t+1} &= \sum_i q_{i,t}(p_{i,t+1} - p_{i,t}) - c_t.
\end{aligned}$$

Here κ_i is the commission rate (0.2 bp on instrument 0, 1.0 bp on the stocks); the symbol avoids r, which elsewhere denotes a daily return. Executed q and executed position changes drive turnover and fees. The absolute change makes commissions apply to buys and sells. In the pinned loop, the fee calculated for decision t is deducted on the next mark, so the last decision day's fee is still inside the scored daily net dollar P&L sequence. The final iteration is a mark-only day: it keeps the prior executed holding and creates no new turnover. This accounting identity is about dollar P&L, not market returns. The accounting sequence follows the pinned evaluator [5].

5.5 Active scoring window and failure boundary

The evaluator starts decisions at $T -$ `numTestDays` and appends P&L only after that initial positioning day. Exactly `numTestDays` daily net dollar P&L observations enter μ, the population standard deviation, Sharpe, and Score. The published rules fix the Testing round's evaluation window at 250 days [3], and it is that window the starter evaluator's `numTestDays` carries [5]; so $N = 250$, and every later statement that evaluates an estimator "at $N = 250$" is evaluating it at the published window length. A wrong output length, runtime failure, or invalid execution is a submission failure; an inactive Testing submission is assigned zero [2, 5].

Scope. Every statement in this section is read from the official pages or from the pinned evaluator source; none of it is estimated, and no quantity here was obtained by running a strategy. The clipping rule, the accounting identities and the active-window definition are transcriptions of executable code, so they fix what the later measurements of Part III are measurements *of*. The one margin worth quantifying — how much of the 600-second budget a submission actually uses — is a measurement and is reported in Part III.

6 Staged Release: Why the Effective Sample Is Limited

At the public 1,500-day endpoint, the price panel has nominal dimension $1{,}500 \times 51$, or 76,500 recorded prices. This figure describes the size of the stored matrix, not the amount of independent evidence. The research problem has a **limited effective sample** because observations and research decisions remain connected across time, assets, market states, and successive information releases.

6.1 The published release schedule

Algothon releases market history in stages. Each decision may use only the history available at that date, and each release changes the history available for research while preserving a distinct forecast and scoring window [1, 2].

Table 5: Published staged information sets and scoring windows [1, 2]

	Testing	General
Observed history	Days 1–500	Initially Days 1–750
Forecast/scoring window	Days 501–750	Days 751–1000
Scored length	250 days	Expands from 50 to 250 days over the round's release schedule

Days 1–1000 were released late in General while the leaderboard window remained Days 751–1000. The observed history could therefore widen without changing that scoring interval [2].

6.2 Sources of dependence

Temporal dependence. Adjacent prices and returns belong to one ordered path. Serial structure, overlapping histories, and persistent states mean that nearby days cannot be treated as fresh replications.

Cross-asset dependence and factor dependence. The 51 instruments share market and latent-factor shocks. A common event can move many columns together, so the assets are **not 51 independent replications** of the forecasting problem. That statement is quantitative rather than rhetorical: for a panel whose pairs correlate at $\bar{\rho}$ on average, the number of independent units available for averaging a common signal is

$$n_{\text{eff}} = \frac{n}{1 + (n-1)\bar{\rho}}, \tag{26}$$

which Section 8 measures on this panel as 4.4 **to** 5.2 **across blocks**, against a nominal fifty. A cross-sectional average over these instruments therefore carries about a tenth of the independence its width suggests.

Regime change. A relationship learned in one period can weaken, reverse, or disappear later. Additional older days may improve estimation while also making the training distribution less representative of the next scoring window.

Repeated selection. Feature, model, parameter, and ensemble choices are made after inspecting related slices of the same realised path. Repeated selection spends information: the best observed result inherits selection bias unless decisions are separated by leakage-controlled validation and dated model locks.

Single-path transfer. Every later stage asks whether a rule learned from an earlier prefix transfers forward along the same realised sequence. It does not provide many independent worlds. Stability across several public-data endpoints is useful reproducible evidence, but single-path transfer is not the same as independent replication or official hidden-window evidence.

Chronological consequence. Uncertainty must therefore be assessed through forward blocks, regimes, and locked decisions, rather than by treating the full matrix as exchangeable.

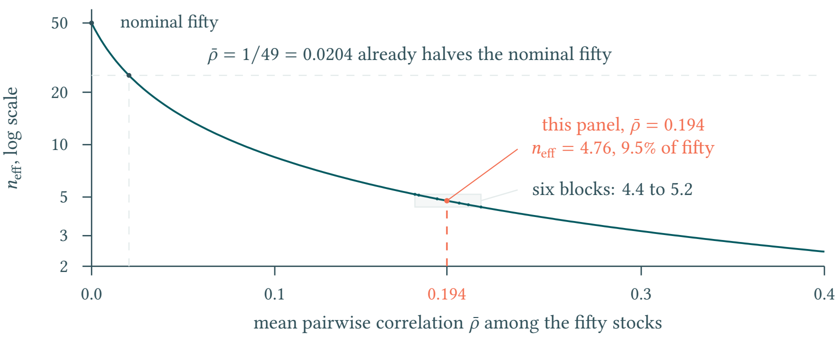


Figure 9: The effective number of independent units among the fifty stocks, $n_{\text{eff}} = n/\big(1 + (n-1)\bar{\rho}\big)$ at $n = 50$, on a logarithmic scale. This panel's measured mean pairwise correlation of 0.194 leaves 4.76 units of the nominal fifty and the six blocks span 4.4 to 5.2, while an average correlation of one part in forty-nine already halves the width.

6.3 The relevant inference unit

Definition (inference unit). The relevant evidence unit is forward transfer from information available at a dated cutoff to a later scoring block. More cells can reduce some estimation noise, but they do not erase temporal dependence, shared factors, non-stationarity, or the cost of repeated inspection.

This conclusion is a design-level inference from the competition's documented staged interface [2] and the approved research specification. It makes no new empirical claim about the realised data.

6.4 Stages beyond the published schedule

The organiser publishes the Testing and General windows. The Finalist and Final round windows used below — observed history through Days 1,000 and 1,500, scored windows Days 1,001–1,500 and 1,501–2,000, 500 scored days each — are **user-reported and unverified**, and no claim in this report may rest on them alone. For the same reason the General and later-round activity thresholds remain unverified and must not inherit the Testing threshold [3].

6.5 Dated information cutoffs and the non-retrospective record

For its sequential research record, this report defines four dated research locks: cutoffs 500, 750, 1,000, and 1,500. These are research locks, not official stage outcomes. Each lock may use only the history available through that dated cutoff; later releases can revise the next strategy, but cannot rewrite what was known at an earlier lock.

Figure 10: Dated information cutoffs for the research record. The markers are research locks, not official stage outcomes or claims about hidden-window results.

Protocol requirement (non-retrospective record). The report must not project the final model backwards. A record kept at a cutoff states what was retained, **what newly released evidence supported or falsified**, **what changed in signals, sizing, constraints, or validation**, and what remained unresolved. No feature, parameter, failure mode, or rationale discovered after a cutoff may be presented as contemporaneous knowledge at that cutoff.

And this report does not present one. No contemporaneous narrative of the four cutoffs was kept, and writing one now — from the finished engine, knowing which directions closed — is precisely the retrospective projection the requirement forbids. The requirement is therefore stated here as the standard this work is held to and *failed* on that axis, not quietly discharged: what survives is the dated artefact record, which fixes what existed when, and not an account of what was believed when. A reader weighing the sequential claims of Part II should weigh them accordingly.

Scope. The release schedule, the window sizes and the stage boundaries are published facts. The dependence taxonomy is an argument about the structure of a single realised price panel, and this section estimates none of the quantities it names. Equation (26) is derived here but evaluated in Part II; where the text quotes "between four and five" against a nominal fifty it is carrying Part II's measurement forward, not making one, and no correlation and no half-life is quantified here at all. The four cutoffs and the non-retrospective protocol are commitments this report makes about its own method; they are auditable against its dated record rather than against data.

7 What the Rules Leave Open

Four published conditions have now been analysed one at a time. Read together they determine more of the solution than any of them does alone, and what they leave undetermined is exactly the research problem.

They fix the objective. The score saturates in Sharpe but never stops paying for mean: the elasticity gap of Equation (11) is one everywhere, so mean work outranks volatility work at every operating point.

They fix the form of the book. Sizing has no interior optimum (Equation (22)), reversal is priced at one basis point of daily expected return (Equation (24)), and the whole cost of the most active possible book is bounded by $104 per day. A fully extended, three-valued, daily-repositioned book is therefore not a design preference but the form the rules select. The volume floor does not bind it.

They leave direction undetermined. Magnitude is settled, activity is settled, and cost is settled. What no published rule supplies is the *sign* to hold on each instrument on each day. That single unknown is what the remaining parts of this report spend their evidence on.

They also fix what is delivered. Official judging is 50% quantitative performance on unseen out-of-sample data and 50% technical presentation, judged on technical maturity and strategy logic, communication and presentation, and team cohesion and Q&A [2]. Half of the assessed weight therefore rests on an argument a reader can follow, which is what this report is.

They leave two leads. The caps and rates single out ALGO as both larger and cheaper, which invites separate treatment rather than a fifty-first column; and the $10,000 cap fixes every equity at $1/60$ of the book, which is the structural reason a wide, near-equally-weighted cross-section is the natural carrier of any signal.

Table 6: Competition rules and required research responses [2, 3]

Rule	Required research response
Score geometry	Prioritise stable positive mean net daily P&L, then improve SR.
Per-asset caps and no total budget	Use cross-sectional diversification and account for daily clipping.
Low but non-zero turnover fees	Test cost-aware signals rather than assuming turnover is free.
Target-holding interface	Model executed holdings and evaluator post-processing, not abstract trade orders.
Activity threshold	Treat zero or inactive baselines as diagnostic only, not as valid Testing submissions.
Staged hidden windows	Use leakage-controlled expanding-history validation, time-stamped model locks, and later chronological strategy development only from newly available data.
Runtime/sandbox	Require an online, self-contained, deterministic implementation.

The table closes the rule-to-method argument. Two obligations follow from its right-hand column: contemporaneous information boundaries must be preserved, and no research lock upgrades a local replay to official hidden-window evidence. Section 6 records that the report fails the first: it keeps the dated artefact record the obligation presumes, but not the contemporaneous account the obligation asks for.

Scope. This section introduces no result of its own. Every entry in the map restates a conclusion established earlier in Part I and names the section that established it, so the whole of it inherits the evidence footing of those sections: closed-form consequences of published constants, with the two declared imports of Section 1.7 — the round cut adopted at SR $= 3$, and the luck scale ε, which is measured only once a strategy exists. Nothing here is measured. Appendix A lists every claim Part I takes from the published pages together with the capture it was read from.

8 Data Integrity: One Common Factor, and Little Left Behind It

This chapter defines the diagnostic work that may be performed at every **information cutoff**. It re-expresses the frozen Gate 2 analysis in English; it does not launch a new analysis, tune a strategy, or establish future performance. Every calculation must be rerun on the available prefix only and stored with its data and code identity. Throughout this chapter, the reported shape, dependence, and transfer quantities are descriptive diagnostics, not causal evidence. That distinction is load-bearing for every later chapter: nothing established here licenses a forecast. The translated source material is analytical rather than independent strategy-performance evidence.

Part I fixes what is worth diagnosing before any of it is computed. The objective pays for direction and barely for dispersion (Section 1.4), the caps already impose the diversification a risk model would otherwise have to argue for (Section 2), and turnover is nearly free (Section 3). **The diagnostics below therefore ask what can be known about direction, and treat the covariance structure as a constraint to be respected rather than a quantity to be optimised.** Part I closes half that survey in advance.

A word on what is transcribed. Several diagnostics below were computed during development and are reported here from a frozen internal record. That record is not published with this report and a reader cannot consult it, so it is not cited as a source: where a figure comes from it, the text says the figure is transcribed and says what it is transcribed from. A reference to a document the reader cannot obtain would look like provenance while supplying none. Everything not marked that way is either a consequence of the published rules or a recomputation against a record this report retains.

8.1 Integrity and schema gate

The first operation at each information cutoff is a fail-closed integrity gate. Table 7 separates observations recorded in the frozen source from the checks that must be repeated on each newly available prefix.

Table 7: Integrity checks repeated at each information cutoff

Check	Frozen-source observation	Required cutoff action
Schema and orientation	Gate 2 records 51 instruments by 1,500 dates after evaluator-aligned orientation.	Assert dimensions, chronological ordering, instrument identifiers, numeric type, and a unique date axis.
Missing / non-finite	The frozen census records zero missing and zero non-finite cells.	Fail closed; never insert, forward-fill, or silently drop an observation.
Nonpositive prices	The frozen census records zero nonpositive cells.	Reject any nonpositive price before returns, ratios, caps, or logarithms are computed.
Return construction	Daily returns begin only after two valid consecutive prices.	Record the exact simple- or log-return convention and alignment; never let a next-day label enter today's feature row.
Release boundary	Later releases extend the same ordered path.	Rerun the gate on the new prefix and retain the earlier gate record unchanged.

These shape and hygiene counts support safe preprocessing only; they do not prove a signal, a stationary law, or transfer to a later block.

8.2 ALGO and the equal-weight index

Instrument 0 is reconstructed as the re-based equal-weight aggregate of the other fifty,

$$\hat{I}(t) = 100 \times \frac{1}{50} \sum_{i=1}^{50} \frac{P_i(t)}{P_i(1)}. \tag{27}$$

Computed on the published panel this reproduces the quoted series with a price correlation of 0.999999972 over all 1,500 days, a daily log-return correlation of 0.999991 over the 1,499 return days, and a maximum absolute discrepancy of 0.008731 index points.

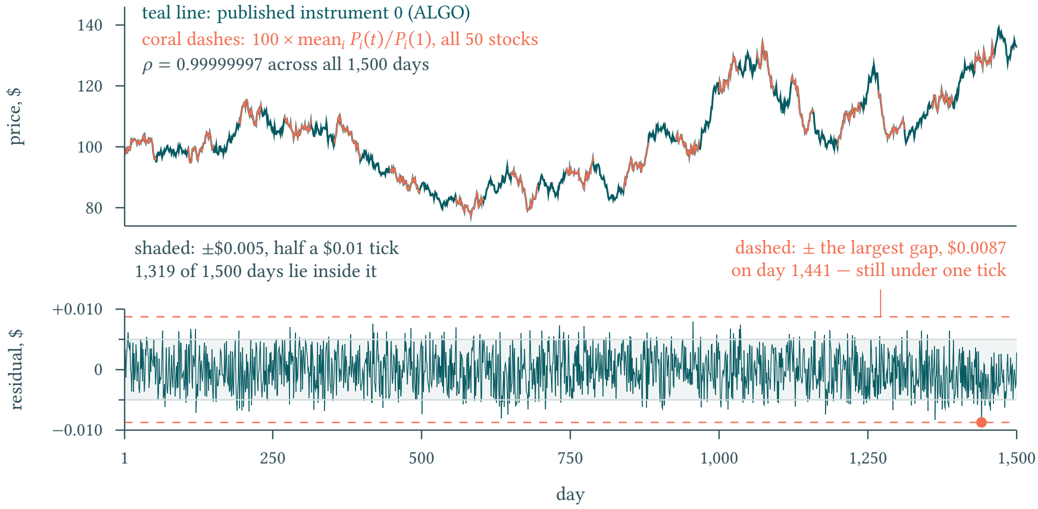


Figure 11: Instrument 0 is not an independent asset but the equal-weight basket of the fifty stocks: the published price and the reconstruction $100 \times \text{mean}_i\, P_i(t)/P_i(1)$ have a price correlation of 0.99999997 across all 1,500 days. Their largest disagreement on any day is $0.0087, smaller than the $0.01 grid both series are quoted on, and rounded back to the cent the reconstruction reproduces the published quote on 1,319 of the 1,500 days and is one tick away on the rest.

The residual is quotation rounding, and its size says so. An identity this close is only worth reporting if the gap it leaves is accounted for rather than merely small. Prices are published to the cent, which separates two rounding sources. The index's own quotation contributes an error uniform on ±$0.005, of mean absolute value $0.0025. The fifty constituents are rounded as well, and that noise propagates through the ratio and the average; redrawing it on the published panel puts its mean absolute contribution at $0.0017. Treating the two as independent predicts a mean absolute discrepancy of $0.0030 against the $0.0028 observed — agreement to about a tenth, on the low side as partial cancellation across the average would give. **The reconstruction is exact to the resolution at which prices are published**, and the residual needs no explanation beyond the quotation grid.

This is an identity diagnostic. It is not a forecast, and it does not claim exact equality before rounding.

What the basket becomes. The index is bought once and never rebalanced, so its composition is itself a price path rather than a fixed rule.

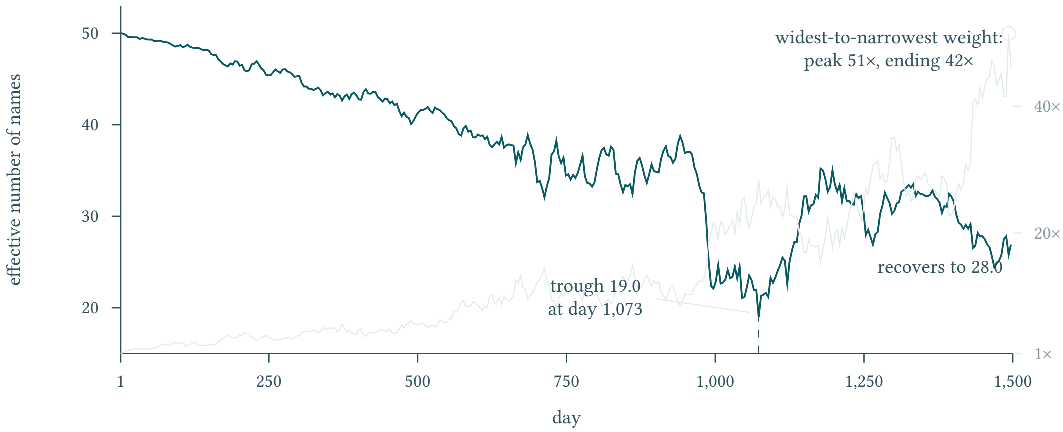


Figure 12: The benchmark's own concentration over the window, computed here from the retained panel. Effective breadth falls from fifty names to nineteen by day 1,073 and recovers only to twenty-eight. The ratio of widest to narrowest weight does not move with it: from 28.4 at the trough it narrows to 15.4 by day 1,177 before climbing to a peak of 51.5 near the end, so breadth and dispersion are not two readings of one quantity. A terminal snapshot understates the peak concentration by a third — nineteen effective names at the trough against twenty-eight at the end. The grey series reads on the right-hand scale.

Two names in the panel therefore mean different things at different dates, and any statement about the index that is read off its final composition is a statement about one date.

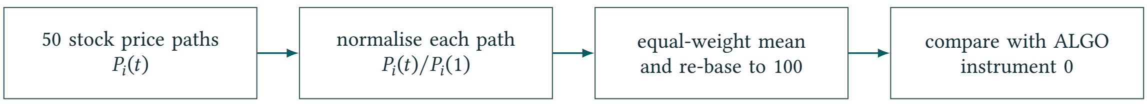


Figure 13: The reconstruction chain used for the instrument-0 identity check, Equation (27). The figure shows the chain only; the agreement it achieves — correlation 0.999999972 and a maximum discrepancy of 0.008731 index points, recomputed here from the retained price panel — is reported in the text above. The chain has no predictive direction and is not independent performance evidence.

8.3 Common factor and residualisation

Raw stock returns share a **common factor**. The diagnostic compares raw returns with a market-beta residualisation followed by cross-sectional de-meaning, and reports the leading eigenvalue of the return correlation matrix [8] together with the mean of the $\binom{50}{2} = 1{,}225$ pairwise correlations, in each case before and after the transformation.

Three of the four figures are recomputed here. On the 1,499 log returns of the fifty stocks in the retained panel, the leading eigenvalue carries 21.55 per cent of the trace and the mean pairwise correlation is 0.1939; after cross-sectional de-meaning the mean pairwise correlation is −0.0161, against the arithmetic floor of $-1/49 = -0.0204$ that Equation (29) fixes. These three are class B and are reproduced by the script the register names. The fourth — the residualised eigenvalue share, recorded as 7.31 per cent — is *not* reproduced: a plain cross-sectional de-meaning of the same panel gives 7.58 per cent, so the market-beta step that precedes it is doing work this report cannot reconstruct, and the figure remains transcribed with that gap stated rather than hidden.

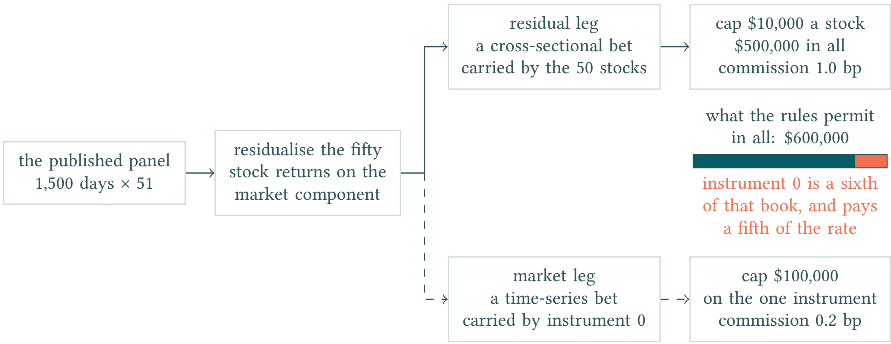


Figure 14: Residualising the fifty stock returns on the market component splits the panel into two legs that the rules then price differently: the residual leg is a cross-sectional bet spread across fifty $10,000 caps, $500,000 in all, charged one basis point, while the market leg is a time-series bet carried by instrument 0 alone under a single $100,000 cap charged 0.2 basis points, and its path through the diagram is drawn dashed. The bar is the $600,000 the caps permit between them, drawn to scale: the index leg is a sixth of the permitted book at a fifth of the commission rate.

Residualisation therefore defines two diagnostic legs: market exposure, represented by instrument 0, and cross-sectional residual exposure. It does not prove that either leg is predictable.

Both ends of that transformation admit a closed-form expectation, and checking them changes how each should be read.

The factor is as simple as a factor can be. If every pair of instruments were correlated at a common $\bar{\rho}$ and nothing else, the leading eigenvalue of the correlation matrix would take a share of its trace equal to

$$\frac{\lambda_1}{n} = \bar{\rho} + \frac{1-\bar{\rho}}{n}, \tag{28}$$

the second term being what a finite panel contributes even with no common component at all. Both sides must be read on one basis or the comparison is between two panels. On the full retained panel, the same 1,499 return days that give the 21.55 per cent share, the mean pairwise correlation is 0.1939 and Equation (28) predicts 21.00 per cent against the 21.55 measured — agreement to about half a percentage point, the measured share being the *larger* of the two. **The panel's leading direction carries what uniform equicorrelation already predicts, and exceeds it by half a percentage point**, so the "common factor" is not a rich factor structure to be modelled but the simplest object that could account for the number: one market, entering everything about equally.

The residual figure is mostly arithmetic, and must not be read as evidence. Subtracting the cross-sectional mean at each date forces the residuals to sum to zero across instruments on every date, and that constraint alone fixes their average pairwise *covariance* at

$$\overline{\mathrm{Cov}}_{\mathrm{resid}} = -\frac{1}{n-1}\,\overline{\mathrm{Var}}_{\mathrm{resid}}, \qquad \text{so} \qquad \bar{\rho}_{\mathrm{resid}} = -\frac{1}{n-1} = -0.0204 \quad (n=50) \tag{29}$$

when the instruments carry equal variances, and approximately otherwise. The covariance form is an identity and holds whatever the data: on this panel both sides evaluate to -8.835×10^{-6}. The correlation form is the one quoted, and it is not an identity — the variances here differ by 63 per cent of their own mean, which is why the recomputed figure is −0.0161 rather than −0.0204.

The fall from 0.194 is therefore not a measurement of how much common variation was removed: an identical fall would follow from de-meaning pure noise. What might carry information is the departure from the floor — but on a panel with heterogeneous variances that departure mixes information with the heterogeneity itself, and this report cannot separate the two. Read as an upper bound on the information available, a departure of 0.004 supports no claim in either direction.

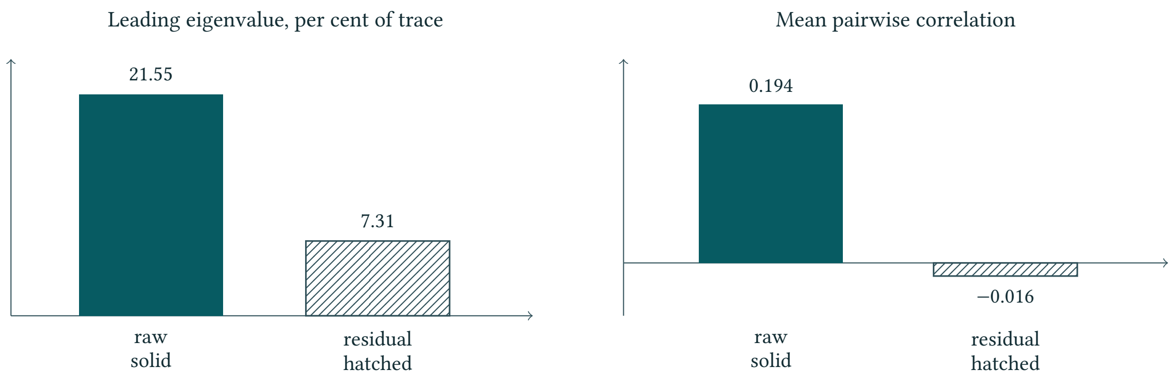


Figure 15: Dependence before and after the residualisation of the frozen source: the leading eigenvalue of the 50-stock return correlation matrix as a share of its trace (left) and the mean of the 1,225 pairwise correlations (right), raw against residual. The two raw values and the residual mean correlation are recomputed here from the retained panel; the residual eigenvalue share alone is transcribed, for the reason given beside it. All four are descriptive diagnostics, not performance evidence.

8.4 Residual and stationarity diagnostics

Residual and pair structure. The common protocol may inspect residual correlations, principal components, cross-sectional ranks, and pair spreads. Pair candidates must be selected from the prefix only with fixed filters, deterministic tie-breaking, and a recorded re-selection cadence. A low spread-stationarity statistic is a descriptive screen, not proof of persistent cointegration or profitable mean reversion. Pair membership learned after a reveal belongs only to the next lock.

Stationarity and regime caveats. Return-scale stationarity can coexist with changing conditional relationships. ADF tests, autocorrelations, beta estimates, and covariance maps are all window-dependent. The lookback, cutoff, multiplicity, and stability across forward blocks must be reported. Synthetic near-Gaussian behaviour does not license a claim about real-market tails, liquidity, impact, or capacity.

8.5 Tradability and information budget

A statistical structure becomes a trading candidate only after evaluator caps, integer holdings, next-mark P&L, fees, turnover, and the activity rule are replayed. Factor removal and pair selection are modelling transformations; they are not themselves evidence of net economic value.

The frozen master separates predictable directional structure from the much larger body of contemporaneous dependence and risk structure. This is an **information budget**: a model may explain covariance without adding next-day directional information. Reported in bits per day, it converts into a bound on accuracy, and the conversion is worth carrying out rather than summarising, because everything Part III prices is denominated in the result.

From bits to a hit rate. A directional decision is one binary symbol. If a decision carries I bits, its achievable accuracy q satisfies

$$I = 1 - H_2(q), \qquad H_2(q) = -q \log_2 q - (1-q)\log_2(1-q), \tag{30}$$

and for the small budgets at issue here the upper branch inverts in closed form,

$$q - \tfrac{1}{2} \;\approx\; \sqrt{\tfrac{I \ln 2}{2}}, \tag{31}$$

which agrees with the exact inversion to two parts in 10^5 at these magnitudes. The measured incremental structure is a panel quantity, so it is divided across the directional decisions taken each day before the inversion is applied. **How many that is, is a choice this report has to make explicitly, because the choice decides the answer.**

Table 8: The information budget converted to an accuracy ceiling by Equation (30). The bit figures are transcribed; the inversion is performed here.

Training endpoint	Same-day structure (bits/day)	Incremental (bits/day)	Share	Accuracy ceiling
Day 500	9.276	0.1020	1.10%	52.66%
Day 750	9.254	0.1230	1.33%	52.92%
Day 1,000	8.817	0.1705	1.93%	53.44%

The divisor, stated as the open question it is. The budget is measured for the panel as a whole. Turning it into a bound on one decision needs the number of *independent* decisions a strategy takes each day, and there are two defensible readings of that, which this chapter has itself supplied.

Taking the nominal fifty — one decision per instrument — puts the ceiling at 52.66, 52.92 and 53.44 per cent at the three endpoints. Taking the effective count instead, the 4.38 to 5.18 independent cross-sectional units that Equation (26) measures on this same panel, puts it at 58.26 to 61.62 per cent.

These land on opposite sides of everything Part III does with them. The accuracy a replacement signal must reach to pay for its own turnover is 55.99 per cent (Section 15). The nominal divisor puts the ceiling below that and closes the direction; the effective divisor puts it above and leaves it open.

This report does not resolve the choice, and says so rather than making it silently. What it can say is that the two are not equally comfortable positions to hold: this chapter argues at length that a panel of fifty instruments carries between four and five independent units, and that an argument quoting the nominal fifty overstates the evidence ninefold. Applying that argument consistently favours the divisor that *reverses* the conclusion. Table 8 therefore reports the nominal reading, because it is the one the transcribed budget was constructed against, and Section 20 is required not to lean on it.

One reading is safe under either divisor. **The predictable temporal structure is one to two per cent of what the panel contains**; the rest is same-day dependence — the risk skeleton — which the scorer does not pay for, a conclusion the transfer comparison of Section 14 reaches independently. That closeness is structural rather than incidental: Equation (31) converts bits into an edge over a coin through a square root, so the growth in the measured budget across the three endpoints moves the ceiling by less than a point.

The inversion assumes each decision is one independent binary symbol and that the budget divides evenly across them; neither holds exactly, and the entropy calculation is specific to the model family that produced it. It is a ceiling for that family on this panel, not a universal impossibility theorem, and Part III treats it as such.

8.6 How many independent things are there

Section 6 argued from the structure of a single realised panel that fifty instruments are not fifty replications. That argument can be made quantitative, and doing so turns out to expose why the question has no single answer.

Write $\lambda_1 \geq \ldots \geq \lambda_n$ for the eigenvalues of the cross-sectional second-moment matrix and $\bar{\rho}$ for the mean pairwise correlation. Three standard summaries count "how many independent directions" and each answers a different question:

$$n_{\mathrm{eff}} = \frac{n}{1 + (n-1)\bar{\rho}}, \qquad \mathrm{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}, \qquad \mathrm{ER} = \exp\Big(-\sum_i \tilde{\lambda}_i \ln \tilde{\lambda}_i\Big), \tag{32}$$

with $\tilde{\lambda}_i = \lambda_i / \sum_j \lambda_j$. The first is the number of independent series that would average a common signal as well as these n do; the second and third count directions carrying variance, the participation ratio weighting the spectrum by λ^2 and the entropy rank by $\ln \lambda$. Because squaring emphasises the leading eigenvalues more than the logarithm does, PR $\leq$ ER whenever the spectrum is not flat, and on this panel n_{eff} is smaller still: it responds to the common component alone, which is exactly the component the leading eigenvalue carries. Under uniform equicorrelation the two are one statement rather than two, since Equation (28) puts the leading eigenvalue at $1 + (n-1)\bar{\rho}$, which is exactly the denominator

n_{eff} divides n by.

Table 9: The three counts on six contiguous return blocks of the public panel, recomputed from a retained record. Nominal dimension is fifty.

Count	Lowest block	Highest block	Answers
n_{eff} (correlation basis)	4.4	5.2	independent units for averaging
PR (covariance basis)	14.5	17.7	directions carrying variance
ER (covariance basis)	27.9	30.3	soft count of the spectrum

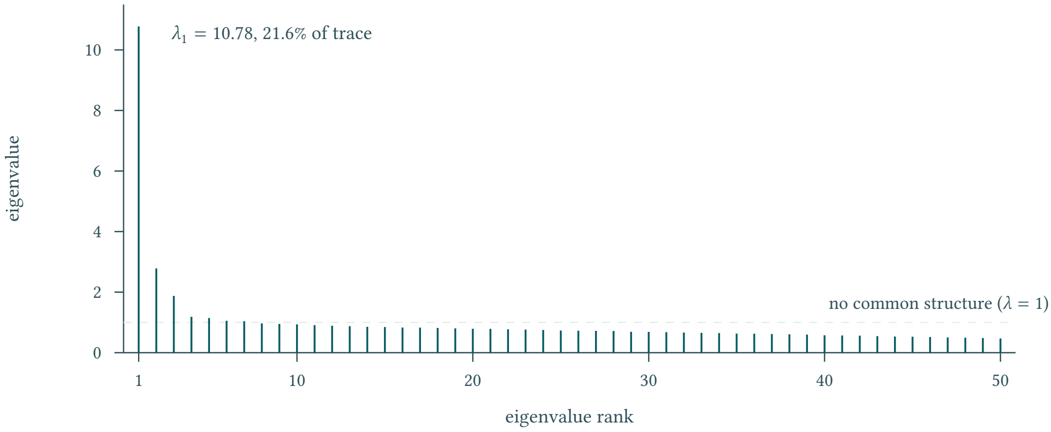


Figure 16: The correlation spectrum of the fifty stocks, computed here from the retained panel. One eigenvalue carries a fifth of the trace; the second and third stand at 2.78 and 1.87 times the structureless level, seven exceed it in all, and the tail then falls away to 0.47. That shape is why the three counts of Table 9 differ: a measure weighted towards the leading eigenvalue reports a handful of units, one weighted across the tail reports about thirty.

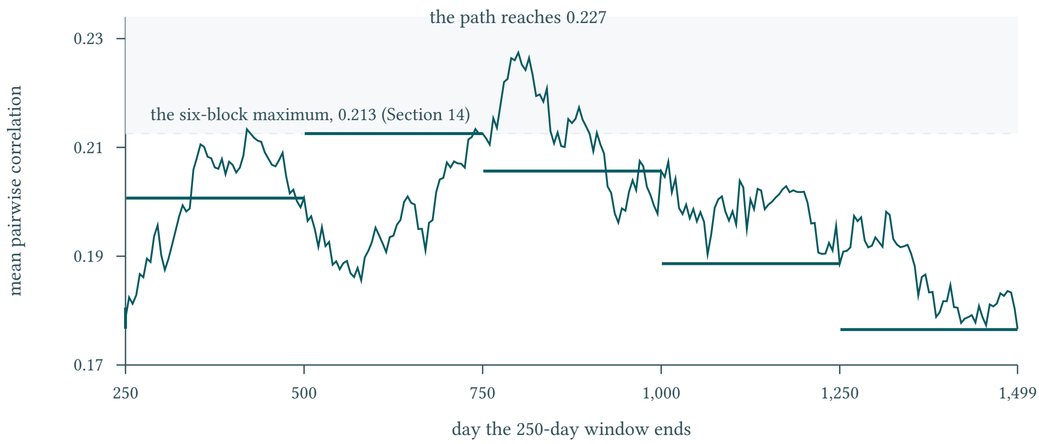


Figure 17: The same estimator and the same 250-day width as the block table, with only the partition's offset removed. The path reaches 0.227 inside a window the partition happens to cut in two, so the table's stated maximum of 0.213 is a property of where the blocks were placed: eleven per cent of the path lies above it, and the rolling swing exceeds the block-to-block swing by a factor of 1.4.

A block table reports the partition as much as the panel. Nothing in this report rests on the block maximum, and the ranges quoted above are used only as ranges. But 0.213 is not the panel's highest correlation, and a reader who takes it as one has taken an artefact of the cut.

The three answers differ by almost an order of magnitude, and the spread is the finding. A panel of fifty instruments behaves like roughly thirty directions when one asks how many are moving at all, like fifteen when one asks how many carry appreciable variance, and like *five* when one asks how many independent replications of a common signal are available. The last is the count that governs inference about a cross-sectional strategy, because such a strategy is estimating one common structure and averaging over the instruments that share it. It is also the smallest of the three, so any argument that quotes a nominal fifty overstates the evidence by a factor between $50/5.2 \approx 9.6$ and $50/4.4 \approx 11.4$, and one that quietly uses the entropy count still overstates it by between $27.9/5.2 \approx 5.4$ and $30.3/4.4 \approx 6.9$.

This does not multiply out into a single effective sample size for the panel: the temporal axis has its own dependence, and this report does not identify the two jointly, so no combined figure is offered. What it does establish is the cross-sectional factor on its own, measured rather than asserted.

8.7 What the daily series is, and where it is not

The score reads two numbers off the daily profit series. Section 1 took those two as given; this subsection asks what else the series contains, because a mean and a deviation summarise it faithfully only under conditions that can be checked.

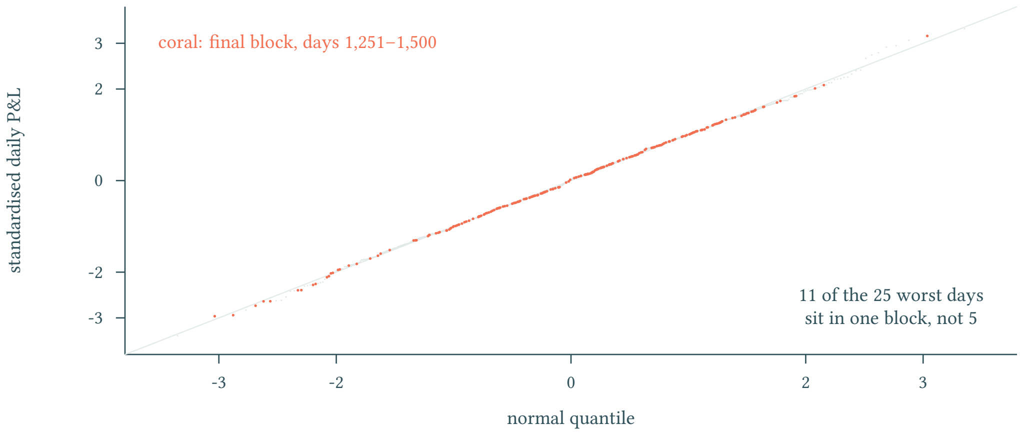


Figure 18: The 1,250 scored days against the normal they are taken to be, computed here from the retained ledger. Pooled, the fit is close to exact — the worst day falls at −3.39 standard deviations where −3.35 is predicted. But the pooled skew of −0.018 is cancellation, not absence: the final block carries skew −0.41 and eleven of the twenty-five worst days where five are expected. The one left tail in this record belongs to the block sitting directly against the out-of-sample boundary. Final-block days are drawn in coral; the grey diagonal is the identity line.

The pooled fit is what a Sharpe-based score assumes, and it holds. The block structure underneath it is what a single Sharpe cannot show, and it is not symmetric across the window: the four earlier blocks have skews between −0.17 and +0.26, the last has −0.41.

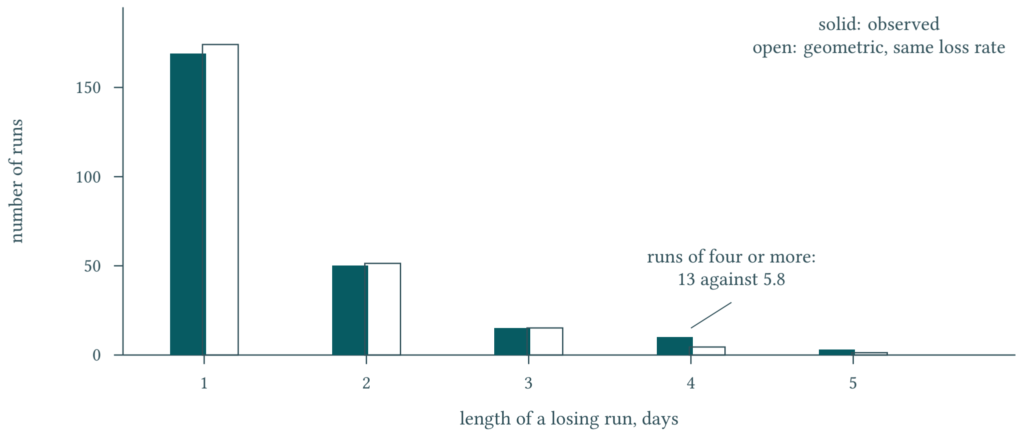


Figure 19: Losing days grouped into consecutive runs, against the geometric distribution the same loss rate would produce if days were independent. The body matches; the tail does not. Runs of four days or more occur thirteen times where 5.8 are predicted, so losses reinforce mildly rather than arriving independently. The longest run, five days, is exactly what independence also predicts — which is why a maximum-streak statistic cannot detect this.

Losses cluster, and the statistic reached for first is blind to it. This matters for the report rather than for the strategy: it is one more reason the five endpoints are not five independent draws, and it is why Section 6 declines to convert them into an effective sample size.

8.8 Forward transfer and inference limits

The frozen transfer diagnostic compares adjacent blocks. Across the four transitions the covariance correlations are 0.524, 0.508, 0.558, and 0.520, while the alpha-map correlations are 0.119, 0.103, 0.122, and 0.087: covariance structure is materially more persistent than the fitted one-step alpha map, while the alpha values remain low in every transition. This supports testing sequential re-estimation and shorter windows; it does not identify a causal signal half-life.

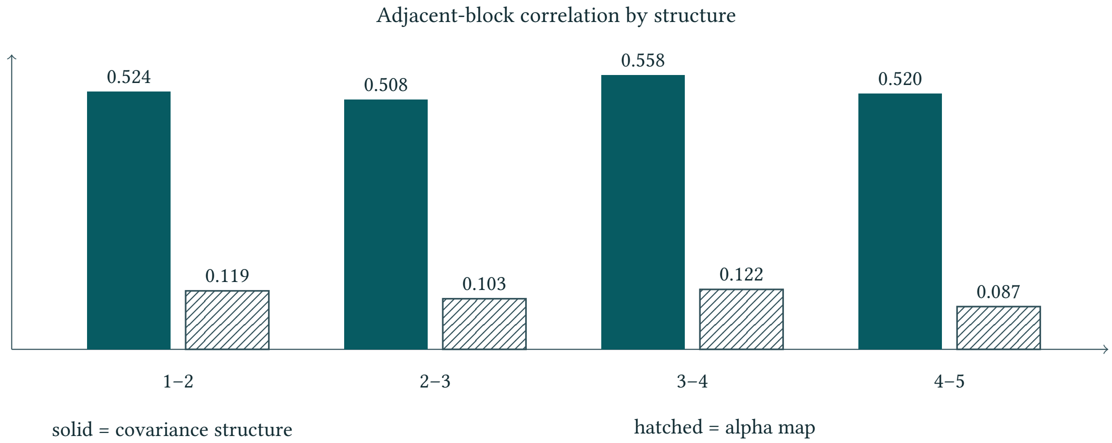


Figure 20: Adjacent-block transfer across five consecutive 250-day blocks spanning 1,250 days. For each adjacent pair the figure reports Pearson r between two objects estimated inside each block: the vector of 1,225 off-diagonal residual-covariance elements (solid), and the vectorised coefficient matrix of a ridge VAR(1) fitted after per-stock z-scoring of the residuals, with $\lambda = 1.60$ (hatched). Both the method and the eight correlations plotted are transcribed: no retained artefact of this report carries the fitted maps, so neither the ridge parameter nor any bar can be recomputed here. This repeatedly inspected diagnostic is not a clean holdout result.

Signal half-life must be estimated prospectively by freezing a map at a cutoff, scoring it at increasing forward horizons, and reporting the decay path with uncertainty. Until that test exists, the signal half-life is not identified.

Factor and residual diagnostics define candidate questions rather than trading decisions. A visually stable structure, a low p-value, or a compelling source chart must first pass a leakage-controlled validation protocol.

Scope. This section mixes two classes of figure and the distinction is load-bearing, so it is stated per quantity, and the basis is named because two different retained records are involved.

Recomputed here from the retained price panel: the instrument-0 identity, the index reconstruction and its rounding decomposition, the raw correlation spectrum and the seven eigenvalues that exceed the structureless level, the raw and de-meaned mean pairwise correlations, and the benchmark's own concentration path. Read from the retained block-diagnostics record rather than recomputed from prices: the effective-dimension counts, whose participation-ratio and entropy-effective-rank columns this report carries but does not re-derive. Performed here on figures taken from elsewhere: the information-ceiling inversion, which is arithmetic applied to transcribed bit counts.

Transcribed, and marked as such where each appears: the residualised eigenvalue share, whose market-beta step this report cannot reconstruct; the information budget in bits, both the same-day totals and the incremental figures; and the block-to-block transfer similarities together with the ridge parameter their figure names. No transcribed figure may be read as this report's own measurement or cited as a local reproduction; no recomputed figure needs to be.

Two families of quantity on the same subjects *are* local reproductions against records this report holds, and they are reported with their own partitions rather than substituted for the figures above. Across six contiguous return blocks the mean pairwise correlation lies between 0.177 and 0.213 and the leading covariance eigenvalue share between 0.181 and 0.215; across five adjacent-block transitions the covariance-structure similarity lies between 0.509 and 0.600 and the one-step fitted-map similarity between 0.105 and 0.157. Section 14 reports why the block-bootstrap intervals those figures carry cannot be read as uncertainty bands around them. These are computed on the raw series under a different partition from the transcribed figures and are not the same numbers restated.

Nothing in this section is evidence that future transfer is proven, that any strategy is superior, or of any result on the hidden window. The descriptive ordering it does support — that covariance structure persists across adjacent blocks more than a fitted one-step map does — is carried at the level of its ordering only; the magnitude of that gap is not established here.

9 Causal Validation: The Contract Must Precede the Candidate

A diagnostic becomes admissible research only after it survives a procedure that respects the order in which information arrived; this chapter defines that procedure. Throughout, *causal* carries its information-set meaning: every input to a quantity was observable at or before that quantity's own timestamp. The word asserts nothing about an identified mechanism. The corresponding failure is **leakage**, any path by which information dated after a decision reaches it. Leakage matters because it changes what a statistic estimates, so that a number computed on available history stops predicting the number the evaluator will compute on history that has not yet been released [2].

9.1 Leakage taxonomy at a cutoff

Table 10 enumerates the channels by which information dated after a decision can reach it at a dated **information cutoff**, and matches each to the control that closes it. Channels are named by mechanism rather than by symptom, because a leaked result looks ordinary from the outside: it is simply better than it should be.

Table 10: Leakage channels at an information cutoff and the control that closes each

Channel	Mechanism at the cutoff	Closing control
Future labels	A feature row absorbs information dated after its own label: a full-sample normaliser, a centred filter, a next-day input.	Prefix-only model input
Overlapping evaluation	Consecutive blocks share days, or the labels of a multi-day horizon, so one shock is scored twice.	Non-overlapping scored blocks; gap only when labels overlap
Post hoc tuning	Window lengths, shrinkage, feature sets, or thresholds are chosen after seeing the block that reports them.	Prefix-only hyperparameter fitting
Retrofitted candidate	A candidate is edited after its own scored result is observed, and the edit inherits the original record.	No same-submission revision
Silent multiplicity	Many candidates are compared on one block; only the maximum is carried forward, and the size of the search goes unrecorded.	Priced max-of-K allowance and mandatory labelling

The first four channels are properties of one experiment; the fifth is a property of the research process around it, which is why closing the first four alone still yields optimistic conclusions.

9.2 The validation contract

The validation contract has six clauses, and they apply to every result reported in Parts II and III.

Prefix-only model input. At a cutoff of T observed days, every input reaching the model at day $t \leq T$ is a function of observations dated t or earlier. Scaling constants, factor loadings, covariance estimates, and winsorisation limits fall under this clause: a statistic computed once over the full panel and reused at every t is leakage even when the model is otherwise well behaved.

Expanding-history fitting. Parameters are re-estimated as the history grows, from observations up to the current decision day and no others. This **expanding-window** design never places a training day after the day being predicted, and a shortened lookback is a modelling choice inside it.

Non-overlapping scored blocks. Out-of-sample assessment is reported on blocks that partition the forward history, never on sliding windows that reuse days and inflate the apparent number of independent tests on which the multiplicity accounting below depends.

Gap only when labels overlap. A purge between training and scored block is inserted when, and only when, the label horizon spans more than one day, so that the two label sets share observations. For a one-day-ahead label the boundary is already clean. The rule is stated in terms of label overlap rather than as a fixed buffer, because a buffer chosen by inspection is itself a tuned parameter.

Prefix-only hyperparameter fitting. Every tunable quantity is selected inside the prefix, by an inner expanding-history split that never touches the scored block. A hyperparameter selected on the block that later reports it turns an out-of-sample statistic into an in-sample one under an unchanged name.

No same-submission revision. Once a candidate has been scored on a block it is closed. Improvements produce a new candidate with a new identity and a new scored block; they never replace the record of the version actually tested, which is what makes the experiment ledger countable.

9.3 The validation chain

Figure 21 draws the chain these clauses define, together with the two stages that close it: the time-stamped lock and the reveal that follows.

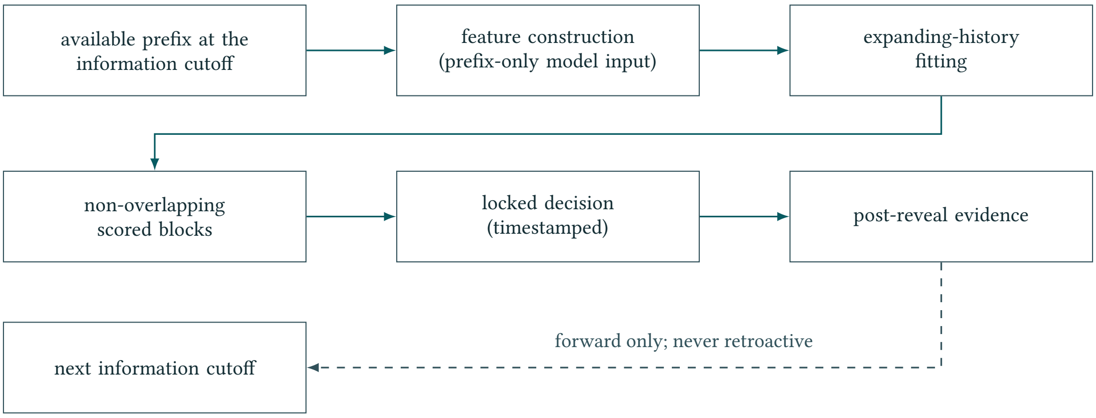


Figure 21: The validation chain at one information cutoff. Solid arrows carry admissible information forward; the dashed arrow is the only route out of a reveal, and no arrow returns to a locked decision.

9.4 Locking and provenance

A contract that exists only as intention is unenforceable. Each decision is therefore fixed by a **timestamped lock** taken before the corresponding forward evidence exists, at the report's four dated locks: **cutoffs 500, 750, 1,000, and 1,500**. Table 11 states its contents. The artefact record retains a contemporaneous lock only at the final cutoff; for the earlier three, conformity is asserted rather than evidenced — the same concession the Abstract and Section 10 record.

Table 11: Objects fixed by a timestamped lock, and the failure each one prevents

Locked object	Content of the record	Failure prevented
Candidate identity	The feature list, parameterisation, initial fitted state, and the deterministic code that turns a prefix into positions.	Retrofitted candidates and untraceable variants
Data prefix	The first and last observation index of the history admissible at that cutoff, and the release that supplied it.	Future labels entering through a wider prefix than declared
Evaluator	The pinned scoring program with its clipping, fee, activity, and scoring-window semantics.	Silent drift of the objective between experiments
Harness	Seeds, package versions, block boundaries, and the order in which blocks are constructed and scored.	Irreproducible reruns and unstated re-selection

The lock is a boundary in time, not a claim of quality: a locked decision that later performs poorly stays in the record on exactly the same terms as one that performs well.

The same discipline governs the figures this report reuses. Three frozen analytical figures from the Gate 2 memorandum are retained in the project figure store: one for the panel and coverage census, one for the instrument-0 identity check, and one for the adjacent-block transfer comparison. Each is a **source-only provenance copy**: the raster is stored unaltered, checked against its recorded digest, and never printed in the reading flow. What the reader sees is a **faithful English adaptation** redrawn from it, matching the original in quantities, orderings, and comparisons while its labels meet the language of this report. A reused analytical figure of this kind is not an evaluator figure and not independent performance evidence; it records what a descriptive analysis observed on an available prefix.

9.5 Multiplicity and multiple testing

Pricing the search. The sample-size limits Section 6 documents make **multiple testing** the dominant threat to inference, because every candidate is scored on slices of one realised path. Let K be the number of candidates compared on a single scored block and σ_b the standard deviation of that block statistic for one candidate under the null of no edge, taken as a common scale across candidates. For draws from that null the expected best value obeys the **max-of-K** bound [7] — independence is not required; the bound holds under any joint dependence with sub-Gaussian margins and is nearest to equality when the draws are independent —

$$\mathbb{E}\left[\max_{1\le k\le K} \hat{s}_k\right] \;\le\; \sigma_b\sqrt{2\ln K},$$

so a search of size K manufactures apparent advantage of that order before any true advantage exists. The bound is stated in score points because Section 1 established that score points are what the objective pays in: with E_μ close to one, a point of score is a point of mean, and the same units price the search and the improvement it is searching for. Positive dependence lowers the expected maximum, so the bound is conservative for variants of one candidate and tightens exactly when the candidates are genuinely different ideas.

Pricing precedes looking. The bound is informative only if K is declared before the candidates are inspected, and only if it counts every comparison actually performed, including candidates abandoned after a single look. A K reconstructed afterwards from the survivors is not a price but a rationalisation, and it is always too small. A candidate is admitted only when its observed block advantage exceeds the allowance recorded when the search was declared.

Three admissible labels. Not every useful number can be produced under these conditions. Instead, every repeated-inspection result is labelled, and only three labels are admissible.

Clean. Produced once, by a rule fixed before the scored block existed, on a block that entered no earlier selection.

Repeatedly inspected. The block, or one adjacent to it, entered an earlier selection, or the candidate survived a comparison. The recorded number of comparisons travels with the statistic, which is then read against the corresponding allowance rather than against zero.

Retrospective. Produced by replaying a later rule on an earlier window with full knowledge of everything in between. A retrospective number is a descriptive benchmark about the data, never evidence about what could have been decided at that cutoff, and never quoted without its label.

9.6 Non-retroactivity

The ordering rule follows directly and is stated once for the whole report: post-reveal evidence may inform the next cutoff, but may never rewrite a prior locked decision. A reveal changes the admissible prefix for every decision that follows it and nothing about any decision that preceded it. In particular, a local replay of a later model on an earlier window is a retrospective benchmark under the rule above, and no such replay is ever upgraded into evidence about an official scored window, a submission acceptance, or a final ranking.

Non-retroactivity binds negatives as tightly as positives. A **negative result** — a candidate that fails its planned test — is archived with the same locked objects, the same label, and the same permanence as a retained one. Discarding negatives would leave a record that looks cleaner than the search that produced it, and would corrupt the count K on which the max-of-K allowance depends. The archive of rejected candidates is therefore not housekeeping; it is the denominator of every claim this report makes.

Two of Part I's results bear directly on how this protocol is set. Because the objective is nearly linear in the mean and almost flat in dispersion (Section 1.4), a validation design that spends its power on variance estimation is spending it where the score does not pay; the contract above therefore tests directional agreement across blocks rather than the stability of a risk estimate. And because combining books is worth under one per cent (Section 1.6), an ensemble may not be admitted on the grounds that it diversifies — it has to earn its place on the sign, which is what the ablation protocol of Section 10 is built to test.

Scope. This section states a protocol, not a result. Nothing in it is measured and nothing in it is estimated: the leakage taxonomy enumerates the ways a cutoff can be violated, the validation contract fixes what a candidate must satisfy before it may be locked, and the multiplicity discussion prices the search rather than reporting one. A protocol is auditable against this report's own dated record and against the archive of rejected candidates, which is why that archive is kept. It is not auditable against price data, and no claim here should be read as evidence about the realised series. Whether the protocol was in fact followed at each cutoff is a question about this report's conduct. It is answered only as far as the dated artefact record answers it — what existed when — and not by any diagnostic reported earlier, nor by a contemporaneous narrative — which, as Section 6 records, this report does not have.

10 Model Selection: Complexity Must Earn Its Place

This chapter fixes what may be proposed, what every proposal must be compared against, and the order in which criteria decide between two survivors. It is a protocol chapter rather than a results chapter: no milestone score, no ranking, and no performance number for the finally submitted model appears here. The rules are stated so that they can be checked against the work rather than recalled after it. What no later chapter supplies is a per-cutoff account of applying them: Part III is organised by question rather than by date, and Section 6 records that no contemporaneous narrative of the **information cutoffs** was kept. This chapter therefore states a standard, and the reader should treat conformity to it as claimed rather than as demonstrated.

10.1 Baseline and candidate families

A candidate is interpretable only against baselines that operate under identical constraints: the same available price prefix, the same price-dependent per-asset caps, the same commissions, and the same obligation to return an integer position vector [3]. Table 12 lists the four baseline families that must be reconstructed at every cutoff before a more elaborate proposal is entertained. A baseline is not a formality: it is the null against which the incremental claim of a candidate is stated.

Table 12: The baseline families the protocol requires at every information cutoff

Family	Mechanism assumed	What the comparison isolates
Static long exposure	Hold a fixed equal-dollar long book in every instrument with no timing decision.	How much of a candidate's daily net dollar P&L is drift already obtainable without any signal, and at what fee cost.
Cross-sectional reversal	Sell recent relative winners and buy recent relative losers on residual ranks, rebalanced daily.	Whether the candidate adds anything beyond the strongest documented cross-sectional effect in this panel.
Time-series momentum	Take the sign of a trailing per-instrument return with one lookback parameter.	Whether the apparent edge is a per-instrument trend already captured by a one-parameter rule.
Pair and spread reversion	Trade the residual of a prefix-selected instrument pair against its own trailing mean.	Whether pair structure contributes beyond a two-leg spread rule, and what the pair-selection search itself costs.

The **candidate families** admitted for development are the proper extensions of those baselines that remain implementable inside the submission interface. A candidate must consume the price prefix only, run single-pass and online, be deterministic given its inputs, express its intent as integer holdings under price-dependent caps, and complete inside the runtime budget without external data or network access. A proposal outside that boundary—a look-ahead filter, a smoother fitted on the whole sample, a stochastic search reseeded at evaluation time, or a configuration whose cost exceeds the runtime budget—is not a weaker candidate but not a candidate at all, and is rejected before any statistic is computed for it.

Two variant axes are admitted inside a family and must be declared when the family is first proposed: a confidence axis, which modulates requested size by a measure of margin, and a risk axis, which throttles gross exposure after adverse realised behaviour. Both axes inherit every rule of the family they modify, and neither may be introduced after a scored comparison has been inspected.

10.2 The submitted architecture

The architecture carried into the final submission is a **seven-feature online non-negative logistic ensemble**. Seven cross-sectional channels—core side, low-rank factor, pair network, pair rank, disagreement overlay, residual level, and level tail—are computed on the same daily panel, each emitting one directional statistic per stock, and a single pooled weight vector combines them into one blended score per stock.

Three constraints of that construction matter for selection. First, fitting is online and prefix-only: the weights are refit at each decision date inside the leakage-controlled **expanding-window** protocol of the preceding chapter, calibrated on a trailing 125-day cross-sectional panel of labels, so that no parameter ever sees a label it is asked to predict. Second, the weights are constrained to be non-negative. A channel can therefore be driven to zero and switched off, but it can never be re-signed; the ensemble cannot rescue a channel whose directional hypothesis is wrong by silently inverting it, and the weight path becomes a readable record of which hypotheses were being paid for at each date. Third, the blended score is used for direction only. Its sign selects the side and the request is taken at full permitted size, so magnitude is governed by the evaluator's caps rather than by a fitted confidence scale, while the index leg, instrument 0, is decided by its own rule because its cap and commission differ from those of the stocks.

None of those three reasons is diversification, and Section 1.6 explains why none could be: by Equation (19), combining two equally good books on this objective is worth seven tenths of one per cent even when they are uncorrelated, because the multiplier that dispersion acts through is spent long before the operating point. **This ensemble is a device for choosing a direction, not for smoothing a return.** Its justification stands or falls on whether pooling the channels improves the sign, which is what the ablation of the next subsection tests.

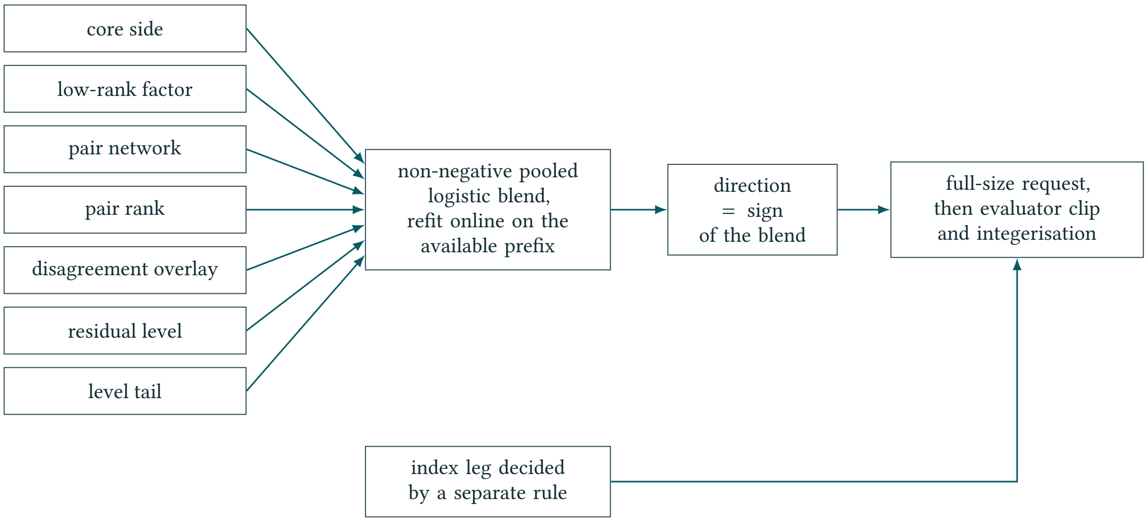


Figure 22: Decision path of the submitted architecture at one information cutoff. The diagram is structural: it records where each degree of freedom sits and where the evaluator, not the model, sets position size. It reports no performance and asserts no predictive direction [5].

The realised weight path of the seven channels over the decision sequence is carried by the frozen weight-path raster and is reproduced later in this report as a faithful English adaptation of that image. It is a source-only provenance copy: it is not an evaluator figure and not independent performance evidence.

10.3 Why complexity must earn its place

Section 8 left Part II with an information budget rather than an impossibility theorem. The frozen record measures that budget in bits per day and compares like with like: against the same-day cross-sectional dependence, the incremental contribution of a lagged network is on the order of one to two per cent of the total structure it finds. The comparison is between two information quantities, not between a signal and a variance, and both are transcribed rather than reproduced here. Covariance structure is meanwhile the more persistent of the two across adjacent blocks, an ordering Section 8 supports without establishing the size of the gap. Complexity is therefore priced against a small budget, and the working rule for the remainder of the report is that **complexity must earn its place**.

The arithmetic of the pooled calibration makes that rule concrete. Seven parameters are fitted on a trailing panel of 125 days by 50 stocks. Those 6,250 labels are not 6,250 independent observations: a single day's cross-section shares a common factor, so the effective sample is closer to 125 day clusters. A per-stock elaboration that fits seven coefficients for each of 50 stocks multiplies the parameter count by fifty against the same clusters. Each added parameter must therefore pay for itself twice—once in the fit, and once in the selection process that retained it after many alternatives had been inspected.

Retention is decided by an **ablation** executed under two protocols on the same decision path. Under column zeroing the channel is removed before the online refit, so the surviving weights re-adapt and absorb whatever they can; the resulting change measures the marginal contribution net of substitution. Under weight zeroing the channel's fitted weight is set to zero at inference while every other weight is held at its fitted value; the resulting change measures the gross contribution, including everything a substitute would have recovered. The two protocols bracket a channel's value and their gap measures substitutability. Neither ordering is universal, so both must be reported. A channel is retained only where the two protocols agree in sign and that agreement holds across blocks; a channel that survives under one protocol alone is recorded as an open question rather than as a component. The magnitudes are not reported here, and no retained artefact of this report carries them: the dual-protocol comparison exists only as a source-only provenance copy adapted into English, so the sign agreement is stated as the rule that governed retention and not as a measurement this report can show.

10.4 Selection policy

Where two proposals both remain admissible, they are compared on an ordered hierarchy applied lexicographically. An earlier criterion is never traded against a later one.

1. **Causal validity.** The candidate and every statistic quoted for it must be reproducible from the prefix available at the cutoff, with a timestamped lock. A leaked comparison is not a small error; it is uninterpretable, and it is discarded rather than discounted.
2. **Deterministic evaluator compatibility.** The candidate must return a conforming integer vector for every day, run inside the runtime budget, and produce the same output on a repeated run of the pinned evaluator. A candidate that times out, raises, or returns a wrong-length vector receives no score at all, so this gate precedes every statistical question [5].
3. **Mean net daily dollar P&L.** The first quantitative criterion is the after-commission mean itself. The objective analysis of Part I showed that the score is this mean multiplied by a factor strictly between zero and one on the regular branch, and that the discount never reaches the other branch: by Equation (2) a loss is scored in full against a discounted equal profit, so a non-positive mean cannot be repaired by any variance improvement [2].
4. **Score and Sharpe.** Only once the mean is established does the composed score discriminate: two candidates with comparable means are separated by the discount their daily-P&L dispersion attracts.
5. **Block stability.** A candidate must hold its advantage across non-overlapping scored blocks. This is the criterion that a limited effective sample makes indispensable, because a single-block advantage is the most probable artefact of repeated inspection.
6. **Turnover, fees, and operational simplicity.** Ties are broken by lower executed turnover, a lower fee load for the same mean, fewer moving parts, and fewer failure modes under the activity and runtime rules.

The order is deliberate. The first two criteria are admissibility gates whose violation costs the whole submission rather than a few points; the third reflects the algebra of the objective, in which the mean is the only channel that cannot be substituted; the fourth prices dispersion; the fifth defends against selection on noise; and the last, while it never justifies a worse objective, reduces the probability of an operational zero and keeps the architecture explicable under examination.

10.5 Robustness and reproducibility protocol

Every robustness check is registered twice. A **planned test** is written before the run and states the hypothesis, the statistic, the decision rule, and the family of comparisons it belongs to. A **realised finding** is written afterwards and states what actually happened, including the case where nothing happened. The two entries are never merged, and the planned entry is never edited to match the outcome. A **negative result** is registered with the same standing as a confirmation: discarding it quietly converts the surviving evidence into a selected maximum and destroys the **multiple testing** accounting that the preceding chapter requires.

Table 13: Robustness checks required of every candidate before a lock

Check	Planned test	The realised finding must state
Ablation	Remove each channel under column zeroing and under weight zeroing on the same decision path.	Both bracket values, their sign agreement, and whether the agreement survives block by block.
Dose path	Inject a graded fraction of oracle-correct next-day signs on a fixed grid and record the realised hit rate.	The fitted slope in score units per percentage point of hit rate, and the fully informed endpoint.
Window sensitivity	Vary the calibration length and the block partition around their chosen values.	Whether the conclusion holds away from the chosen window, or is a window artefact.
Factor and residual legs	Score the index leg and the residual stock leg separately under their own caps and fees.	Which leg carries the effect, since an aggregate gain concentrated in one leg is a different claim.
Fee and cap stress	Re-evaluate with commissions scaled upward and per-asset caps tightened.	Whether the advantage is signal-financed or merely turnover-financed and fee-fragile.

The **dose path** deserves one explicit note, because it supplies the measuring rod that the other checks are read against. On a fixed grid of injected oracle fractions the realised hit rate q is recorded together with the resulting score, and a linear ruler of the form Score $\approx aq + b$ is fitted across that grid. Its slope converts an observed change in hit rate into score units; the fully informed endpoint gives the upper bound the architecture can reach under the same caps and commissions. The ruler is a measurement instrument, not an achievement, and the frozen skill-curve raster is again a provenance copy rather than an evaluator result. **Window sensitivity**, the **factor and residual legs** decomposition, and the **fee and cap stress** are held to the same registration discipline.

No finding is admissible unless a **deterministic rerun** reproduces it. Each registered row carries four identities: a **candidate hash** over the strategy source and its configuration, a **data hash** over the exact price prefix consumed, an **evaluator hash** over the pinned scoring program, and a **harness hash** over the driver that assembles the three. Re-running the same quadruple must reproduce the reported numbers exactly; a mismatch invalidates the finding rather than the record. Any change to one identity opens a new row instead of editing an old one, seeds are fixed, and no branch may depend on wall-clock time or on the machine that runs it [5].

10.6 Common milestone protocol

The protocol fixes five fields, in a fixed order, so that a sequence of dated records reads as one evolving account rather than as a series of independent essays. It is stated in full because it is the standard the work is held to; Section 6 records that this report keeps the dated artefacts the protocol presumes but not the narrative it prescribes.

1. the **retained components** carried over from the previous lock, and those retired;
2. what **new evidence supported or falsified**, stated as a claim about the newly released prefix alone;
3. the **changes in signals, sizing, constraints, or validation** that the evidence justified;
4. the **unresolved questions** carried forward, including every check that ended in a negative or inconclusive result;
5. the **locked decision**, with the timestamp under which it was sealed.

The protocol orders the record strictly by the information **cutoffs 500, 750, 1,000, and 1,500**, and it must not project the final submission — COE/Simple2, the causal online ensemble in its pruned Simple2 form, the architecture of Figure 22 — backwards: the causal online ensemble that eventually became the submitted candidate did not exist at the earlier cutoffs and may not be written about as though it did. It requires that a component absent from an early lock is described as absent rather than as provisionally present, and that a question open at an early cutoff stays open until the evidence which closed it was actually available. Where the final model is replayed on an earlier window, that replay is an explicitly retrospective benchmark computed after the fact: it is never presented as what was known at the earlier cutoff, and no local replay is ever upgraded into evidence about official hidden-window performance, submission acceptance, or final rank [1, 2].

Scope. Two kinds of statement sit in this section and they carry different footings. The architecture, the selection policy and the robustness protocol describe what was built and what rule was applied; they are auditable against the frozen submission source, whose digest is recorded, and against this report's own record. The comparative judgements — that one candidate family was preferred to another, and on what margin — are not established here. Exactly one candidate carries a retained, protocol-bound replay artefact; the others are registered by source hash and authorisation state only, and their relative performance is taken up in Part III, where it is graded and attributed rather than asserted. Nothing in this section is evidence about official hidden-window performance, and no policy stated here is evidence that the policy produced a good outcome.

11 Endpoint Stability: The Result Does Not Move When the History Grows

What is being varied. The staged release of Section 6 supplies a natural robustness axis: the same submission can be scored at successive endpoints, each using every public day up to that endpoint and scoring the final 250. Five endpoints are available on the public panel — Days 500, 750, 1000, 1250 and 1500.

One property makes this a stability measurement rather than a development record. Across all five runs the candidate bytes carry one digest, and the evaluator and price file are likewise one digest each; only the price prefix handed to the strategy differs. **Nothing about the strategy changes between the five rows.** They are five readings of one frozen object, not five stages of an evolving one, so the question they answer is narrow and well posed: does the score depend on where the history happens to end?

Table 14: One frozen submission at five endpoints. Each score is the row's own μ times the multiplier of Equation (3) evaluated at the row's own Sharpe [5].

Endpoint	Scored days	μ	σ	SR	Score
Day 500	251–500	\$1,079.54	\$2,121.89	8.04	\$1,063.11
Day 750	501–750	\$1,174.26	\$2,049.67	9.06	\$1,160.12
Day 1,000	751–1,000	\$1,152.50	\$2,137.33	8.53	\$1,136.86
Day 1,250	1,001–1,250	\$1,212.95	\$2,192.36	8.75	\$1,197.30
Day 1,500	1,251–1,500	\$1,183.59	\$2,227.87	8.40	\$1,167.05

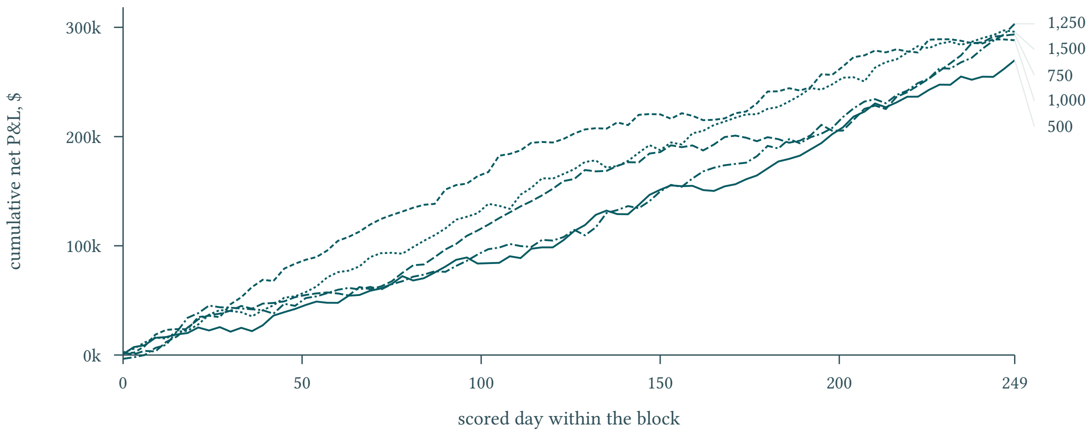


Figure 23: The scored P&L of one frozen submission accumulating on each of the five endpoints, computed here from the retained daily ledger. The table records where each path ends; the paths show that none of them gets there by a different route — there is no endpoint whose result rests on one stretch of the block.

The replay reproduces the published rule. Each row is an independent check on the accounting, because Score is not reported by the strategy but computed by the evaluator. Forming μM from the row's own μ and SR reproduces the recorded score to the cent at every endpoint. The arithmetic of Section 1 and the arithmetic of the pinned evaluator are therefore the same arithmetic, verified five times over.

Every endpoint lands inside the research range. Sharpe ranges over 8.04 to 9.06, so all five rows sit far above the SR ≥ 3 boundary of Equation (20), and the shortfall $1 - M$ they actually realise is

$$1.52\%, \quad 1.20\%, \quad 1.36\%, \quad 1.29\%, \quad 1.40\%,$$

against the 1.6% that Section 1.7 attaches to SR $= 8$ from published constants alone. **The operating point of the submission is the landmark Part I derived before any price was examined.** The first constraint of Section 1.8 — that engineering effort belongs on the mean — is therefore not a prediction awaiting test; it is the condition the book is measured to be in, at every endpoint on the public panel. The same Sharpe range places every endpoint above the SR $= \sqrt{3}$ sign change of Equation (18), so the score is concave in both inputs at the measured operating point and the ranking of the two is settled by the elasticities rather than by curvature.

The upper boundary, evaluated. Section 1.7 left its second boundary as a function awaiting one quantity it could not supply: the luck scale ε of a single scored block, the amount a score moves with the strategy untouched because different days were drawn. These five rows can supply it, but not in the way that first suggests itself.

Two quantities, and only one of them is the luck scale. The dispersion of the five block scores is 50.55 on the sample convention and 45.22 on the population one. It answers how much this strategy's score differs from one quarter to the next, which mixes sampling noise with real differences between market periods. What ε denotes is narrower: how much one block's score would move with the strategy untouched because different days were drawn. That is a standard error, and **five observations cannot estimate it**. On four degrees of freedom the 50.55 carries a 95 per cent interval of $[30.3, 145.3]$ — **a factor of nearly five** — so a boundary built on it is not a loose bound but an unresolved one.

The same records hold two hundred and fifty times more data. Each block carries 250 daily profits, and the standard error of its score follows from them directly: $\mathrm{se}(S) = \mathrm{se}(\mu)\,\partial S/\partial\mu$ with $\mathrm{se}(\mu) = \sigma/\sqrt{250}$, which by Equation (15) is μ/SR — the Sharpe the evaluator reports *is* the t-statistic of the mean, so the standard error needs no separate estimation. Evaluated block by block that gives 136.2, 131.2, 137.0, 140.4 and 142.8 — **a span of** 1.09**, against the five-observation estimator's** 4.8. A moving-block bootstrap [6] on the same daily series, which makes no normal-theory assumption and respects serial dependence, returns 126 to 164 and agrees with the analytic route to twenty per cent. The two are not in conflict with the five-block figure either: 137 lies inside its interval. The five-block figure simply cannot resolve anything, and is not used as ε here.

With ε between 131.2 and 142.8 and a mean daily profit of 1,160.57, the boundary of Equation (21) evaluates to

$$S^* = \sqrt{\frac{1{,}160.57}{\varepsilon} - 1} \;\in\; [\,2.67,\;2.80\,].$$

That the headroom at S^* equals ε is a restatement of the definition rather than a result. The result is where the book sits relative to it.

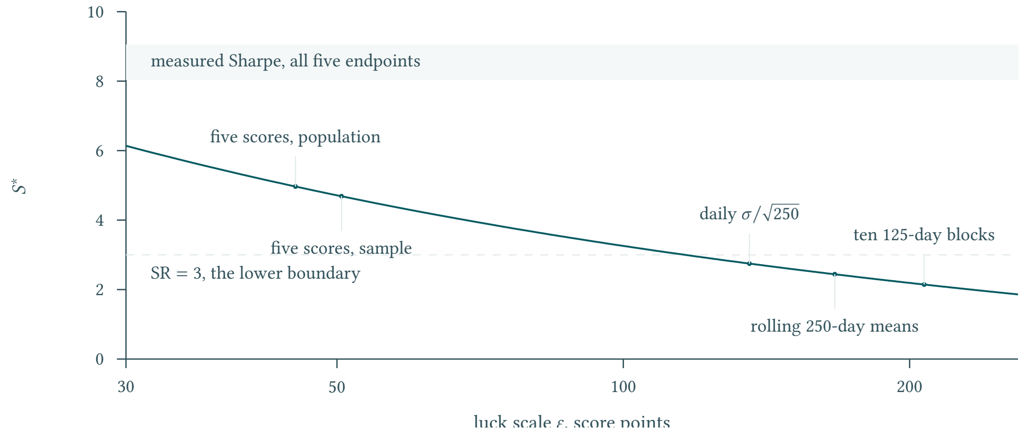


Figure 24: The boundary against the luck scale that determines it, with five estimates of ε taken from the retained records, on a logarithmic horizontal axis. They span a factor of four and a half and put S^* anywhere between 2.1 and 5.0. Every one of them lies below the measured Sharpe: the conclusion does not depend on the choice, and under most choices it is stronger than the printed figure suggests.

The verdict is robust, and it no longer rests on a choice. The earlier reading of these records took $\varepsilon = 45.22$ and reported $S^* = 4.93$, then defended the conclusion by showing it survived four other readings that put S^* anywhere between 2.1 and 4.9. That defence is no longer needed. The estimator used above is not one reading among several but the one built on the data the question requires: 250 daily observations per block rather than five block totals, and it is confirmed by a bootstrap that assumes nothing about the shape of the daily distribution. Its span is nine per cent.

What survives is the comparison that matters, and by a wider margin than before: the measured Sharpe of 8.04 to 9.06 stands **2.9 times** above the worst corner of the boundary, and S^* would reach the measured Sharpe only if ε fell to 17.7 — under a seventh of the smallest value the daily series supports. The dispersion of every endpoint likewise lies inside the tolerance $\sigma \le \sqrt{250}\,\mu/S^*$ under every block's own estimate.

Why this matters beyond the boundary. The 45.22 was used elsewhere in Part III as the scale an improvement must beat before it can be demonstrated. It is the wrong scale for that question by a factor of three, and Sections 15 and 20 now use the standard error of a block's score instead. In one place the substitution reverses a conclusion, which is recorded where it occurs.

The consequence is the one the boundary was constructed to deliver. Evaluating Equation (5) at the operating point, the multiplier still withholds $\mu/(1+\mathrm{SR}^2)$. Evaluated block by block that is 16.43, 14.14, 15.64, 15.65 and 16.54 points of score — identically dollars a day, because Equation (4) puts $\partial S/\partial\mu$ within two per cent of unity at any qualifying Sharpe — so it runs 14.14 to 16.54, or 1.20 to 1.52 per cent of the mean. The figure is quoted as an interval rather than as its largest member because the five blocks are the sample. That is the whole of what remains on the axis, not a step along it: no improvement in Sharpe, however large, can collect more. Against a standard error of 131 to 143 on a single block's score, **the entire remaining return to Sharpe work is about an eighth of the noise it must be demonstrated against**, so pushing Sharpe further cannot be shown to have helped even if it did. Part I fixed that criterion from published constants before a strategy existed; the criterion is now evaluated, and it says the Sharpe axis is closed.

Dispersion, not the mean, absorbs the growth. Between the first and last endpoint μ rises by 9.6% and σ by 5.0%, while SR ends within 4.5% of where it started. The elasticity identity of Section 1.4 accounts for the pattern without further argument: SR responds to the ratio and is blind to common scale (Equation (8)), so a book whose two moments drift together in the same direction moves along the flat part of the multiplier and carries its score with μ.

Boundary. The five rows are not a continuous backtest and they are not independent out-of-sample blocks. Each endpoint scores its own final 250 days using all public history before it, so successive rows share observed history and overlap in the information they condition on; the effective sample argument of Section 6 applies to them in full. They are local public-data replays.

Scope. Every figure in the table is a local reproduction against a retained record that carries the candidate, evaluator, data and harness digests, so each row can be recomputed from the public panel. Two independent repeats at each endpoint returned identical position fingerprints, and every run finished inside the published 600-second budget [5]. The shortfall percentages and the drift figures are arithmetic on the table.

One boundary, stated once for the whole part. Nothing in Part III is evidence about official hidden-window performance, submission, acceptance, or final rank [1, 2]. Every measurement below is a local replay on the published panel, and every figure taken from the frozen technical record is attributed where it appears. Each section's own scope paragraph states only what is particular to it; this sentence is not repeated. Appendix C lists every claim in this part that is a reproduction rather than a transcription, and Appendix D the records they are reproduced against.

12 The Ruler: What a Percentage Point Is Worth

Why a ruler comes first. Everything that follows in Part III is a comparison: a candidate against the submission, a predecessor against its successor, an outside method against the one in place. A comparison needs a unit. Section 1.4 derived that unit from the published rule alone — the elasticities E_μ and E_σ, which convert a percentage change in either input into a percentage change in score. This section measures them where the book actually sits, so that every later claim can be quoted in a unit the objective function itself defines.

The measurement. At each of the five endpoints of Section 11 the score's partial derivatives were recomputed from the recorded μ and σ, against the same pinned evaluator and price panel as the replay itself.

Table 15: Local elasticities of the score at the replayed operating point, and the two identities of Section 1.4 evaluated on the same rows [5].

Endpoint	SR	E_μ	$\lvert E_\sigma \rvert$	$E_\mu - \lvert E_\sigma \rvert$	$E_\mu / \lvert E_\sigma \rvert$
Day 500	8.04	1.0304	0.0304	1.000	33.85
Day 750	9.06	1.0241	0.0241	1.000	42.53
Day 1,000	8.53	1.0271	0.0271	1.000	37.85
Day 1,250	8.75	1.0258	0.0258	1.000	39.76
Day 1,500	8.40	1.0279	0.0279	1.000	36.78

The derived identities hold exactly. Two results of Section 1.4 are checkable here rather than assumed. The gap $E_\mu - |E_\sigma|$ is 1.000 on every row, as Equation (11) requires. Evaluating $|E_\sigma| = 2/(1 + \mathrm{SR}^2)$ at each row's own Sharpe reproduces the recomputed value to within 4×10^{-13} — floating-point agreement, not approximate agreement. **The algebra of Part I and the arithmetic of the evaluator are the same object, confirmed at five independent operating points.**

The ruler, stated. Reading the last column: at the measured operating point **one per cent of mean is worth roughly 34 to 43 times as much score as one per cent of dispersion**. Section 1.4 predicted the ratio $(\mathrm{SR}^2 + 3)/2$ and named 33.5 as its landmark at $\mathrm{SR} = 8$ (Equation (13)); the book is measured at 33.85 there. The prediction was made from published constants, before any price was examined.

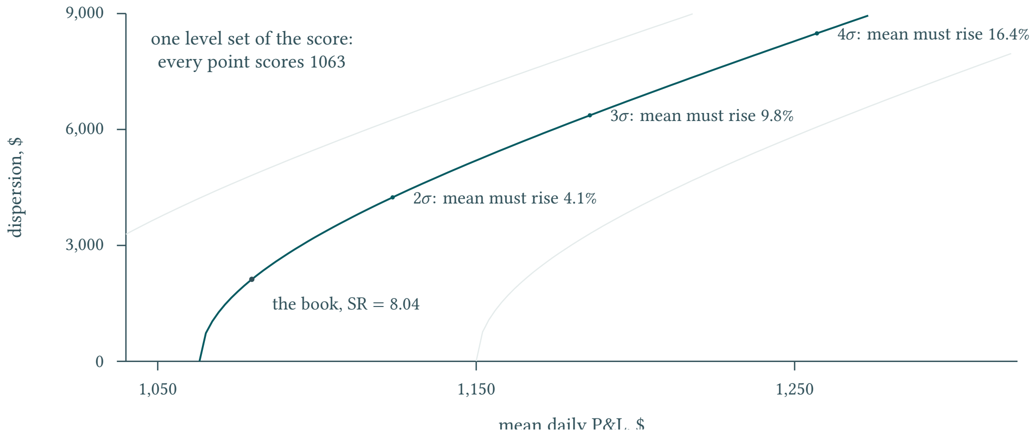


Figure 25: One level set of the published score through the measured operating point, computed in closed form; the two fainter grey curves are the neighbouring level sets. The elasticity above is the slope of this curve at that point, and the curve leaves it quickly: doubling the dispersion requires a mean four per cent higher to hold the score, where the tangent rate over the same log 2 move quotes two. **The exchange rate is a tangent, and it under-prices every finite move by roughly the size of the move.**

Two consequences fix the unit for the rest of Part III.

1. **Dispersion work is priced out, not merely deprioritised.** A variance reduction of a full ten per cent — a large change to any risk model — is a dispersion reduction of only $\sqrt{0.9}$, or 5.13%, and the elasticity applies to dispersion rather than to variance. It therefore moves the score by about 0.16%. On the Day 500 book that is under $2 of score. Any candidate whose entire mechanism is a dispersion change is competing for that $2.
2. **Mean improvements pass through almost undamped.** With $E_\mu \approx 1.03$, a one per cent gain in mean daily P&L is a one per cent gain in score and a little more. The multiplier neither amplifies nor absorbs it.

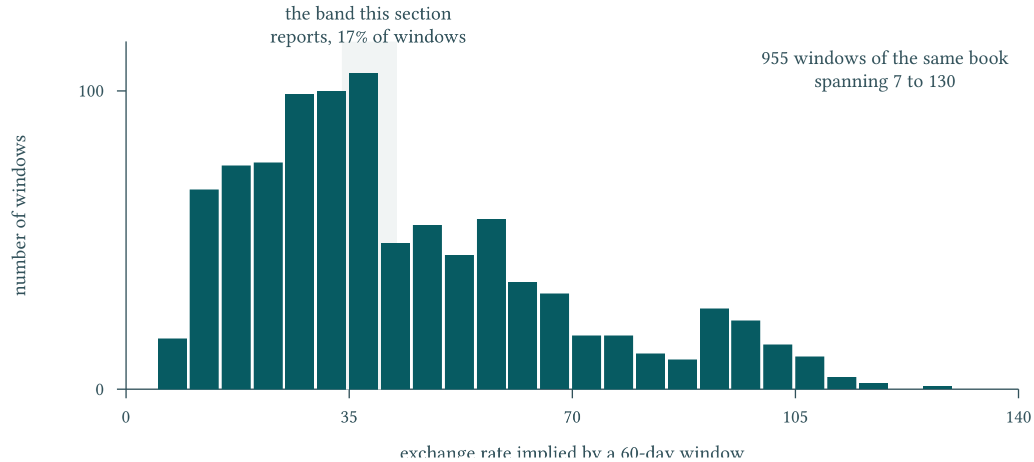


Figure 26: The same exchange rate recomputed on every 60-day window inside the same five blocks. The band this section reports holds seventeen per cent of them; the rest run from seven to a hundred and thirty. **The ruler is a property of the 250-day window the rules fix, not of the book** — which is what makes it usable as a unit, since the competition scores that window and no other.

Boundary. These are *local* slopes, each holding the other input fixed at the replayed operating point. They do not say that risk does not matter or that volatility is irrelevant: σ enters SR with unit elasticity throughout, and the ratio above is large only because the book sits high on the multiplier, where Section 1.7 already places it. At SR = 1 the same formula gives a ratio of 2. Nor do the two columns describe a jointly feasible path: no strategy moves μ while holding σ fixed, and this section makes no claim that one could.

Scope. The elasticities are local reproductions carrying the candidate, evaluator, data and replay digests of Section 11; the identity checks are arithmetic on the published formula. The dollar figure in the first consequence is arithmetic on Table 14.

13 The Engine Audit: Every Published Constant, Measured

What is being tested. Part I read the published rules and derived consequences from them without touching a price. This section puts every one of those consequences against the replay. The instrument is a shadow ledger: a second, independent accounting of the same five endpoints, whose daily figure was compared against the official evaluator's own internal value by tracing the evaluator's execution to the single line where it records a day's profit and loss.

The comparison is exact. Across the 1,250 scored rows the largest absolute discrepancy between shadow and official daily P&L is 0 — not a tolerance met, but agreement to the last representable digit. Everything below is therefore a measurement of the official engine, not of a reimplementation of it.

Table 16: Part I derived each of these from published constants alone. The right column is what the replay records [5].

Derived in	Claim	Measured across five endpoints
Section 3	Fee is the published rate times executed dollar turnover	Identity holds on all 255 instrument-endpoint rows; largest residual 4×10^{-13}
Section 5	Turnover is price times *absolute* share change	On each opening day, where the prior holding is zero, turnover equals gross exposure to the cent
Section 5	The fee for decision t is charged on the next mark	$\text{charged}_t = \text{generated}_{t-1}$ on all 1,250 consecutive pairs, zero violations, maximum difference 0
Section 5	The final iteration is mark-only: no call, no turnover	The single zero-volume row in each 251-row block, and the only row generating no fee
Section 5	Exactly `numTestDays` observations enter the score, with a population deviation	$N = 250$; recomputing μ, σ at `ddof=0`, Sharpe and score from the raw ledger reproduces all four to 10^{-10}
Section 2	Long and short are both recordable	Gross exposure exceeds $\lvert\text{net}\rvert$ on 251 of 251 rows at every endpoint
Section 2	The integer constraint keeps the book about \$1,614 under its ceiling	Mean gross exposure \$598,374 against \$600,000: a shortfall of \$1,626, or 1.007 times the predicted floor
Section 4	The activity floor never binds an extended book	Realised volume \$116.3–\$122.7 million against the \$25,000 floor: a margin of about 4,700×

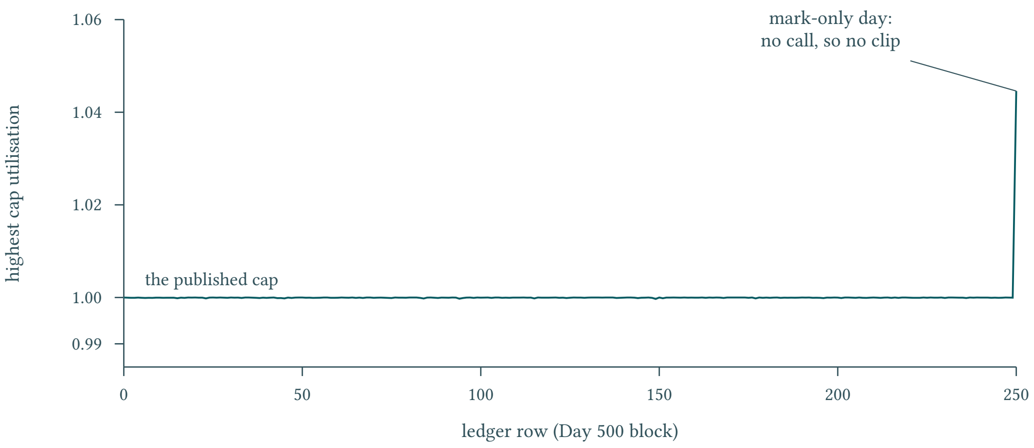


Figure 27: Highest cap utilisation on every row of one block, from the retained ledger. The book sits against the published cap on every decision day and the evaluator's clip never has to act; the single excursion above it is the terminal mark, where no call is made and a price move alone carries a compliant holding over the line.

The sample estimator is excluded, and by how much. One line of Table 16 is worth separating because it is the kind of detail that silently changes a reported score. At the Day 500 endpoint the population deviation is σ_{pop} = \$2,121.894 and the sample deviation is σ_{samp} = \$2,126.151. Using the wrong one moves Sharpe by 0.016 and the reported score by \$0.065 — small, but one-directional, and it would recur at every endpoint. The published rule specifies the population form and the evaluator applies it; the audit confirms the report has read it correctly rather than assumed it.

The cost result, tested where it can actually fail. Section 3 concluded that the research direction *trade less to save cost* is eliminated before any data is examined. That conclusion is not tested by observing that fees are small; it is tested by removing them. Rescoring the same position path with every commission set to zero — positions untouched, so the counterfactual is the strategy's own gross series — gives

Table 17: The whole of what perfect execution at zero cost would return, priced in the objective's own units. Each column is rounded independently from the unrounded record, so a difference recomputed from displayed values can differ in the final cent.

Endpoint	Score as scored	Score at zero cost	Gain	Gain
Day 500	\$1,063.11	\$1,103.49	\$40.38	3.80%
Day 750	\$1,160.12	\$1,200.01	\$39.89	3.44%
Day 1,000	\$1,136.86	\$1,178.41	\$41.55	3.66%
Day 1,250	\$1,197.30	\$1,238.26	\$40.95	3.42%
Day 1,500	\$1,167.05	\$1,207.13	\$40.07	3.43%

Abolishing commission entirely is worth about three and a half per cent of score. That is the ceiling on the entire cost dimension — not the return to a clever turnover schedule, but the return to paying nothing at all. Against it, Section 12 priced one per cent of mean at rather more than one per cent of score, so the whole cost dimension is worth roughly what a three and a half per cent improvement in the mean is worth, and unlike the mean it cannot be exceeded. The Part I conclusion survives the strongest test available to it.

Where the audit contradicted Part I. Two derived claims did not survive, and both have been corrected in place rather than defended. The caps are rechecked only on days the strategy is called, so the terminal mark-only day — which is scored — can carry holdings above their cap after a price move; between 10 and 34 of the 51 instruments do so at every endpoint, the worst by 5.33%. And because turnover charges the pre-trade holding as well as the post-trade one, the \$104 daily ceiling of Section 3 is an order-of-magnitude figure rather than a strict bound. Neither changes a research conclusion, but a derivation that is wrong in the third decimal is still wrong, and a report that only reported its confirmations would not be worth auditing.

Boundary. The evaluator's clipping transformation was never observed to act: the recorded position vectors before and after clipping share one digest at all five endpoints, so in 1,250 calls the clip altered nothing. This audit therefore confirms that the strategy sized itself within the published caps; it does *not* confirm that the evaluator would enforce them against a request that exceeded them. That mechanism is transcribed from the pinned source in Section 5.3 and is not measured here.

Scope. Every figure in this section is a class B local reproduction against a retained record carrying the candidate, evaluator, data and harness digests, and each is recomputed from the raw daily ledger rather than read from a summary. The zero-cost column is a rescoring of the recorded gross series under the published formula, not a replay of a different strategy: no position was changed, and it is not a claim that a zero-cost venue exists.

14 Causality Executed: What Survives Mutating the Future

From taxonomy to test. Section 9 set out the ways a strategy can see across a cutoff and fixed a protocol against them. A protocol is a commitment, and a commitment is not evidence. This section reports what happens when the commitment is executed against the submitted bytes: six behavioural contracts run on the frozen candidate in an isolated interpreter, with environment variables stripped and every threading control pinned to a single thread so that a difference in output cannot be blamed on the machine.

The decisive one. Leakage means the decision depends on information the strategy should not have. That is directly falsifiable: change the information and see whether the decision moves. The price file was mutated after the decision day — a different file, a different digest — while the prefix handed to the strategy was held identical, and the strategy was rerun.

Table 18: Behavioural contracts executed against the frozen submission in an isolated runtime. Each names the failure it would expose.

Contract	What its failure would mean	Result
Future-suffix invariance	The decision depends on prices after the decision day: leakage	pass
Cold sequential prefix parity	A cold start on a prefix disagrees with arriving there sequentially: hidden state	pass
Repeated prefix idempotence	The same prefix twice gives two answers: order dependence	pass
Jump and rewind reset	Moving the history backwards leaves residue from the abandoned path	pass
Deterministic fresh process	A new process gives a different answer on identical input	pass
Return length and integer type	The interface contract of Section 5 is violated	pass

Under mutation the returned position array was **identical, element for element and digest for digest**, while the received prefix digests likewise matched — confirming the mutation lay strictly in the future rather than in what the strategy was shown. A separate scan of the submitted source for calls that could reach outside the passed prefix returned none.

What this does not establish, in the record's own words. The contract enumerates what it declines to claim, and the list is worth reproducing rather than paraphrasing: it is not a proof of no leakage; not a demonstration that all inputs are causal; not cross-environment determinism; not an identity of internal state. Its permitted scope is the tested public prefixes under one fixed local runtime. The source scan covers the submitted file's own calls and explicitly does not cover the internals of the libraries it imports. A passed contract forecloses a failure mode; it does not certify the absence of every one.

What persists between blocks. The second question this section can answer from a retained record is which structure survives from one period to the next. Splitting the public returns into six contiguous blocks and comparing each adjacent pair, the similarity of the covariance structure runs 0.509 to 0.600, while the similarity of a fitted one-step map runs 0.105 to 0.157. **The risk skeleton persists between adjacent blocks; the fitted map barely does.** That ordering is what licenses the architecture of Section 10: a covariance estimate may be carried forward, a fitted directional map may not, and must be refreshed online.

A second caution belongs beside the first. Both similarity families are computed on residual objects, and Equation (29) showed that cross-sectional de-meaning forces an average pairwise *covariance* of $-\overline{\mathrm{Var}}/(n-1)$ whatever the data contains, and an average pairwise correlation of $-1/(n-1)$ when the variances are equal. The similarities here compare structure *between* blocks rather than levels within one, so that floor does not enter them; it is noted because a reader moving between the two sections would otherwise have to establish for themselves that the two quantities are different objects.

A caution about the intervals, which are not what they look like. Each similarity carries a moving-block bootstrap interval at the 95 per cent level. The retained table holds three similarity families — the correlation-matrix structure, its scale-sensitive covariance counterpart, and the fitted one-step map — across the five adjacent transitions, hence fifteen rows. On 12 of the 15 rows *the point estimate lies above its own upper bound.* That is not an anomaly to be reported around: resampling blocks of ten days breaks the very cross-sectional structure the statistic measures, so the resampled distribution sits systematically below the full-sample value. The intervals therefore describe how the estimator behaves under block resampling, and must not be read as an uncertainty band around the estimate; quoted as confidence intervals they would tell a reader the opposite of what the point estimates say. This report uses the point estimates for the ordering above and does not attach an uncertainty statement to their magnitudes.

Boundary. The similarities are descriptive comparisons between adjacent blocks of one realised public panel. They are not evidence that transfer to a future block is proven, not evidence that any strategy is superior, and not evidence about the hidden window. The ordering between the two families is the claim; the size of the gap between them is not established here.

Scope. The contract results are a behavioural record of one pinned local runtime against the frozen submission bytes, retained with its digests and its own boundary statement. The block similarities are a class B local reproduction against a retained record, computed on the raw public series under a six-way partition, with the estimator, block length, replicate count and seed all recorded.

15 Perturbing the Decisions: The Evaluator as a Measuring Instrument

A second instrument, on an axis the algebra cannot reach. Section 12 converted a percentage change in μ or σ into a percentage change in score, and could do so because that conversion is a property of the published formula. It cannot answer the question a strategy actually faces: what is a percentage point of directional accuracy worth? That depends on how much a correct day earns against how much an incorrect day loses, which is a property of the data and appears nowhere in the rule.

The method the record uses to reach it is worth stating as a method, because it generalises beyond this competition. **The strategy's decisions are perturbed by a known amount and the evaluator is read as the instrument.** The perturbation runs in two directions and each answers a different question.

Table 19: One lever, two directions. Both hold the grid, the caps, the fees and the settlement fixed and change only the decisions.

Direction	Operation	What it measures
Inject information	Flip a share of *wrong* cells to correct, using knowledge the strategy does not have	The exchange rate between accuracy and score
Destroy information	Reverse the sign of a share of cells at random	The dispersion of score under no skill, and hence the price of selection (Section 17)

Injecting information is the unusual half. An oracle is not a strategy and could never be submitted; the point is not to build it but to use it as a ruler. By importing exactly the amount of outside knowledge one wishes to price, and reading what the published scoring rule returns, one obtains the conversion factor between research progress and competition score — before deciding whether any particular piece of research is worth doing.

Why the response is linear, and what the two coefficients are. Each decision cell contributes to the daily total independently of the others, so changing N cells moves the score by an amount proportional to $N(2q-1)$, where q is the accuracy of whatever is written into them, less a term proportional to N for the turnover the changes themselves generate. The functional form is fixed by that argument; the sweep only measures the two coefficients. Measured here, on the same validated harness Section 17 uses,

$$\frac{\text{Gain}}{N} = 0.76074\,(2q-1) - 0.09116, \tag{33}$$

across seven injected accuracies from $q = 0.4$ to $q = 1$ in each of the five scored blocks. Setting the left side to zero gives $q^* = \frac{1}{2}(1 + 0.09116/0.76074) = 55.99$ per cent.

What that break-even is the break-even of. It is not the accuracy at which the book stops making money. It is the accuracy **a replacement signal must reach before it is worth switching to**, because overwriting a cell pays turnover whether or not that cell was already right. The book's own directional accuracy, measured over every decided cell in the five scored blocks, is 54.07 per cent. **A proposal that overwrites decisions must therefore beat the existing book by nearly two full points of accuracy before it breaks even.** That is a measured cost, not a bound on what the panel contains: whether this panel holds two more points of accuracy depends on a divisor Section 8 leaves open, and this report does not close it. What the measurement does settle is the price of entry, and it settles it without reference to the ceiling.

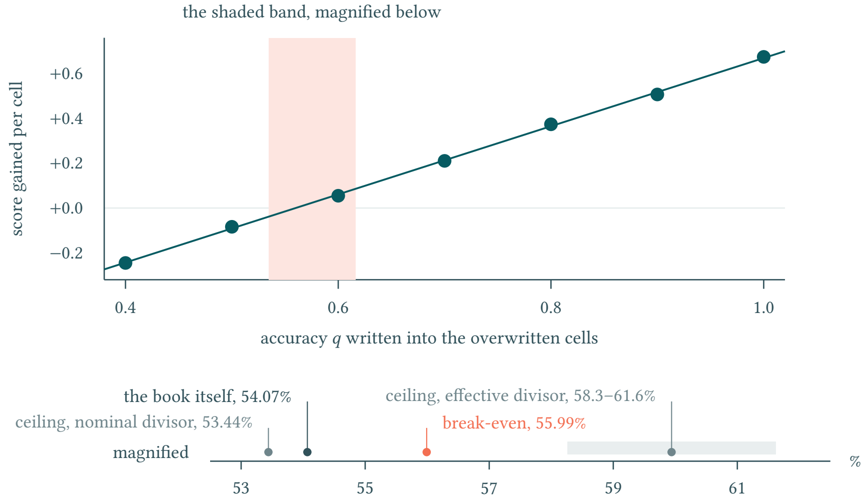


Figure 28: The ruler, measured. Seven injected accuracies in each of the five scored blocks, and the line fitted through them. The three accuracies that decide the argument fall within two and a half points of each other, so they are magnified in the inset. The book sits at 54.07 per cent and a replacement signal must reach 55.99 to pay for the turnover it creates. The panel's information ceiling is drawn twice because its divisor is unsettled: the nominal count of fifty daily decisions puts it at 53.44, and the four to five effective units Part II measures put it at 58.3 to 61.6. **Break-even lies between the two readings, so whether the panel can supply the two points is the open question, not a settled one.**

Two constructions that must agree, and do. The same lever run the other way corrects cells the book got *wrong*, which is the ruler for research that improves an existing decision rather than replacing it. Measured, a corrected cell is worth 1.4696 points. These two numbers are not independent: overwriting a random cell with a perfect signal only helps on the share that was wrong, so it must be worth $1.4696 \times (1 - 0.5407) = 0.6750$ points, and the overwriting sweep at $q = 1$ measures 0.6747. **Two constructions, agreeing to a part in two thousand**, which is the check that neither number is an artefact of how its cells were chosen. The fitted line does not pass exactly through that datum — Equation (33) returns 0.6696 at $q = 1$, about eight parts in a thousand below the measured point — because it is a least-squares fit across seven injected accuracies rather than an interpolation through any one of them, and that residual is the size of the departure from linearity.

What the ruler then says about the whole competition. One percentage point of accuracy moves $2q-1$ by 0.02, so Equation (33) prices it at $2 \times 0.76074 \times 0.01 \times 12{,}750 = 194$ points of block score. The correcting construction gives the same figure by a different route — one point of accuracy moves 127.5 of the block's cells from wrong to right, at 1.4696 points each, or 187 — so the exchange rate is about 190 points per point of accuracy, the two constructions bracketing it within four per cent. Half a point of accuracy is therefore worth about 95 points of block score. Whether that can be demonstrated depends on which experiment is available, and the two answers differ by two orders of magnitude. Run as a paired experiment — the same path, the same days, only those cells changed — the improvement is measured against a paired standard error of about 12 points, so half an accuracy point returns $t \approx 8$ and is positive in every draw. In a backtest on the same days it is easy to see. Run as a comparison of levels — one strategy scored on days nobody has seen — the comparator is the standard error of a block's score, 131 to 143 points (Section 11), and half an accuracy point disappears into it. **The competition asks the second question.** A submission is scored once, on a window, with no paired control; an improvement that is unmistakable in a paired backtest and invisible in a level comparison is exactly the thing this report has to be careful about, because the first is what a developer sees and the second is what the leaderboard reports. The quarter-to-quarter dispersion of 50.55 answers a third question — how much the same strategy's score moves between quarters — and is not the comparator for either. Every proposal, every candidate and every claimed improvement in Part III is converted into accuracy points before it is weighed, which is what makes the comparisons in Sections 17 and 18 commensurable at all.

15.1 What accuracy does not account for

The ruler prices direction, and it is tempting to read it as pricing everything: convert a proposal into accuracy points and weigh it there. That reading is too strong, and the same harness measures by how much.

Build a null that *matches* the book's own directional accuracy but chooses which cells are correct at random: keep every position size, keep the share of correct cells at the book's own q_0, and scatter the correct ones. A thousand draws in each of the five blocks puts the book 362 **to** 493 **points above that null**, four to five null standard deviations clear, with none of the five thousand draws reaching it.

Accuracy alone therefore reproduces between 58 **and** 69 **per cent of the score.** The remainder is not accuracy but placement: the book's correct cells are larger than its incorrect ones. Averaged over a block, a correct cell carries about $198 to $202 of price move against $183 to $186 on an incorrect one, an asymmetry of roughly eight per cent that a randomly placed null cannot have.

This bounds what the exchange rate can be asked to do. It converts a change in accuracy into score correctly, and Section 20 uses it that way. It does not follow that two strategies with the same accuracy score the same, and no claim in this report may be made on that basis.

15.2 What getting the sizes wrong costs

Part I closed the sizing axis only for a *common* scale factor: multiplying every position by one number leaves the Sharpe untouched, so the optimum is the cap. It said nothing about relative size, which the same harness can price. Multiplying a random subset of cells by a factor, leaving every direction intact, gives a null in the magnitude dimension to set beside the sign-reversal null in the directional one.

At 2,550 cells the sign-reversal null produces a score dispersion of 102.45 points on this batch of draws (the batch behind Table 21 puts the same cell count at 99.71; the gap is seed-to-seed variation of the kind Section 17 records). The same number of cells halved produces 24.93, and a scale-neutral pairing of 0.5 and 1.5 produces 27.68. **Getting the relative sizes wrong costs about a quarter of what getting the directions wrong costs**, at the same number of cells disturbed.

This is Part I's conclusion confirmed on the axis it could not reach, and it is also the reason Section 2 can report that the submitted book holds every cell exactly at its cap without that being a lost opportunity: the dimension it declines to use is worth a quarter of the one it does.

15.3 The horizon: what one day of delay costs

A perturbation that changes no decision and no size, only *when* each is acted on, isolates the horizon of the signal. Acting on every decision one day late takes the five-block mean score from 1,144.9 to 296.2: **a single day of delay removes seventy-four per cent of the score**, and no block retains more than 36 per cent. Directional accuracy falls from 54.07 to 51.49 per cent over the same step while the commission bill is unchanged, so what collapses is direction rather than cost. Beyond the first day the curve flattens: lags of one to ten days average 179.7, about a sixth of the unlagged score.

This is a one-day signal. The consequence for Part III is not about performance but about fragility: the failure mode with the shortest fuse is not a gradual loss of edge but a delay in acting on it, and nothing in the published rules guarantees the execution timing the replay assumes.

15.4 The same lever read as erosion

Reversing decisions at random is not only a null. Read forwards rather than sideways it answers the question Part III has to ask about unseen data: how much directional edge can the book lose before it stops working? Reversing a fraction f of decisions moves accuracy from q to $q - f(2q - 1)$, so f is an accuracy scale, and $f = \frac{1}{2}$ destroys the edge exactly.

Measured on the same harness, the block score falls **linearly** at 25.0 points per one per cent of decisions reversed (the least-squares slope; the printed endpoints, from 1,145 at none to −95 at half, imply 24.8 on their own). Block by block it falls to the standard error of that block's own score at 38.7 to 40.9 per cent reversed, and reaches zero at 46.4 to 47.7 per cent — which, on the accuracy scale above, is the book's 54.07 per cent falling to between 50.19 and 50.30. Both crossings are intervals across the five blocks rather than single points, because the standard error they are measured against is itself a per-block quantity.

There is no cliff. The strategy has no critical level of lost edge at which it breaks; it degrades in proportion, and remains profitable until its directional accuracy is within two or three tenths of a point of a coin. The failure mode to plan against is therefore slow decay rather than a regime break through a threshold, and the quantity to monitor is accuracy itself, for which the ruler above gives the exchange rate.

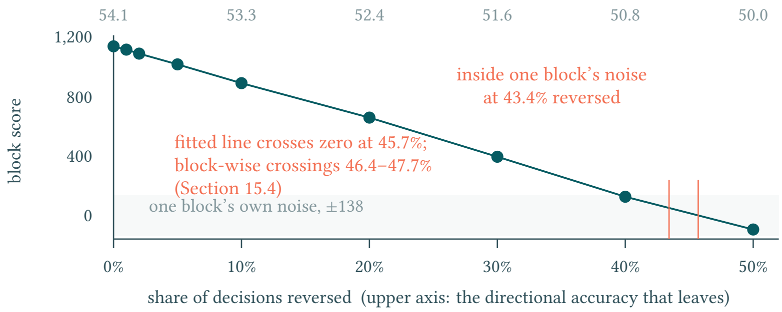


Figure 29: The failure envelope. Score against the share of decisions reversed, with the upper axis reading the same sweep as directional accuracy. The shape is the finding: a straight line, not a cliff.

Boundary. Random sign reversal is a model of lost edge, not of any particular regime change: it erodes direction uniformly, where a real regime break would concentrate the loss in time or in a subset of instruments. The envelope bounds how much accuracy the book can lose, not how quickly it might. The injection is a counterfactual on the decision cells of one recorded path; it prices accuracy *at the operating point*, and the linear form is a local one. It is not a claim that the accuracy is attainable, and the oracle is not a strategy: nothing here says the ceiling can be approached, only what the approach is worth per point gained. The cell count includes the index leg, because the measured path holds a position in it on every day and the settlement charges it a commission like any other: a scored block therefore carries $250 \times 51 = 12{,}750$ cells, and across the five blocks 63,252 of the nominal 63,750 carry a decision at all.

Scope. Both coefficients, the break-even accuracy, the per-cell values of both constructions and the book's own accuracy are measured here from the retained price panel and position path, on the settlement that Section 17 validates against the evaluator's own daily record. The arithmetic cross-check between the two constructions is a check on the linear form, not an independent confirmation of the data.

16 What the Edge Is Made Of, and What It Is Not

Four ways a score like this is usually an artefact. A book that scores well over five quarters can be doing so for reasons that have nothing to do with a repeatable edge: the whole result may sit in one instrument, or in a handful of days, or in a single leg that is really a market bet in disguise. Each of those is a specific alternative explanation, and each can be settled on the validated settlement of Section 17 by switching part of the book off and rescoring. This section reports four such tests. None of them is a performance claim; each is an attempt to make the measured score go away.

16.1 Which leg carries it

Rescoring with one leg zeroed separates the fifty single-name legs from the index instrument. The stock legs alone score 977.47, 1,131.43, 1,053.30, 985.08 and 965.66 against the whole book's 1,063.11, 1,160.12, 1,136.86, 1,197.30 and 1,167.05; the index leg alone reaches 61.11, 6.96, 54.45, 194.23 and 186.67. **The stock legs hold between** 82 **and** 98 **per cent of the score in every block**, and the ordering never reverses.

The parts do not sum to the whole, and the reason is instructive. The two legs' daily profits recompose the book's daily profit exactly — to 2.7×10^{-12} dollars, since profit is a sum over instruments and the fee is a per-column charge — and rescoring that sum returns the book's own score. But the two legs' separate *scores* fall short of it by 14.72 to 29.11 points, because the score divides a mean by a dispersion and each leg alone pays a heavier Sharpe haircut than the pair does together. **Score is not additive in profit**, which is why every ablation here is rescored rather than apportioned.

Table 20: Every ablation, rescored. Each row switches part of the submitted book off and puts the remainder through the evaluator's settlement; the score is not additive in profit, so the rows do not sum to the first.

Scored block	251–500	501–750	751–1,000	1,001–1,250	1,251–1,500
The whole book	1,063.11	1,160.12	1,136.86	1,197.30	1,167.05
Stock legs only	977.47	1,131.43	1,053.30	985.08	965.66
Index leg only	61.11	6.96	54.45	194.23	186.67
Long cells only	247.05	628.48	807.83	654.35	692.11
Short cells only	695.38	414.29	205.65	432.76	375.32

Where the retained data cannot settle it. Against a null built by scrambling the same book's own directions, the stock legs and the whole book stand more than eleven null standard deviations clear in every block, with no draw in five thousand reaching the observed score. The index leg does not. In the first three blocks its standalone score falls inside its own scrambled null — 7.2, 29.5 and 10.0 per cent of draws reach or exceed it — and only in the last two does it stand clear at 0.18 and 0.10 per cent. **On the first three blocks the retained data cannot distinguish a weak index-leg edge from incidental index profit**, and this report does not claim one.

16.2 Whether it sits in a few names

Zeroing one instrument at a time, across all fifty-one and all five blocks, gives 255 rescores and a contribution spectrum. The largest single contributor accounts for 5.1 to 7.5 per cent of the summed drop, the largest five for 23.7 to 33.4 per cent, and **between eleven and fifteen instruments must be removed before a block's score halves**. Read as an effective count — the participation-ratio construction of

Section 8, applied here to the contribution shares — the spectrum carries 28.6 to 34.6 names of a nominal fifty-one.

Directional accuracy is spread the same way: across instruments within a block it runs from 44.9 to 63.1 per cent with a standard deviation of about four points, and no instrument is close to being the book. **The edge is broad.** That is the opposite of what a lucky concentration looks like, and it is also what Part II's dependence structure predicts: a panel whose independent cross-sectional count is four to five does not offer a way to be right about one name and wrong about the rest.

16.3 Whether it sits in a few days

Replacing the k best days of each block with that block's median day and rescoring gives a retention curve. Averaged over the five blocks the score retains 98 per cent at $k = 1$, 96 at $k = 2$, 91 at $k = 5$, 84 at $k = 10$ and 68 at $k = 25$ — a tenth of the block. **No block loses a majority of its score to any k up to ten**, and the book is profitable on 173 to 181 of the 250 days in each block.

The same curve computed for an accuracy-matched null puts the book at the 53rd to 87th percentile of that null in every block, never below it, so the book is if anything *less* dependent on its best days than the accuracy-matched null of Section 15 is.

16.4 Whether it is a market bet in disguise

Splitting the book into its long and short cells and rescoring each (Table 20) shows both legs earning in every block, and **which of them leads changes across blocks** — the short leg leads the first, the long leg the third — so neither is the book.

Taken alone each leg is unmistakably a market position: the index return explains between 0.78 and 0.88 of its daily profit variance. **The market neutrality is a property of the pair, not of either half**, which is what a cross-sectional strategy on a panel with a 21.55 per cent common factor should look like, and it is the reason Section 8 treats that factor as a constraint to be cancelled rather than a signal to be traded.

Boundary. These four tests remove alternative explanations; they do not supply a positive one. Each is an ablation of one recorded path on published days, so each inherits every limit Section 11 states about that path, and none of them speaks to the hidden window. A test that fails to make a result disappear is weaker evidence than a test that predicts one.

Scope. Every figure in this section is measured here by rescoring the retained position path through the settlement Section 17 validates against the evaluator's own daily record, and each is reproduced by the script the register names for it. The nulls are built from the same book by scrambling its directions, so they carry its dependence structure rather than an assumed distribution.

17 The Price of Selection: What Luck Alone Is Worth

A threshold before a comparison. The sections that follow weigh candidates against the submission. Any such comparison faces an objection that no amount of care in the measurement answers: with enough candidates, one of them leads on the scoreboard for no reason at all. A margin only means something once the margin luck alone would produce has been named — and named *first*, before any candidate has been looked at, so that the threshold cannot be chosen to suit the answer.

17.1 The harness, and the check it has to pass first

The experiment below is run here, from records this report retains: the published price panel, the submitted integer position path, and the fee and cap constants of the pinned rules contract. Nothing else enters it. The scoring rule is the branched form of Equation (1), of which Equation (10) is the positive branch the book never leaves, and the settlement is the evaluator's, read out of the retained daily ledger rather than assumed.

Three details of that settlement were read rather than assumed, and each one moves the answer. A scored block opens from a flat book, so its first day is unscored and its only event is the turnover of establishing the position — the same fact Section 4 uses to show the activity floor cannot bind. A commission is generated when a trade prices and charged on the following day, so the fee subtracted on day D is the fee generated on $D - 1$. And the holding decided on day D earns the price change into $D + 1$.

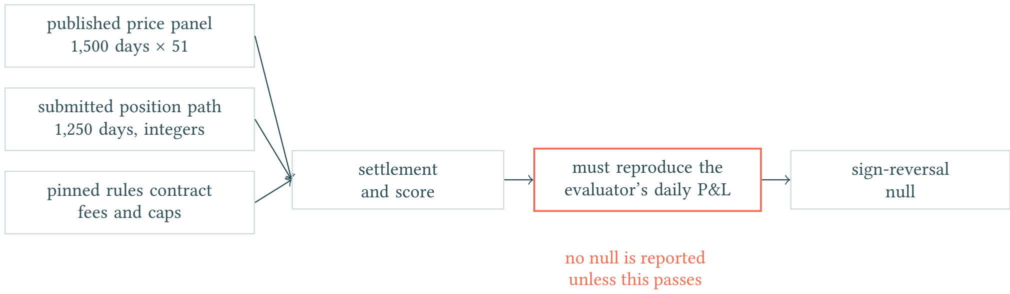


Figure 30: What the null is computed from. Every input is a record this report retains, and the chain is gated: the settlement must reproduce the evaluator's own daily profit and loss before it is allowed to measure anything.

That gate is not decorative. Replaying the unperturbed path across all five scored blocks — 1,250 days — the reconstruction agrees with the evaluator's recorded daily profit to within 1.9×10^{-10} dollars, and the five block scores it produces,

$$1{,}063.11, \quad 1{,}160.12, \quad 1{,}136.86, \quad 1{,}197.30, \quad 1{,}167.05,$$

reproduce to six decimal places the endpoint scores Section 11 obtained by an independent route. A settlement that agrees on one block and drifts on another is still the wrong settlement, and a null measured by a wrong settlement is a wrong null; the script refuses to report one until this check passes.

Those five figures also fix the scale everything below is read against. Their standard deviation is 50.55 points: **the same strategy, on five consecutive quarters of the same panel, varies by about fifty points for reasons that have nothing to do with any choice a researcher made.**

17.2 The null that fits the question

A zero-skill candidate is not a random strategy; it is *this* strategy with its directional information destroyed. The construction takes the submitted position path and reverses the sign of N decision cells drawn at random, leaving the grid, the caps, the fees and the settlement untouched, then re-settles and rescores. Reversing a holding changes the trade required to reach it, so the commission is recomputed too — the perturbation is charged for itself. What this measures is the price of the act of choosing, with everything except direction held fixed.

Table 21: Measured here. Reversing a random subset of decisions, 48 draws in each of the five scored blocks. The mean falls because destroyed direction is destroyed profit; it is the *spread* that prices selection.

Share of decisions reversed	1%	2%	5%	10%	20%
Cells reversed, of 12,750	128	255	638	1,275	2,550
Mean score	1,122.6	1,095.5	1,022.7	895.5	662.3
Score dispersion σ	24.97	35.97	56.31	80.98	99.71

At five per cent of decisions the dispersion is $\sigma = 56.31$ points. Repeating the measurement on disjoint seeds gives 51.64, so the quantity is a property of the perturbation rather than of one random stream. Sweeping N gives a power law with a measured exponent of 0.473 against the 0.500 a square-root rule would produce — close, and slightly sublinear, which is what dependence between neighbouring decisions predicts.

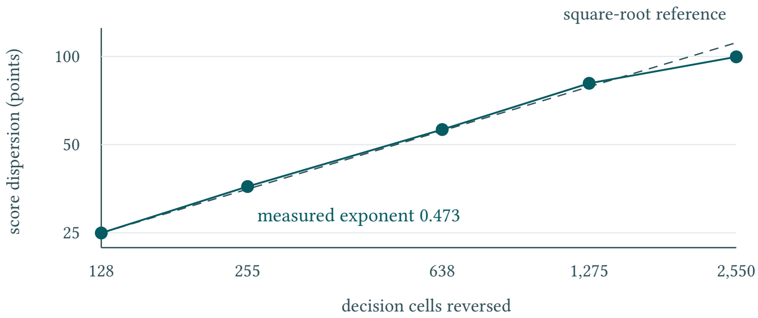


Figure 31: Dispersion of the null against the number of decisions destroyed, on log axes so the exponent is checkable by eye. The dashed line is the square-root rule anchored at the smallest perturbation.

17.3 Pricing the maximum

If a search produced K such candidates, the expected best of them is the expected maximum of K zero-mean draws. The expectation is taken here by order statistics on the measured distribution rather than on an assumed normal one, and the draws are centred within each block: the differences between block means are differences between market periods, not dispersion a search could exploit. The closed-form bound $\sigma\sqrt{2\ln K}$ is reported beside it.

Table 22: Measured here: what the act of picking the best of K zero-skill candidates is worth, before any candidate is examined. Three counts of the development search are carried because the conclusion is insensitive to which is used.

Search size K	248	600	1,681
Expected maximum, order statistics	+158.6	+166.2	+168.9
Bound $\sigma\sqrt{2\ln K}$	+187.0	+201.4	+217.0

Picking the best of a development search is worth about a hundred and sixty points even when nothing in the search has any skill. Against the submitted book's mean block score of 1,145, that is a fourteen per cent lead available for free. It is also three times the block-to-block dispersion of the strategy itself, and four times the entire cost dimension that Section 13 priced at about forty points by abolishing commission altogether.

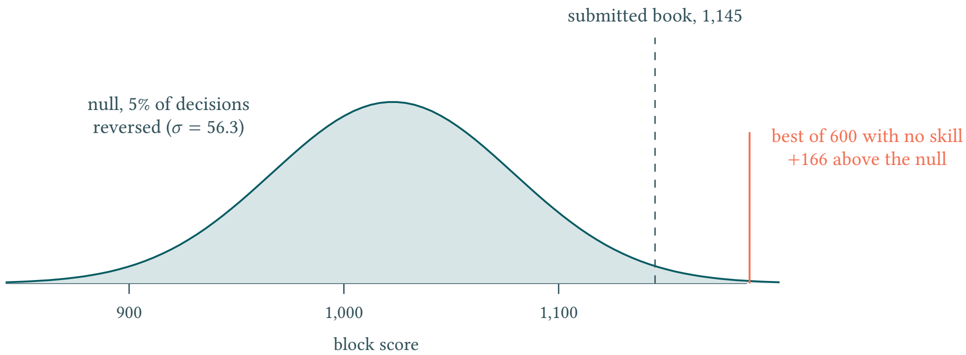


Figure 32: The measured null beside the quantity it judges. The distance between the null's centre and the best-of-600 mark is what a search buys with no skill at all.

17.4 What this licenses, and what it does not

The threshold is used in one direction only: a candidate whose margin over the submission falls below it is *not distinguishable from zero skill by this test*, and no ranking may be built on such a margin. The converse does not follow. A margin above the threshold is not thereby established as skill; it would merely have survived this particular objection, and would still owe the paired test and the out-of-sample evidence that Section 9 requires.

One qualification comes from Part II rather than from the construction. The max-of-K bound assumes the K candidates are independent draws, and Equation (32) measures this panel as carrying between four and five independent cross-sectional units against a nominal fifty. Candidates built on it inherit that dependence, so they are not K independent draws either. Positive dependence lowers the expected maximum, so the printed threshold overstates the price of selection. The conclusion does not turn on that assumption, because it does not rest on the threshold's exact level: the largest margin on the only available account is +1.14, two orders of magnitude below +166, and no plausible dependence correction brings the price of selection within reach of it.

Boundary. The null is a sign-reversal perturbation of one path, so it prices the selection luck available *within the family the search actually explored.* It says nothing about a candidate whose mechanism differs structurally rather than directionally, and it is not a significance test of any individual comparison. The three values of K are three ways of counting the development search rather than a measured quantity.

Scope. Every figure in this section is recomputed here from retained records and can be reproduced by running the script the evidence register names for it; none is carried from elsewhere. The section introduces no claim about performance — the perturbed paths are not strategies anyone proposed, and their scores are used only as a ruler.

18 The Candidate Ledger: The Right Tail Is Empty

The question, and the two registers that answer it differently. The development period left a family of near relatives to the submitted engine. Did any of them beat it?

Two records speak to this and they must not be merged. The lineage register this report retains is a source-hash and authorisation ledger: it records nine candidate identities, binds eight of them to source files whose digests recompute, and marks the state of each as proposed, historical, not evaluated, research-only or unavailable. It carries no scores at all, and the process that produced it declares that it executed nothing. Its *not run* entries are therefore statements about authorisation, not observations that no run occurred.

The frozen technical record does carry scores, reporting a common-grid replay of every candidate. No artefact of that replay survives in any tree available to this report. The table below is that record's account, transcribed.

Table 23: As reported by the frozen record: seven near relatives replayed on the same grid alongside the submitted engine, with each difference taken from it. No retained artefact reproduces any row.

Candidate	Difference from the submitted engine	Five-block mean	Δ
Confidence-cap variant	Position geometry only	1,146.03	+1.14
Overlay trigger variant	Trigger timing	1,145.58	+0.69
Dual-engine variant	Two engines, pre-registered protocol	1,144.94	+0.05
Submitted engine	—	**1,144.89**	—
Cap-geometry variant	Confidence factor and caps	1,142.33	−2.56
Two-book variant	One core, two calibrations averaged	966.79	−178.10
Soft-prior variant	Pairing information demoted to a prior	952.20	−192.69
Early variant	Mechanism not archived	835.06	−309.83

Reading the table against the threshold, not against zero. Section 17 priced the act of picking the best of this search at about a hundred and sixty points under no skill at all. **The largest margin any relative achieved is +1.14 — seven tenths of one per cent of that price.** The right tail is not thin; on this evidence it is empty. Three candidates lose by more than a hundred and seventy points, which places them outside the question entirely.

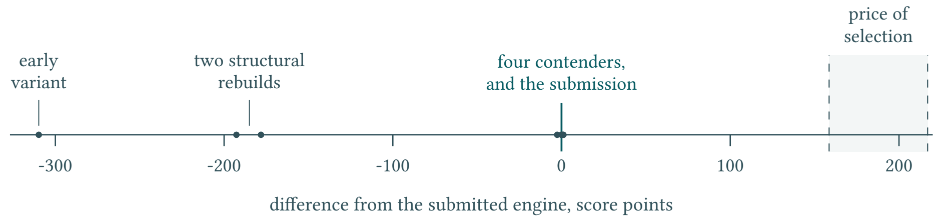


Figure 33: Every relative on one axis, against the price of selection. The four contenders are indistinguishable at this scale, which is the point: the largest of them stands at +1.14 where picking the best of a zero-skill search is expected to be worth +159 to +169; the band is drawn out to +217, the closed-form bound at the largest search size. Candidate differences are transcribed (Table 23); the band is measured in Section 17.

The nominal leader, replayed here. One of the nine identities is not transcribed after all. The archive the perturbation work of Section 15 already loads carries a second complete position path, and it had never been scored. Scoring it on the validated settlement of Section 17 gives a five-block mean of +1.115 above the submitted engine — within three hundredths of the +1.14 the frozen record reports for its leading relative.

That is where the agreement ends. The blocks the mean is taken over are −35.74, −7.43, +22.65, −0.75 and +26.85: **the candidate loses three of the five**, and the paired statistic across them is $t = 0.099$ against a standard error of 11.32. A mean of +1.1 on a block-to-block swing of ±30 is not a small consistent lead; it is the average of a coin.

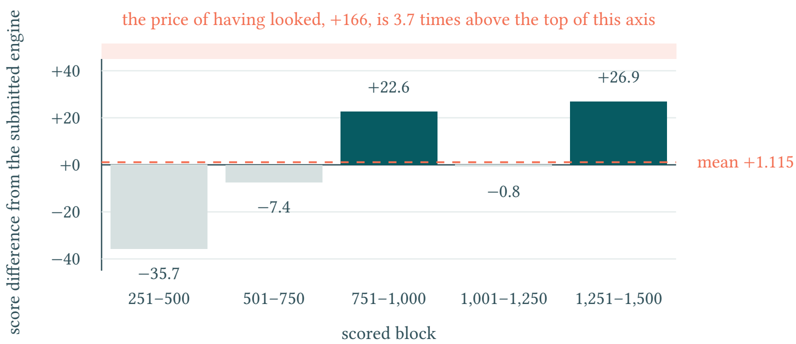


Figure 34: The replayed candidate's margin, block by block, measured here on the retained position path. The mean is positive and three of the five blocks are not. The threshold this margin has to clear does not fit on the axis: the price of picking the best of a zero-skill search is +166, and the whole plot spans ±45.

The dissection does not describe this path. The frozen record's account of its leading relative is a candidate differing from the submission in 19 of 63,750 decision cells, all in the index leg, with no change of direction anywhere. The retained path differs in 1,838 **cells across all fifty-one instruments and 848 days, every one of them a sign reversal**, of which the index leg carries 25. Either the retained path is that candidate and the dissection is wrong about it, or it is a different candidate whose five-block mean coincides to three hundredths of a point. **This report cannot tell which, and does not choose.**

Two things follow. The first is that an argument this section previously rested on has to be withdrawn: it held that the leading relative changed no direction and therefore competed only in the magnitude dimension Part I had already closed, and no retained artefact supports that. The second is that the conclusion does not depend on it. The margin is measured here rather than reported: +1.1 points ($t = 0.099$) against a threshold of +166 points, so it is falsified by the measurement directly, without needing any claim about what kind of difference it was.

One candidate has no identity. The dual-engine variant appears in the frozen record but in no lineage artefact: no candidate identifier, no source path, no digest. It is carried here as a named claim of that record and is excluded from every count sourced to the register. The register itself, with the gate recorded against each identity, is Appendix B.

Boundary. This section does not establish that the relatives were never executed — the register cannot witness that — nor that the submitted engine is the best of them. It establishes something weaker and more defensible: that on the only account available, no relative cleared the price of selection, and that the one whose path this report retains does not clear it when replayed here either. The coverage is also partial. The seven near relatives (the table's eighth row is the submitted engine itself, at zero) vary synthesiser geometry, book structure, feature weighting, trigger timing and engine count; they do not vary window length or the feature set itself, and nothing here speaks to those.

Scope. Every figure in the table is transcribed from the frozen technical record. The replay of the retained path is not: its five-block differences, its paired statistic and its cell-level comparison with the submitted book are measured here on the settlement Section 17 validates, and are reproduced by the script the register records for them. No retained artefact of this report reproduces any of them, the harness named for them is absent from every available tree, and no candidate, evaluator or data digest is recorded against those runs, so comparability between rows cannot be checked. They are reported as that record's account, attributed. The lineage facts — nine identities, eight source digests recomputed, one unavailable — are a class B reproduction against a record this report retains, and are the only part of this section that is.

19 The Predecessor: Why the Higher Nominal Score Was Not Submitted

The uncomfortable fact this section exists to answer. The leaderboard position this report can point to was not set by the submitted engine. A retained snapshot of the standings records the entry at the top with a score of 984.62, and that row belongs to the predecessor, not to the strategy described in Section 10. Choosing to submit the successor therefore requires a reason, and the reason cannot be a better nominal number.

What the record reports. Replayed on the same grid as every other candidate, the predecessor's five block scores are 1,126.91, 1,275.47, 1,246.81, 1,006.76 and 930.95 — a mean of 1,117.38, below the submitted engine's 1,144.89. Its block-to-block swing — the population standard deviation of the five block scores — is 130.68 against the submitted engine's 45.22 on the same convention, a factor of 2.9. Its lead is not spread across the window: it wins the first three blocks and loses the last two. Under the control that separates contaminated from clean windows, its advantage of +96.37 in its own strongest window becomes −213.32 in the control.

A test the predecessor could have passed. A strategy that decays across a window invites an obvious objection: perhaps the market changed, and any strategy would have shown the same break. The record's answer is the part of this section worth borrowing.

Four mechanical rules were constructed as anchors — momentum over twenty days, one-day reversal, buy-and-hold, and a sixty-day volatility rule — none of which had any contact with the development process, and none of which was ever a candidate. Their only purpose is to be strategies whose behaviour cannot have been shaped by this data. Six daily profit sequences — the submitted engine, the predecessor and the four anchors — were then standardised, the common shape across the other five removed from each so that a market-wide move could not be mistaken for an individual break, and each regressed on an indicator for days after the public/hidden data boundary at Day 1,000, with standard errors robust to autocorrelation.

Table 24: As reported by the frozen record: the break test at the Day 1,000 boundary. The anchors are mechanical rules with no contact with the development process; they exist to say what a market-wide break would look like. Negative z: the series' level is lower after the boundary. The smoothing behind the second column is the record's; its window is not retained.

Series	Daily z	Smoothed z
Predecessor	−2.17	−3.00
Submitted engine	+1.20	+1.33
Four mechanical anchors	all $\lvert z\rvert < 1$	

The only series that breaks at the boundary is the predecessor. Four rules that never saw the development process pass through it untroubled, and the submitted engine does not break either. That is what the anchors are for: a change in the data registers in series that never saw the development process, and these four register none. What broke was one mechanism's fit to the stretch of history it had been developed against.

Why the boundary was chosen there, and not found. Day 1,000 is where the organiser's release schedule separates the fully published history from what was still hidden. It is given by the structure of the competition rather than selected by inspecting the series for the point that maximises the break, which is the difference between a test and an artefact.

The conclusion, stated at the strength the evidence supports. The predecessor's nominal advantage does not travel: its mean is lower, its swing is nearly three times larger, it reverses under the window control, and it is the sole series to break at a boundary the data's own release structure supplies. That is a reason to prefer the successor. It is not a demonstration that the successor will perform better on unseen data, and the leaderboard row remains what it is — a position achieved by the predecessor's realised sequence of holdings, not by its mechanism. This report states that openly rather than inheriting the number.

Boundary. A break test on one realised path with four anchors is evidence about *this* series at *this* boundary. It does not establish that the predecessor would fail on other data, nor that the submitted engine is stable in any general sense; Section 11 is the only stability evidence in this report that rests on a retained record, and it covers the submitted engine alone.

Scope. The leaderboard standing is a transcription of an external system's display, retained as a row-by-row record whose original capture was not kept; it is not reproducible and is reported as transcribed. The block scores, the swing figures, the window control and the break statistics are likewise transcribed from the frozen technical record, which no retained artefact of this report reproduces. The predecessor's identity — its source digest — is a class B fact against a record this report holds, and is the only part of this section that is.

20 Verdict: Three Lines, and What Survives Removing Any One

What independence means here, stated before it is claimed. Three lines of argument run through Part III. They are independent in method — none uses another's conclusion as a premise — but they are not independent in infrastructure: all three read the same replay. That shared foundation is not underwritten by any of them. It is underwritten separately, by the parity of Section 13, where the shadow ledger reproduced the official evaluator's own daily figure exactly across 1,250 scored rows. The independent settlement reconstruction of Section 17 agrees to within 1.9×10^{-10} dollars over the same rows, which is floating-point noise; the two are different objects and are not merged here. A reader who doubts one line may discard it; a reader who doubts the replay discards all three at once, and Section 13 is where that doubt has to be taken up.

Table 25: The three lines, and what remains if any one is struck out.

Line	What it establishes	Struck out
Price	Picking the best of the search is worth +159 to +169 under no skill; converted at about 190 points of score per accuracy point, that is a lead of 0.8 to 0.9 accuracy points available for nothing	The ledger still finds the tail empty: no relative cleared the threshold, and the one path replayed here posts $t = 0.099$
Experiment	No relative cleared the threshold; the predecessor is the only series to break at the release boundary	The price still puts a margin of this size inside what a zero-skill search buys
Information	Usable incremental structure is one to two per cent of the total. The ceiling it converts to depends on a divisor Section 8 leaves open, and the two readings fall on opposite sides of break-even	**This line is already struck out**: nothing below rests on it, and the price and the ledger still falsify each candidate individually

What the information line can no longer be asked to do. An earlier version of this section multiplied two quantities: the room between the engine's measured accuracy and the panel's accuracy ceiling, and the exchange rate that converts accuracy into score. The product bounded the remaining improvement at about forty points and was the decisive number of the report.

That product is withdrawn. The ceiling depends on how many independent directional decisions a day is taken to contain, and Section 8 now records that the two defensible answers — the nominal fifty and the four to five independent units the same chapter measures — put the ceiling at 53.4 and at 58.3–61.6 per cent respectively, on opposite sides of the 55.99 per cent a replacement signal must reach. **An upper bound that moves eight accuracy points with a choice the report has not settled is not an upper bound.**

What survives without it. Three things, none of which uses the ceiling.

The Sharpe axis is closed. The entire remaining value of risk-adjustment work is $\mu/(1 + \mathrm{SR}^2)$ by Equation (5) — by Equation (16) the sampling-noise share of the mean, which is what the evaluator withholds and nothing a strategy can trade for — which Section 11 evaluates at 14.1 to 16.5 points against a standard error of 131 to 143 on a single block's score. That is about an eighth of the noise, and the criterion was fixed from published constants before the strategy existed.

A replacement must clear a measured threshold. Equation (33) puts break-even at 55.99 per cent against the book's own 54.07: a signal that overwrites decisions must beat the book by nearly two accuracy points before it pays for the turnover it creates. This is a measured cost, not a bound on what the panel contains, and it does not depend on the divisor.

Nothing demonstrated an improvement. The one candidate whose path this report retains replays to +1.115 with a paired t of 0.099, losing three of five blocks, against a selection price of +166 (Sections 18, 17).

What is lost with the ceiling is the claim that the room is small. The report can say that no improvement was demonstrated, that any replacement must clear two accuracy points, and that the Sharpe axis is closed. It can no longer say how much room remains, and it does not.

The submission does not itself collect a selection premium. The threshold of Section 17 prices picking the maximum of a search. The submitted engine was not that maximum: three relatives posted higher nominal five-block means, and the predecessor posted a higher leaderboard score still. It was chosen on structural grounds — fewest unverified moving parts — and the higher nominal figures were rejected for the reasons Sections 18 and 19 record. Whatever else is true, the selection premium the threshold prices is not embedded in this choice.

How the parts fit. Part I is deduction: from the published rules alone, any qualifying strategy must maximise the mean, run the book extended, treat turnover as nearly free and refresh direction online. Part II states the problem those rules define and the protocol used against it. Part III checks the instance against every one of those consequences and prices each remaining avenue. The verdict is where the three prices are compared with the three dispersions.

The specific parameters inside that shape — the feature combination, the window length — are products of a development history, and this report does not manufacture derivations for them. Their defence is of two kinds and neither is a derivation: every substitution actually tried was weighed and none was better, and the dimensions never tried were not measured at all. This report makes no claim about their value in either direction: the bound that once capped them is withdrawn, and nothing replaces it.

The verdict. **The claim is not that this strategy won.** It is that with no candidate demonstrating an improvement, with any replacement owing nearly two accuracy points before it breaks even, and with the Sharpe axis closed by a criterion fixed before the strategy existed, the version with the simplest structure and the fewest unverified parts is the only defensible choice. **No upper bound on the remaining room is claimed**, because the report withdrew the one it had. Its weakness is stated rather than discovered: *the submitted engine has no out-of-sample record of its own.* The leaderboard position belongs to the predecessor, the five endpoints are public-data replays, and nothing in this report establishes performance on the hidden window.

Scope. This section introduces no measurement. The dispersion figures are recomputed here from the retained five-endpoint record: across the five blocks the submitted engine's score has a population standard deviation of 45.22 and its mean daily profit one of 44.93, and the choice of estimator matters — on the sample convention the score figure is 50.55, which is why each is named with its basis rather than quoted bare. The measured accuracy and the exchange rate are recomputed from retained records by Section 15 and carry its boundaries; the accuracy ceiling rests on the information budget, whose bit figures Section 8 marks as transcribed while performing the inversion here.

21 Limitations, and What the Report Establishes

The limitations are specific, so they are listed specifically. A limitations section that describes its subject as "preliminary" tells a reader nothing. The ones below are the places where this report's own machinery says a claim stops, each with the section that says it.

Table 26: Where the evidence ends. The right column is the section that states the limit, not a promise to address it later.

Limit	What it means for the claims	Stated in
The submitted engine has no out-of-sample record	Every performance figure here is a replay on published days. The leaderboard position belongs to the predecessor	§11, §19
The candidate comparison has no retained artefact	The lineage register binds eight of its nine identities to source files, but every score in the comparison is transcribed and no candidate, evaluator or data digest accompanies those runs, so row-to-row comparability cannot be checked	§18
The clip was never observed to act	Cap enforcement is read from the pinned source, not measured; the strategy stayed inside the caps on all 1,250 calls	§13
The five endpoints share history	They are not independent out-of-sample blocks; successive rows condition on overlapping evidence	§11
The information ceiling rests on an unsettled divisor	Converting the budget to a per-decision bound needs a count of independent daily decisions. The nominal fifty and the four to five effective units this report measures give ceilings of 53.4 and 58.3–61.6 per cent, which fall on opposite sides of break-even. No conclusion in Part III now rests on it	§8, §20
The block-bootstrap intervals are not uncertainty bands	Point estimates support the ordering; no uncertainty statement attaches to their magnitudes	§14
The effective-sample factor is cross-sectional only	The temporal axis is not combined with it into a single number, because the combination is not identified here	§6

One limitation is structural rather than evidential. Part II describes diagnostics performed during development against scripts that no longer exist. Where a retained input allowed it, those diagnostics were recomputed and the figures now stand as reproductions; where it did not, they are attributed to the frozen record and marked. The report cannot convert the second class into the first, and does not pretend the distinction is cosmetic: a reader who accepts only the reproducible half still has Part I entire, the engine audit, the causality contracts, the endpoint replay and the elasticity measurements, and would lose the candidate comparison and the transcribed half of the panel diagnostics.

One limitation is about how the report was written. Language models were used throughout the research and throughout the drafting. No claim here rests on a model's assertion: every number is a closed-form consequence of published constants, a recomputation against a retained record, or a transcription marked as such. What is worth reporting is narrower — the model-produced claims that were caught being *wrong*. Three are recorded, and all three are of one kind. None was a wrong number. Each was a number correctly reported and wrongly scoped: attached to the predecessor's leaderboard position rather than the submitted engine, or to a stronger class of evidence than the one that produced it. **That is the failure mode this report's evidence classes exist to resist**, and it is why a class is a property of a retained record rather than a judgement a drafter makes. A record of caught errors bounds nothing about uncaught ones, and no claim is made that the three are exhaustive.

What is established. Three things, in decreasing order of how firmly.

First, what the rules require. From published constants alone and without any price data, a qualifying strategy must maximise the mean rather than the risk-adjusted mean, must run the book fully extended, may treat turnover as nearly free, and cannot be excluded by the activity floor at any competitive size. These are closed-form consequences, and Section 13 confirmed each against the evaluator's own arithmetic.

Second, where the objective stops paying. The research range that Part I stated as a function evaluates, on the standard error of a single scored block's score, to S^* between 2.7 and 2.8. The book runs at a Sharpe between 8.0 and 9.1, nearly three times that, and the entire remaining return to Sharpe work is 14.1 to 16.5 points of score against a standard error of 131 to 143 on the score it must be demonstrated in — about an eighth of it. **The risk-adjustment axis is closed, and closed by a criterion fixed before the strategy existed.**

Third, that no better instance was demonstrable. Under a threshold measured on the validated settlement, no relative cleared the price of selection and the predecessor's higher nominal score does not travel: the largest margin any relative achieved is +1.14 as transcribed from the frozen record (+1.115 on the one retained path that replays) against a selection price of +166. This is the weakest of the three claims and the one most dependent on transcribed evidence, which is why Section 20 states it as an absence of demonstrable improvement rather than as optimality.

What is not established. That the strategy will perform well on unseen data. Nothing in this report speaks to the hidden window, and the honest summary of its contribution is a method rather than a result: a way of deciding, before looking at data, which research directions the published rules have already closed — and then of pricing every remaining direction in the objective's own units, so that an improvement too small to measure is recognised as such instead of being pursued.

Scope. This section introduces no measurement. Every limit in the table restates one already stated where the evidence for it appears, and every figure quoted in the conclusion is carried from the section named beside it.

Postscript: The Verified Outcome

The body of this report was closed before any hidden-window result existed, and it promised nothing about one. This page records the one outcome that has since become checkable, and reads it against rulers the report had already fixed.

The fact. At the awards presentation on 18 August 2026 the organisers displayed the official Technical Score Leaderboard for the Final round — the round scored on the hidden Days 1,501–2,000 release. The submission of this report, **abc123, placed third**, with a technical score of **1,085.169047**. The board is transcribed row by row in the retained capture (Appendix D); technical score is the quantitative half of the published judging split.

Table 27: The top of the official Final-round Technical Score Leaderboard, as displayed at the awards presentation and transcribed in the retained capture. Ten teams were shown; the full transcription is the retained record.

Rank	Team	Technical score
1	I_m_going_to_quit	1,247.727312
2	Axion_Labs	1,088.898739
3	**abc123 (this report)**	**1,085.169047**
4	Crank_the_Money_Machine	1,026.5222
5	Markov Monkeys	1,000.43887

Read against the report's own rulers. Every number below is arithmetic on quantities this report had already printed; nothing is measured here. The official hidden-window score sits 59.72 points below the public-data replay mean of 1,144.89 — 0.42 to 0.46 of the 131 to 143 standard error of a single block's score, and about 0.31 accuracy points at the exchange rate of Section 15. **The out-of-sample result landed inside half a block's noise of the replay mean**: the transfer this report declined to promise arrived within the tolerance its own noise analysis had priced. The gap to the runner-up is 3.73 points — three hundredths of that standard error, indistinguishable from a tie on the report's ruler. The gap to the leader is 162.56 points, roughly 1.2 standard errors, or about 0.86 accuracy points at the same exchange rate.

What this does and does not verify. It verifies that the submission was accepted, ran on the hidden window, and scored within the noise band the public replay implied — the erosion-not-cliff reading of Section 15 survived contact with unseen data. It does not verify any mechanism claim: one scored window cannot separate skill from a favourable draw, for exactly the reasons Part III priced, and a 1.2-standard-error gap to the leader is likewise not evidence that the leader's edge is real. The body's evidence grades are unchanged; nothing upstream of this page is upgraded by the outcome.

Scope. The board was photographed from the projected display and transcribed row by row; the photograph is retained by the author and the transcription is a retained record with its digest in Appendix D. The window designation for the Final round is taken from the published schedule and the event's announcement, consistent with how Section 6 treated later-stage windows as reported rather than verified.

References

[1] Algothon 2026 Organising Committee, *Competition Schedule*, Algothon 2026 Wiki, 2026. `https://wiki.algothon.au/schedule/` (accessed 11 August 2026).

[2] Algothon 2026 Organising Committee, *Scoring and Evaluation*, Algothon 2026 Wiki, 2026. `https://wiki.algothon.au/evaluation/` (accessed 11 August 2026).

[3] Algothon 2026 Organising Committee, *Competition Rules*, Algothon 2026 Wiki, 2026. `https://wiki.algothon.au/rules/` (accessed 11 August 2026).

[4] Algothon 2026 Organising Committee, *Submission Requirements*, Algothon 2026 Wiki, 2026. `https://wiki.algothon.au/submission/` (accessed 11 August 2026).

[5] UNSW FinTech Society IT, *Algothon 2026 Starter Code: Official Evaluation Program*, *eval.py*, commit dated 16 July 2026. `https://github.com/UNSW-FinTech-Society-IT/algothon26-starter-code/blob/8c04f39e2ae730495bc3030cf012295b32d16244/eval.py` (accessed 11 August 2026).

[6] H. R. Künsch, *The jackknife and the bootstrap for general stationary observations*, Annals of Statistics 17(3), 1217–1241, 1989. `https://doi.org/10.1214/aos/1176347265`

[7] S. Boucheron, G. Lugosi and P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence*, Oxford University Press, 2013. `https://doi.org/10.1093/acprof:oso/9780199535255.001.0001`

[8] L. Laloux, P. Cizeau, J.-P. Bouchaud and M. Potters, *Noise dressing of financial correlation matrices*, Physical Review Letters 83(7), 1467–1470, 1999. `https://doi.org/10.1103/PhysRevLett.83.1467`

A Rules Evidence

Every claim Part I takes from the published pages. Class A claims carry the capture they were read from — a digest identifies the retained bytes, and the capture itself is held in the report's evidence tree; class D rows are descriptive records with no capture, shown with an em-dash.

Table 28: Official rule claims and their captures.

Claim	Class	Statement	Digest
RULE-UNIVERSE	A	The competition universe contains ALGO plus 50 synthetic instruments.	ed11c9eec4dc7912
RULE-INTERFACE	A	Submissions provide 51 desired integer end-of-day holdings through getMyPosition(prcSoFar).	ed11c9eec4dc7912
RULE-CAPS-FEES	A	ALGO has a $100,000 position cap, other instruments have $10,000 caps, and commissions differ by instrument class.	ed11c9eec4dc7912
RULE-SCORE	A	The evaluator scores mean daily P&L with a Sharpe-derived multiplier using a fixed sqrt(250) annualiser.	22c06d2618a6d174
RULE-TESTING-ACTIVITY	A	The Testing Round requires at least $25,000 total dollar volume.	ed11c9eec4dc7912
STAGE-TESTING	A	Testing observes Days 1–500 and scores Days 501–750 over 250 days.	22c06d2618a6d174
STAGE-GENERAL	A	General initially observes Days 1–750 and scores Days 751–1000; its leaderboard window expands from 50 to 250 days.	22c06d2618a6d174
STAGE-FINALIST	D	Finalist is described as Days 1–1000 forecasting Days 1001–1500 over 500 days.	—
STAGE-FINAL	D	Final is described as Days 1–1500 forecasting Days 1501–2000 over 500 days.	—

B Candidate Lineage and Authorisation Gates

The nine registered candidate identities with the state recorded against each. The register carries no scores: it records what was authorised, not what performed. Section 18 explains why its NOT_EVALUATED (*not run*) entries are permission states rather than observations.

Table 29: Candidate identities and the gate each is held behind.

Identity	Decision state	Grade	Missing gate
Simple2	PROPOSED_CURRENT	A_RECOMPUTE	Finals OOS, upload, and acceptance remain unverified
Final2	HISTORICAL_CONTEXT_ONLY	D_RECORD	Separate execution authorisation and common-protocol replay
Simple4	NOT_EVALUATED	D_RECORD	Separate execution authorisation and common-protocol replay
Simple4-B	RESEARCH_ONLY	D_RECORD	Research-only source; no execution, submission, or promotion authority
Simple6	NOT_EVALUATED	D_RECORD	Separate execution authorisation and common-protocol replay
Simple11	NOT_EVALUATED	D_RECORD	Separate execution authorisation and common-protocol replay
V8	NOT_EVALUATED	D_RECORD	Separate execution authorisation and common-protocol replay
PRIOR_OVERLAY_P1	RESEARCH_ONLY	D_RECORD	Research-only source; no execution, submission, or promotion authority
Simple2_2	UNAVAILABLE	D_RECORD	Exact source bytes

C Reproduced Measurements

Every claim the report grades class B: a measurement recomputed here against a record the report retains. These are the claims a reader can check without trusting any narrative in this document.

Table 30: Class B claims and what each reproduces.

Claim	Class	Statement
LOCAL-REPLAY-500	B	A deterministic local public-data replay reaches endpoint 500 over Days 251–500.
LOCAL-REPLAY-750	B	A deterministic local public-data replay reaches endpoint 750 over Days 501–750.
LOCAL-REPLAY-1000	B	A deterministic local public-data replay reaches endpoint 1000 over Days 751–1000.
LOCAL-REPLAY-1250	B	A deterministic local public-data replay reaches endpoint 1250 over Days 1001–1250.
LOCAL-REPLAY-1500	B	A deterministic local public-data replay reaches endpoint 1500 over Days 1251–1500.
REPLAY-FIVE-ENDPOINT	B	One frozen submission scored at five endpoints returns mean daily P&L 1079.54 to 1212.95 dollars with annualised Sharpe ratios between 8.04 and 9.06.
REPLAY-ELASTICITY	B	Local score elasticities at the replayed operating point satisfy the derived identity to within four parts in ten trillion.
REPLAY-SHADOW-LEDGER	B	A shadow ledger reproduces the official evaluator daily profit and loss with maximum absolute discrepancy zero across all scored days at the five endpoints.
REPLAY-RISK-METRICS	B	Per-endpoint exposure, turnover, fee and cap-utilisation metrics are recorded for the replayed book.
REGIME-BLOCK-METRICS	B	Average pairwise correlation across six contiguous return blocks lies between 0.177 and 0.213, and the top covariance eigenvalue share between 0.181 and 0.215.
TRANSFER-BLOCK-SIMILARITY	B	Adjacent-block covariance structure similarity is 0.509 to 0.600 and one-step ridge map similarity 0.105 to 0.157.
PANEL-INDEX-IDENTITY	B	Instrument 0 is the re-based equal-weight aggregate of the other fifty: reconstructing it from the published panel reproduces it with price correlation 0.999999972 and maximum absolute discrepancy 0.008731 index points.
PANEL-LEADING-EIGENVALUE	B	The leading eigenvalue of the fifty-stock return correlation matrix carries 21.55 per cent of the trace, against the 2 per cent a structureless panel would assign any single direction.
PANEL-DEMEANING-FLOOR	B	Cross-sectional de-meaning takes the mean pairwise correlation from 0.1939 to -0.0161, against the arithmetic floor of -1/49 = -0.0204.

D Retained Records and Digests

The complete manifest. Digests are SHA-256 over the file's bytes, truncated to the first 16 hexadecimal digits. Recomputing the digest of any file listed here reproduces the value shown; the contract suite performs exactly that check on every build, and fails if any retained record has drifted from what the register claims about it.

Table 31: Every retained record with its digest.

Retained record	Digest
rules.html	ed11c9eec4dc7912
evaluation.html	22c06d2618a6d174
submission.html	6d7917610da0f2aa
schedule.html	4c3f2d8427d434e5
rules_contract.json	c526988373c53558
evaluation_summary.json	0895109257b10eae
lineage_manifest.csv	b7dba5e356b6648d
leaderboard_metrics.json	5a404e2ceae8d447
before_final_leaderboard_snapshot.csv	ffdd33dce247c8db
fused.html	c7918dcb874416bf
index.html	df36e87c9801b83f
claude_fused_master.pdf	7689b596211d279e
claude_gate_master.pdf	9828586b441da2ee
five_block_evaluator_summary.csv	02c21b45710d925c
score_local_sensitivity.csv	5c7e4bd15738bcfb
simple2_daily_ledger.csv	63d5c94ef9263168
simple2_risk_metrics.json	112d979f3608a671
simple2_positions_summary.csv	deba2f834e06a4f7
data_regime_metrics.csv	c3c795d0119b067d
transfer_metrics_v2.csv	9e49bde92d2722ed
candidate_comparison.csv	5b0ad7a2dca16579
evidence_manifest.json	6d85170338680fc1
claims_registry.csv	82dd6a42c85c62e7
public_prices_1_1500.txt	66c21813b394dd3b
session 77c9b294 (digest over its first 7,972 records)	0b5f4b85a9024ab5
session dca2a9fe (digest over its first 1,827 records)	11d71b9f8948c44f
final_technical_leaderboard_transcription.csv	cb58b7210fb138ef